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The integration of renewable energies into the electricity systems of North Africa

De integratie van duurzame energie in het
Noordafrikaanse elektriciteitssysteem

(met een samenvatting in het Nederlands)

Proefschrift

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door

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geboren op 6 Juni 1974
te Würzburg, Duitsland

Promotor:
Prof. dr. K. Blok

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I. Introduction

1. Background and motivation

Stretching over a length of 4000 km, from the Strait of Gibraltar to the Sinai, the North African region has for centuries been a cross-bridge of trade, migration and cultural exchange between Europe, Africa and Asia. With the discovery of large oil and gas fields in the 1950s, the region also started to emerge as a player on the global energy markets. Algeria, Libya, and Egypt are today large producers (and partially exporters) of oil and natural gas, but also Tunisia and Morocco, less endowed with hydrocarbon reserves, have become important energy transit countries, as their territories are crossed by a network of natural gas pipelines interlinking North Africa with the European continent. But fossil fuels are not the only energy asset of North Africa. It is widely known that the region is also endowed with abundant renewable energy resources. The North African coastlines, particularly those of the Atlantic and the Red Sea, feature excellent wind conditions, while the region's solar potential is virtually endless due to the vast areas of undeveloped, sun-drenched land (see Figure I-1). It is surprising that despite this abundance, North Africa still significantly lags behind other regions of the world in terms of renewable energy deployment. As illustrated in the diagram in Table I-1) the non-hydro share in the electricity generation mix - i.e., wind and solar power - remains at a very low level across all North Africa countries.

What can be proposed to improve this picture? High hopes have been pinned on the 'Desertec' concept, the idea of tapping North Africa's wind and solar potential for renewable electricity exports to Europe. The project, which announced a long-term goal of meeting 15% of Europe's electricity demand by renewable power from the MENA region (Desertec, 2009), brought a veritable boost to the (media) popularity of North African renewable energies. However, although the concept has been intensively advocated by a variety of public and private stake-

holders¹, it still remains to be proven whether power exports can actually become a true driving force for the expansion of renewable technologies in the region.

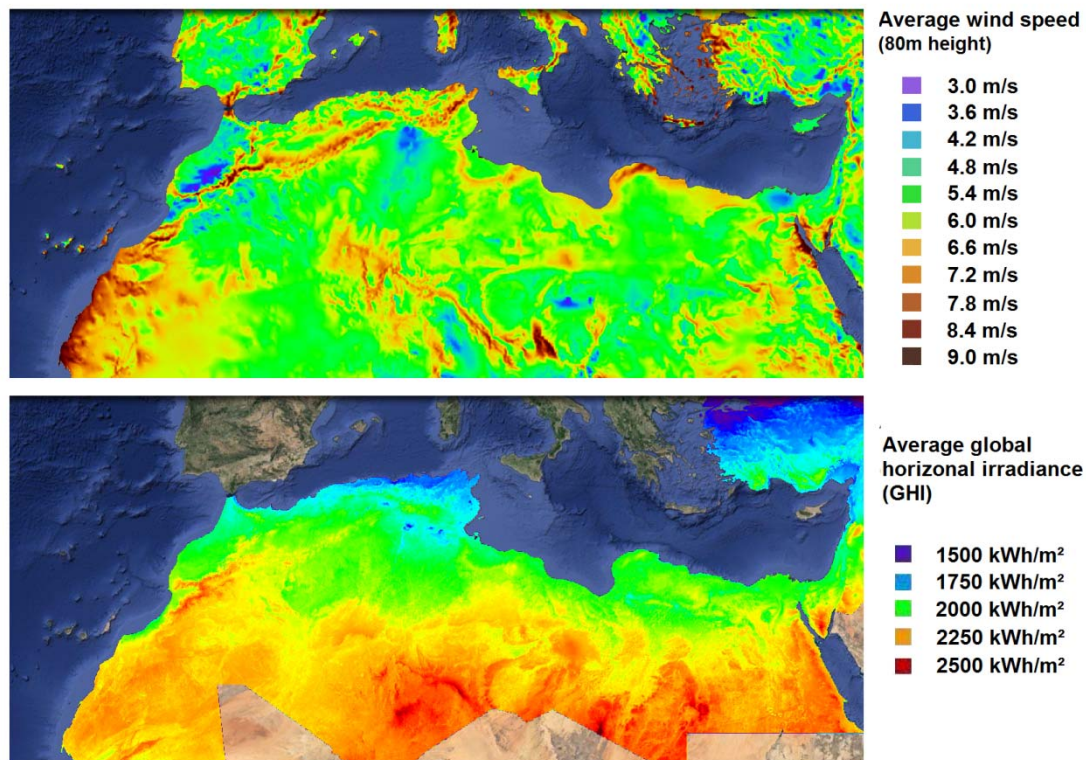


Figure I-1: North Africa: Average wind speed and solar irradiance. Source: IRENA Global Atlas for Renewable Energy (IRENA, 2014). Map data: 3Tier (wind speed), SolarMedAtlas (GHI).

Unlike fossil fuels, for which transport infrastructures (e.g., pipeline systems) are already available between North Africa and Europe, electricity must be conveyed from the production sites to the end consumer via electricity networks. These are yet to be built, as no suitable infrastructure for bulk electricity exchange exists so far between the two sides of the Mediterranean Sea. Missing

¹ The Desertec project is primarily promoted by the Germany-based Desertec Foundation and the Desertec Industrial Initiative (Dii) GmbH, but it also receives support from political institutions such as the EU, the Union for the Mediterranean (UfM, 2012) and other industry advocacy groups like the French-based Medgrid consortium (EC, 2011).

regulatory frameworks and the absence of a common transnational electricity market among North Africa and Europe are further barriers to the export paradigm.

An additional obstacle is the political instability in many North African countries following the events of the Arab Spring and the associated European concerns about energy supply security (Lilliestam and Ellenbeck, 2011). Likewise from the European side, an increasing reluctance can be observed - particularly by the Southern European countries - to support green energy imports in a situation where home markets and industries are suffering from an economic crisis². North Africa's possibilities to realize massive electricity exports are also impeded by the fact that the entire region itself is short of electricity. At the moment, due to the tremendously growing demand (see demand growth rates in Table I-1), many countries in the region are rather more inclined to import than to export electricity. For example, Morocco, the only country that actually holds an electricity connection to Europe, procures around 20% of its electricity demand from Spain. At present, all North African states counter the threat of supply shortages by accelerating the construction of fossil power plants. This business-as-usual approach still dominates the reality of North African power system expansion, despite the assertions of many North African governments to implement new energy roadmaps.

The present thesis takes the perspective that renewable energy technologies can very well form an integral part of the North African electricity supply, going far beyond the role of being an "add-on" to the conventional power system. Renewable energies can also endogenously expand into North African power systems

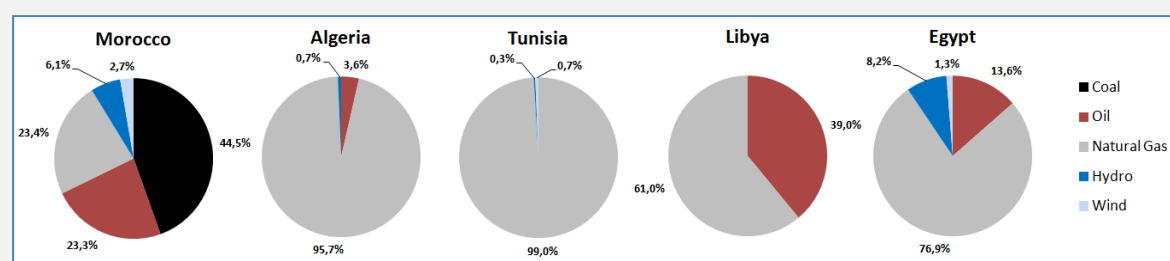
² None of the EU Member states has so far undertaken efforts to apply Article 9 EU Renewables Directive (2009/28/EC) (Directive, 2009) for joint renewable electricity import projects with North African countries. The prospects for transcontinental renewable energy transfers received a further backlash in Dec. 2013, when southern European countries (in particular Spain) overtly opposed the adoption of the 'MSP master plan' of the Union for the Mediterranean due to concerns about cost-competition from North African renewable electricity and lacking interconnectedness with the Central European electricity markets (Ansamed, 2013).

without necessarily being driven by the prospects of energy exports. Like anywhere else in the world, North African decision makers must find answers for the key challenges of the energy supply, which are to ensure supply security while maintaining the economic viability, environmental and social sustainability of the energy system. Renewable energies must prove that they can address these key requirements if integrated into the North African power systems. Providing a better understanding of the effects of such integration is the overarching objective of my thesis.

Table I-1. Key energy data (2012) of the North African countries. Source: IEA (2014), EIA (2014).

		Morocco	Algeria	Tunisia	Libya	Egypt
General	Inhabitants (mill.)	32.5	38.5	10.8	6.2	80.7
	GDP (\$ ²⁰⁰⁵)	83.2	123.6	40.8	36.9	125.9
	Total final energy consumption (Mtoe)	14.3	30.6	7.3	11.8	55.9
Oil production (thousand barrels/day)	Consumption	206	368	89	239	725
	Production	5	1875	67	1483	711
	Net exports	-201	1507	-22	1244	-13
Natural Gas (billion cubic feet/y)	Consumption	38	3053	130	202	1882
	Production	2	1323	66	430	2141
	Net exports	-36	1730	-64	228	259
Electricity	Total demand (TWh/a)	27.6	40.8	14.4	28.3	140.3
	Average annual demand growth rate since 2000 (%)	6.1	6.7	4.0	7.9	6.7
	Total net generation (TWh/a)	26.2	51.9	17.2	33.0	160.8
	Net exports (TWh/a)	-4.8	0.0	0.0	0.0	0.4

Electricity production by fuel (percentage shares, 2012)



2. State of research

In the year 2009, at the beginning of my research project, the topic of renewable energy integration into North African electricity systems was addressed only in a relatively cursory fashion by few publications. Most of the early studies had a strong focus on assessing North Africa's solar/wind potential; but instead of investigating how these resources could be accommodated by the local electricity infrastructures and markets (to satisfy domestic demand), most researchers primarily looked at the aspect of renewable power exports from North Africa to Europe. Staiss et al. (1994) and Knies et al. (1999), for instance, were the first to propose large-scale export schemes via trans-Mediterranean electricity interconnections. This idea was later developed by Czisch (2005) in a comprehensive study about a large, fully renewable-driven electricity system connecting Europe with its southern neighbors by means of high voltage direct current (HVDC) transmission lines. Czisch assessed the feasibility of this concept with a linear modeling approach, which already included the patterns of North African electricity demand, though only in a very rough and coarse manner. A more comprehensive consideration of North Africa's domestic energy demand can be found in the "MED-CSP" study of the German Aerospace Center (DLR, 2005). This publication dealt with scenarios how renewable power plants - particularly concentrated solar power (CSP) - could replace conventional generation in Southern Mediterranean countries by 2050. However, the study did not perform a thorough bottom-up modeling considering the particularities of the North African power generation systems. The same is the case with the follow-up study, "TRANS-CSP" (DLR, 2006)^{3,4}, which primarily explored the consequences of solar electricity imports for Europe, but likewise refrained from carrying out a detailed analysis of renewable integration into the domestic North African electricity supply and demand structures. A further study, likewise published by the

³ The TRANS-CSP study can be seen as one of the key study for the Desertec concept. It postulates the feasibility of around 15% renewable electricity imports of Europe from the MENA region by 2050; the same number was later taken up by the Desertec Industrial Initiative Dii (Desertec 2009).

⁴ TRANS-CSP was followed by a third study, AQUA-CSP (DLR, 2007), featuring the potential of concentrated solar power plants for seawater desalination.

German Aerospace Center (DLR, 2009) specifically explored transmission corridors for the export of CSP-generated electricity from North Africa to Europe. To these above-mentioned publications may be added a spate of other export-centered studies, like Egerer et al. (2009), Pfluger et al. (2009), Lilliestam and Ellenbeck (2011), EWI/Energynautics (2011), Trieb et al. (2012) and Brancucci et al. (2013). These studies also analyzed the effects of North African renewable electricity for the European power system - but only peripherally addressing questions about the impact on the North African electricity systems. Altogether, this illustrates the still prevailing “European bias” of many research papers conceptualizing the role of renewable energies in the MENA region: renewable power expansion is primarily perceived in terms of opportunities and risks for the European energy supply (contribution to renewable targets of EU countries, reduction of carbon emissions, energy security), while issues revolving around the impact for the North African countries receive significantly less attention.

Nevertheless, most recently, there has emerged another strand of publications placing more emphasis on the needs of the North African power systems. These studies attempt to explore in higher detail how renewable technologies can contribute (in a cost-efficient way) to the increasing electricity demand in the region. Regrettably, it must be noted that only very few scholars from the MENA region have carried out these analysis themselves; therefore, again, most publications described in the following are largely the outcome of European research initiatives. In 2012 and 2013, the Germany-based Desertec Industrial Initiative (Dii) released two reports, “Desert Power: 2050” (Dii, 2012) and “Desert Power: Getting Started” (Dii, 2013). Both exerted considerable efforts to simulate renewable energy expansion in an EU-MENA power system, consisting of 42 interconnected model regions. Although only a sub-part of this system, the five North African electricity systems were modeled with a relatively high accuracy - for instance, the optimization of the power plant dispatch was performed in hourly time intervals and by using actual load data of the electricity utilities. Similar studies, also being more considerate of the regional supply/demand patterns, and also simulating these with bottom-up models, have been published by Haller et al. (2013) and Boie et al. (2014). Like Dii, they examined scenarios with very high renewable penetrations by 2050, assuming an embedding of North African power systems into a large, transnational electricity system with

connections to Europe and the Middle East. The results of the last four studies have demonstrated that (almost) fully renewable-supplied power systems are principally feasible in North Africa - but they also indicated that for a cost-efficient achievement of these ambitious schemes a significant expansion of cross-border interconnectors is needed to enable electricity exchanges and power balancing across large geographical areas. To complete the literature overview about quantitative, model-based studies on MENA power systems, one must also add a couple of studies portraying more moderate renewable expansion pathways. Here, substantial shares of the power mix are still allotted to conventional generation with coal, oil- or natural gas-fired power plants. Often, these studies look at a short-term time horizon (e.g., until 2030) and take the existing power generation portfolio as a starting point for the analysis of possible system transformation trajectories with renewable energies. Mostly, these studies are also geographically more limited, covering only sub-regions or single countries instead of large transcontinental power systems. Fragkos et al. (2012), for instance, used modeling techniques to explore the impact of four different macroeconomic and political high level trends on the energy systems of South Mediterranean countries (North Africa, as well as Jordan, Syria, Lebanon and Israel). For the power generation mix of North Africa in the year 2030, the authors calculated renewable shares between 5% and 57%, depending on scenarios' assumptions with regard to renewable energy expansion. A similar study by the Observatoire Méditerranéen de l'Energie (OME, 2011)⁵ contrasting two different policy scenarios, resulted in renewable penetrations between 9% and 22% by 2030. Another study, commissioned by the World Bank and the European Commission (Mercados, 2011) portrayed renewable penetration of approximately 15% in the year 2030. This study, however, did not cover the full set of all five North African countries; it was limited to the Maghreb region, Algeria, Morocco, and Tunisia.

It is remarkable that none of the above-mentioned publications has so far attempted to scrutinize in detail the national renewable expansion targets of the North African governments. All North African countries have administered

⁵ OME's study, as well as the results of the aforementioned studies by Dii, Fragkos et al. and Haller et al. are discussed in detail in chapter VI of this thesis

themselves ‘official’ renewable energy goals, but as this thesis will show, many open questions still remain related to the actual consequences of an implementation of these goals in practice: how would the existing conventional power system interact with the increasing renewable feed-in? And: are the renewable targets of the North African governments actually cost-optimized? The last question particularly concerns the proportions of the three key renewable technologies (wind power, PV and CSP) in the targeted generation mixes. It appears that at the moment many of the North African renewable strategies tend to favor CSP technology – mainly because of its inherent technical advantages such as the capability to store solar energy and control the power output (dispatchability). Whether these qualities actually have an economic value justifying large-scale CSP expansion into North African power systems, has not yet been addressed deeply enough by any model-based scenario study. A further, also widely unexplored topic touches the decision-making process on North African (renewable) electricity strategies: what importance do North African decision-makers actually attribute to aspects of security of supply, social and environmental impacts, or the expansion of cross-country transmission infrastructures in their deliberations about future power system strategies? These and other questions (see next chapter) motivated the research for the present thesis.

3. Research questions

As mentioned earlier, the currently most pressing challenge for electricity system planners in North Africa is how to deliver sufficient electricity amid an ever-growing demand. Both the overall electric energy consumption, but also the peak load events reach new records every year. For decades, the electricity utilities in the region have been relying on new fossil power plants to counteract the permanent threat of supply shortages. This routine, but likewise fossil fuel subsidies and the dominating role of the gas and oil sector on energy decisions are reasons why North Africa’s achievements in solar and wind power expansion are still so moderate today (see Diagram in Table I-1)

However, there is good reason to believe that some advancement will occur in the future. In the first place, there is the generally rising economic attractiveness of renewable energies, which, due to falling technology costs, are expanding

worldwide. Many North African governments have recognized this trend and try to implement their own strategies for domestic renewable energy expansion, regardless of the uncertain prospects for electricity exports to Europe. The motivation of my work is mainly related to this aspect, i.e., to take the “North African perspective” and to examine how the integration of renewable energy sources can be realized from the standpoint of regional power system planning.

Therefore, the guiding question of my research project is formulated as follows:

How can renewable energy sources for electricity generation (RES-E) be cost-efficiently integrated into the North African electricity systems?

I decided to explore this question with quantitative electricity system modeling methods. Large parts of the thesis are built around a linear optimization tool which calculates cost-minimized expansion pathways for different mid-term scenarios (usually until 2030) for the North African electricity systems. The model results allow conclusions about cost-optimized power plant portfolios, the commissioning/retirements of power plant capacities, evolution of the fuel mix in power generation, and even projections of needed transmission capacities to interconnect North African countries among themselves and abroad. With these results, a second, “sub-research question”, concerning the impact of renewable integration into North African power systems, can be addressed:

What are the consequences of RES-E expansion, not only economically, but also in terms of other aspects for decision making on energy strategies in North African countries: infrastructural changes, security of supply, social and environmental impact, or the role of cross-country electricity transmission?

The outcome of my work - a series of five essays published in scientific journals - constitutes the present PhD thesis. It should be noted that the publications are not intended to give any policy prescriptions for renewable energy expansion in North Africa. Rather, they should unfold some of the major effects and consequences that would come along with rising RES-E penetration in North African power systems and might, in this context, be helpful for macro-planners and interested decision makers in the region.

3. Approach and Methodology

As mentioned previously, the overarching approach of my thesis is to analyze North African power system transformations with quantitative modeling methodologies. The majority of the articles (Chapters II-V) use cost-minimizing electricity supply-and-demand models to optimize RES-E integration into North African power systems. Only the last paper (Chapter VI) pursues a different approach: here, I decided to provide a comparison with other researchers' modeling/optimization attempts. The following points elaborate in more detail the different methodologies featured in the five articles (see also the summary of the research objectives in Table I-2):

The first article (Chapter II) uses a simplified bottom-up power market model to analyze electricity scenarios for the three Maghreb countries: Morocco, Algeria and Tunisia. The research question of this paper is how an increased penetration of renewable energies would influence the configuration of the conventional power plant fleet in these countries and to what extent fuel and investment costs can be avoided. Renewable energies were introduced exogenously by assuming that their expansion takes place according to official renewable energy targets of the three countries. The model subsequently optimizes long-term investments into conventional (thermal) power plant capacities, as well as their short-term electricity dispatch. The model was formulated as a linear optimization problem (LP), solved in the GAMS programming environment⁶, with the objective function being the total system costs, i.e., the (discounted) sum of all fixed and variable costs incurred by the electricity generation system. The central boundary condition of the model is the balance of supply and demand: conventional power plants, e.g., gas-, coal-, oil-fired, but also hydro power stations must follow - on an hourly basis - the residual load (the difference of the actual, hourly electricity consumption and the solar and wind power feed-in). Regional power exchanges are likewise part of the simulation, as the model assumes that by 2025 an interconnected Maghreb power system will be realized. Transmission capacities are

⁶ The simulations in Chapter II were carried out with the DIME model at the Institute for Energy Economics at the University of Cologne (EWI, 2010).

not optimized endogenously, but remain fixed, exogenous model parameters, following the expansion trajectories foreseen by national electricity transmission system planning.

Contrary to the first article, where the optimization was limited to conventional power plants, the second article (Chapter III) sets out to include renewable energies endogenously into the linear modeling process. The residual load approach of the previously used model was replaced by a method in which all technologies, including fluctuating renewables, can actively participate in the power plant dispatch. Investments in renewable power plant capacities are likewise modeled endogenously. Improving the model with these additional features was particularly necessary to address the key question of this paper, how CSP technologies can be integrated cost-efficiently into the North African electricity systems. CSP power plants have the particular feature that they can be outfitted with thermal storage systems, granting them (a limited) control of the electricity output during daily operation. The ability to dispatch electricity offers some economic advantages. For instance, CSP power generation can be shifted to the evening hours, when electricity demand peaks in many North African countries. To properly analyze the cost advantages of CSP dispatchability, the model had to be specifically tailored to address the various technical features and constraints inherent to parabolic trough power plant technology - currently the most widespread type of CSP power plants being built in North Africa. Morocco and Algeria, two structurally very distinct power systems with different load patterns and a different conventional power plant fleet, were the subject of the analysis.

The third article (chapter IV) of my dissertation deals with electricity mix scenarios for Tunisia. The background was a research project on strategies for the national electricity sector⁷ which started in 2011 right after the Tunisian revolution. This event, which triggered a still ongoing phase of political and economic troubles in the entire MENA region, has also marked the style of decision making in the Tunisia energy sector. In many countries, deliberations about energy

⁷ The project, “Strategic study for the electricity mix in Tunisia“ (Wuppertal Institute/Alcor, 2012).

strategies, previously from the top-down perspective and mostly focusing on cost aspects, have become more considerate of aspects of political participation, social welfare and sustainability (Brand and Fritzsche, 2012). In Tunisia, where the project was carried out in collaboration with local stakeholders, it became apparent that electricity strategies cannot be decided exclusively on the basis of economic costs as the sole criterion, but must also consider socio-economic issues and sustainability aspects. The article presents a two-step method to analyze different Tunisian electricity mix scenarios until 2030, both from the technical, as well as from the socio-economical side. In the first step, a linear, least-cost optimization model (as described in the previous articles) ascertains technically and economically optimized power system transformation pathways; subsequently, in a second step, these pathways are subjected to a multi-criteria decision analysis (MCDA), selecting a ‘best-performing’ scenario on the basis of a wider set of decision criteria, considering not only cost aspects but also security of supply, energy dependency, emission of pollutants - as well as the prospects for job creation and the social acceptability of power generation technologies in Tunisia. The weighting criteria that had been assigned to the different criteria within the MCDA process were obtained in a stakeholder survey with Tunisian energy experts and decision makers.

The fourth article (Chapter V) extends the linear models of the previous articles (Chapter II - IV) so that it additionally optimizes the expansion of the transnational transmission system. Investments in transmission capacity as well as the power exchanges between two countries, become part of the optimization exercise. Model-wise, this was done by adding the costs of interconnectors to the objective function of the linear problem. By doing so, the model optimizes not only the generation but (simultaneously) also the upgrades of transnational interconnection capacity. This combined modeling was necessary to address one key question of the paper: how does the penetration of renewable energy impact power exchange patterns between North African countries (and further beyond with satellite regions in Europe and the Middle East)? Furthermore, the article explores the extent of economic benefits that could be achieved through more cooperation and joint electricity system planning among North African countries. This evaluation was quantitatively carried out by comparing an uncon-

strained scenario (high ability to interconnect and exchange electricity) with a case where the countries' willingness to mutually use interconnectors is constrained. A further issue addressed by Chapter IV is whether - and if yes, under which conditions - renewable energy expansion in North Africa could trigger electricity exports to adjacent regions like Europe and the Middle East.

The fifth article (chapter VI) draws its motivation from the fact that, in the last years, other researchers have likewise attempted to analyze MENA electricity systems with modeling techniques. I decided to conclude my thesis with a 'meta-analysis' summarizing the results of the most pertinent publications in this field. In order to provide coherency with the geographical scope of my research project, the analysis was restricted to studies that cover all five North African countries: Morocco, Algeria, Tunisia, Libya and Egypt. By using a uniform comparison framework, the paper analyzes the key scenario-building approaches, modeling methodologies and model outcomes of five publications dedicated to renewable energy expansion in the North African region. The article particularly addresses the question of how the studies have assessed pathways for power system transformation, and which technology mix (renewable, but also conventional) they imagine being the best mix for the North African region in the future. An additional objective of this meta-analysis was to scrutinize the studies' projections for electricity exports from North Africa to Europe. The assessment encompasses mid-term (2030) as well as long-term (2050) electricity scenario studies.

Table I-2. Summary of research objectives and the key methodologies of the thesis.

Chapter N°	Geogr. scope	Scope of research	Methodology / Particular features.
II	MA, AL, TN	• Impact of RES-E penetration on the conventional power generation system • Financial impact of national RES-E goals	• Linear optimization, time horizon 2025 • Modeling of investment and dispatch of conventional power plants • Residual load approach: RES-E are exogenous model parameters
III	MA, AL	• Integration of CSP power plants • Optimized CSP configuration for different solar penetration scenarios in Morocco and Algeria • Monetary value of CSP dispatchability / storage	• Linear optimization, time horizon 2030 • Modeling of investments and dispatch of all power plants (including RES-E) • Focus on CSP dispatch modeling
IV	TN	• Evaluation of different electricity mix scenarios for Tunisia until 2030 • selection of best-performing scenario with regards to economic, social and environmental criteria	• Linear optimization combined with Multi-criteria decision analysis (MCA) for subsequent scenario evaluation, time horizon 2030 • Stakeholder survey • Scenario ranking
V	MA, AL, TN, LY, EG	• Impact of RES-E integration on transmission infrastructures and cross-country power exchanges • Economic benefits of multi-lateral cooperation on the power sector • Conditions for electricity export to Europe	• Linear optimization, time horizon 2030 • Modeling of investments and dispatch of all power plants, as well as transnational interconnector capacities and cross-country power exchanges
VI	MA, AL, TN, LY, EG	• Meta-analysis of existing model-based studies on North African electricity system scenarios • RES-E expansion scenarios of other studies • Assessment of electricity export options for North Africa	• Comparison of different studies using electricity models to analyze RES-E expansion scenarios in North Africa • Time horizon of the studies: 2030 and 2050

*MA: Morocco, AL: Algeria, TN: Tunisia, LY: Libya, EG: Egypt.

It must also be mentioned that my work occurred in a very dynamic environment, concerning both the political situation in North Africa and the general, worldwide market conditions on the renewable energy sector. During the course of my research, these changes required some reorientation. As mentioned above, one major disruption to cope with was the event of the Arab Spring. The Tunisian Revolution of 2011, which later spread to Egypt and Libya, has left North Africa in heavy political and economic difficulties - with consequences also for the energy sector. Scenario building, especially for renewable energies, is very difficult in this context. For instance, one very dynamic element I had to deal with during my research was the volatility of North African renewable energy strategies. Many governments have started to develop their own national targets or roadmaps for renewable energies (REN21, 2013), but these goals have often been modified or were, occasionally, even completely revised. This impacted some of the scenario assumptions in my thesis: while the first article (Chapter II) still is based on the North African renewable energy targets back in the year 2010, the later scenario assumptions (e.g., in Chapter III and Chapter IV) already include a more recent set of government goals.

Furthermore, it must also be noted that the renewable energy sector itself has seen many changes between the start of my research in 2009 and today. The most dramatic shift—unexpected even by insiders—was the dramatic cost decreases in the photovoltaic (PV) sector. With system prices having sharply dropped by more than 50% in only a couple of years (Candelise et al., 2014), input parameters and assumptions for cost developments in my models had to be revised several times between the first article published in 2010 (see Chapter II) and the later publications. The issue of falling PV prices, for instance, increasingly questions the economic viability of concentrated solar power (CSP), which for a long time has been believed to be the ‘first choice’ technology for renewable power expansion in MENA. Chapter III of my thesis explores these challenges by taking a closer look at the comparative advantages of storage-bound CSP power plants when integrated into the North African electricity systems.

It can be expected that the volatile political and economic framework conditions, as well as the dynamic situation of the worldwide renewable energy markets, will remain a challenge for any research conducted about renewable energy expansion worldwide, particularly in the Middle East and North Africa. The pre-

sent thesis can certainly only provide a jigsaw piece to the understanding of the transformation processes ruling the MENA energy systems. Nevertheless, I believe that, thanks to the quantitative, model-based approach, some important patterns of renewable energy integration into the region's power systems are now understood in a better, more robust manner.

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II. The renewable energy targets of the Maghreb countries: Impact on electricity supply and conventional power markets⁸

Abstract

Morocco, Algeria and Tunisia, the three countries of the North African Maghreb region, are showing increased efforts to integrate renewable electricity into their power markets. Like many other countries, they have pronounced renewable energy targets, defining future shares of “green” electricity in their national generation mixes. The individual national targets are relatively varied, reflecting the different availability of renewable resources in each country, but also the different political ambitions for renewable electricity in the Maghreb states. Open questions remain regarding the targets’ economic impact on the power markets. Our article addresses this issue by applying a linear electricity market optimization model to the North African countries. Assuming a competitive, regional electricity market in the Maghreb, the model minimizes dispatch and investment costs and simulates the impact of the renewable energy targets on the conventional generation system until 2025. Special emphasis is put on investment decisions and overall system costs.

1. Introduction

The North African electricity markets are in a phase of rapid transformation. Soaring electricity demand, caused by economic growth, demographic changes and progressing urbanization, urges the countries in the region to massively increase their power generation capacity and upgrade their electric grids. However, electric infrastructure projects still face strong barriers due to inflexible elec-

⁸ This chapter is based on a peer-reviewed article published in the journal *Energy Policy*: Brand, B., Zingerle, J., The renewable energy targets of the Maghreb countries: Impact on electricity supply and conventional power markets. *Energy Policy* 39 (2011) 4411-4419, doi: 10.1016/j.enpol.2010.10.010.

tricity market structures prevailing in most of the North African countries. Mainly unliberalized, non-competitive and dominated by cumbersome state monopolies, they often encounter difficulties when seeking project financing at the needed scale. Over the past years a consensus has emerged among the countries that more competitive market rules need to be applied. Moreover, integration and harmonization of the different national electricity markets has been placed on the agenda. Particularly progressive signs show the Maghreb states Morocco, Algeria and Tunisia⁹. In 2003 they signed a protocol for the stepwise integration of their power markets with the long-term objective of a common electricity market with the European Union. Already today, the three Maghreb countries are electrically interconnected with each other and are likewise synchronized with the European electricity network via an undersea interlink between Morocco and Spain. Further projects for transmediterranean interconnections, as well as ongoing construction of new interconnectors between the Maghreb countries indicates that an integrated electricity market might become a realistic scenario in the future.

As an additional aspect, renewable energies have entered the discussions. The high natural potential for wind and solar energy has recently fueled a massive interest for RES-E (Electricity from Renewable Energy Sources) technologies in North Africa. Renewable electricity export scenarios, promoted by the industrial initiative Desertec (Desertec, 2010) or the Mediterranean Solar Plan (MSP, 2010) have become prominent in Europe, but it is less known that the North African countries themselves have set up their own goals to integrate RES-E technologies into their national electricity supply schemes. In particular, Morocco, Algeria and Tunisia, seem to give RES-E technologies a very active role in electricity supply, as announcements of relatively well-defined national renewable electricity goals show.

The aim of the present study is to analyze the potential impacts of these goals on a regional electricity market formed by the three Maghreb countries and adjacent EU countries. The time horizon of the analysis is 2025. It should be stressed

⁹ In the literature, the term Maghreb is in some cases viewed in a wider sense and also includes Libya and Mauritania. In this article we use the narrow definition of Maghreb, meaning the countries Morocco, Algeria and Tunisia.

that this short-term analysis only covers the actual planning perspective of the North African countries - large-scale renewable export projects via High-voltage direct current (HVDC) lines from North Africa to Europe are not considered. The work addresses the following main questions: (1) How do the current national RES-E goals influence the electricity mix and the conventional power plant structure in the Maghreb countries? (2) How large are the financial advantages of Maghreb RES-E goals with regards to lower fuel costs or avoided investments in conventional power plants?

The article is structured in three parts. We start with an overview of the currently ongoing transformations in the Maghreb electricity markets and describe each country's renewable electricity goals. From these findings we derive two scenarios for 2025. The subsequent part provides a description of the model, the assumptions and the input parameters used. In the third part, the model results are outlined, analyzed and discussed.

2. Background and Scenarios

We limit our study to Morocco, Algeria and Tunisia for three main reasons: (1) Contrary to their neighbors Libya and Mauritania, these three states show relatively advanced policies with regards to liberalization of their electricity markets. (2) The three countries have expressed intentions to form a Maghreb-integrated electricity market, which is intended to be harmonized with the European electricity system in the future. (3) Additionally, as mentioned above, all three countries strive for integrating RES-E capacity to their current fossil-dominated electricity generation systems. In this section we will describe these observations in more detail and draft two scenarios for our subsequent model analysis.

2.1. Electricity market reforms in the Maghreb countries

Algeria's reform objectives of bringing its market closer in line with international standards are built around an electricity law enacted in 2002. As a direct consequence of the law, the state electricity and gas monopolist Sonelgaz was forced to unbundle its activities, and an independent regulatory body was estab-

lished. In the years following Algeria's electricity reform, several projects of independent power producers (IPP) – some even with international equity participation – emerged in the country. Tunisia's electricity sector also underwent a liberalization process concerning the market segment of electricity generation. In 1996, the monopoly of power production was withdrawn from the state-owned gas and power utility STEG (Société Tunisienne de l'Electricité et du Gaz), which subsequently resulted in private IPP companies competing in tenders for generation licenses issued by the Tunisian state (CAIMED 2006). In Morocco, the opening of the electricity market started even earlier, in 1994, when a decree opened up the possibility for private IPPs to act as concessionaires for the national electricity utility ONE (Office National d'Electricité). A further liberalization step came in 2008 when IPPs were given a more general right to access the Moroccan electricity generation market, particularly if they operate smaller power plants with capacities of up to 50 MW. Another step towards more liberalization is a planned split of the Moroccan electricity market into one regulated segment and one open market segment, where certain industrial customers are allowed to freely choose their electricity suppliers (MEMEE, 2008, GTZ 2009).

Despite these examples of reform efforts, it should be stated that the Maghreb states still have far to go in order to achieve a fully liberalized electricity markets. As of now, the 'market' operations are still almost exclusively performed by the aforementioned, omnipresent state utilities that hold monopsonies on electricity purchase while acting as (quasi-) monopolists in terms of electricity transmission and distribution. Formal trading platforms, where competing generation companies offer their electricity to independent retailers, are in the planning stages, but do not yet exist in any of the three Maghreb states. There is nevertheless evidence that the countries are on a clear pathway towards a competitive electricity market and that this trend will continue and even accelerate in the coming years.

2.2. Goals for transnational market integration

The aforementioned belief in a future common electricity market is also sustained by the Maghreb states' efforts to merge their different national electricity

systems into a larger, regional market. The political will for this target was expressed by the governments of the Arab Maghreb Union¹⁰ which in 1989 created the Maghreb electricity committee COMELEC (Comité Maghrebin de l'Electricité). Besides the realization of a common internal electricity market, COMELEC envisages, as a long-term goal, a gradual integration and regulatory harmonization with the European electricity market (Chouireb, 2008). Particularly COMELEC members Morocco, Algeria and Tunisia have subscribed to this idea: Already technically interconnected with the European Network of Transmission System Operators for Electricity (ENTSO-E) since 1997, they signed in 2003 an official declaration with the European Commission to further support the integration of their electricity markets into those of the European Union (Athens Declaration 2003). There is strong evidence that this kind of market integration is actually desired by the Maghreb states—since 2001, for instance, the public electric utilities ONE (Morocco) and Sonelgaz (Algeria) have acted as licensed traders on the Spanish electricity exchange platform Operador del Mercado Ibérico de Energía (OMEL).

2.3. Renewable energy targets

Over the past years, as in other regions of the world, wind and solar energy has received significant attention in North Africa. As hydropower faces stagnating expansion potential in the region due to geographical limitations, the governments are increasingly considering wind and solar technologies as the future RES-E technologies for their countries. However, a look at the currently-installed RES-E capacities in Morocco, Algeria and Tunisia, reveals that the actual contribution of these sources of energy is still very low: Compared to the overall installed electricity capacity of 16 GW, wind power, with a mere 304 MW cumulated in the Maghreb states by the end of 2009, is at the moment the only noteworthy new RES-E source (GWEC, 2010). The only solar electricity facilities worth mentioning are two concentrating solar thermal plants currently under construction in Morocco and Algeria. Once finalized, they will contribute

¹⁰ Besides Algeria, Morocco and Tunisia, the Arab Maghreb Union (and thus COMELEC) also encompasses Libya and Mauritania. As of now, only Morocco, Algeria and Tunisia are electrically interconnected and exchange electricity.

to approximately 50 MW of the Maghreb states' generation capacity. Grid-connected photovoltaic (PV) electricity is at the moment negligible in all countries in the region. Against the background of these relatively moderate achievements, the recent, hereafter listed national renewable development goals of Morocco, Algeria and Tunisia sound relatively ambitious:

2.3.1. Morocco's renewable electricity goals

Morocco's goals target a strong increase in the number of both wind and solar power plants. In a detailed wind electricity development program (ONE, 2008), the national electricity utility, ONE, intends to increase the installed capacity from the current 253 MW to up to 2 GW until 2016. Most of the projects will be located alongside the country's Southern Atlantic coastline which features excellent wind conditions comparable to off-shore sites in Europe (CDER, 2007). Concentrating solar power (CSP) is the second axis of the Moroccan government's RES-E development plans. In 2009, a multi-billion Euro investment program was announced, likewise targeting 2 GW CSP plants, for which precisely defined project sites already exist. The project, called the 'Moroccan Solar Plan,' will be carried out until 2020, and is promoted by a government agency being exclusively established for that purpose (Masen 2010). For grid-connected photovoltaic (PV) electricity, a smaller program was set out by ONE in 2007 targeting 150 MW of distributed PV capacity by 2015. This program, however, has suffered from delays, making realization of the goals by 2015 seem unlikely (Hirshman, 2009).

2.3.2. Algeria's renewable electricity goals

Algeria's renewable electricity goals are set out as percentage values of overall power generation. As a short-term goal, for 2017, the Algerian electricity regulatory commission (CREG, 2008) published a 5 percent renewable electricity target. In the long run, by 2030, Algeria expects to reach 20 percent overall renewable coverage, of which 70 percent is generated by CSP, 20 percent by wind and 10 percent by PV (CIF, 2009).

2.3.3. Tunisia's renewable electricity goals

In 2009, the Tunisian government released a “Tunisian Solar Plan” containing several detailed renewable RES-E projects. Moreover the plan includes energy efficiency measures, solar water heating technologies, and to a minor extent biomass development projects. Compared to the Moroccan and Algerian solar plans, the Tunisian Solar Plan remains relatively modest with regards to solar capacity additions—until 2016, projects in CSP and PV plants will only add up to a total capacity of 120 MW. In terms of wind capacity, around 330 MW of installed capacity are foreseen by 2016, while 1200 MW shall be reached in 2020 and 1800 MW by 2030 (ANME, 2009; Ounalli, 2007).

2.4. Scenarios

It is clear that the above mentioned RES-E deployment plans face strong barriers, especially if looking at the political and financial realities in most of the Maghreb countries. Resistance against the renewable energy goals can for instance be expected by those parts of the political elite, which have vested interests with the power sector or the national oil and gas industry. It should be mentioned that in the past, many publicly-announced renewable energy projects in the region have been postponed or were never realized. Therefore, doubts are justified as to whether the recent renewable energy goals will actually stand the test of time. On the other hand, the climate for funding renewable power projects in the region has noticeably improved in recent years (Masen, 2009; CIF, 2009). In addition, there is a certain competition for prestige between the North African governments with regards to renewable energies. This might accelerate the pace of RES-E penetrating the Maghreb's electricity markets. Against this background, we decided to draft two scenarios. Both consider the situation of an integrated, competitive electricity market as given - while they differentiate between two contrasting situations regarding renewable energies.

- *RES-E scenario*: The Maghreb countries fulfill their renewable targets and build renewable power plants according to the published RES-E development plans.

- *Business as usual (BAU) scenario*: Here, the assumption is that the countries do not build any new RES-E at all and continue a conventional pathway until 2025.

It should be stressed that, apart from the different RES-E integration, all other remaining technical input parameters, e.g. the conventional power plant data, investment costs or fuel prices, and assumptions, e.g. on the countries' demand growth and political strategies for the use of fossil fuels, stay unchanged for both scenarios.

3. Methodology and input parameters

Our analysis uses the linear optimization model DIME (EWI, 2010), a bottom-up power market simulation tool, which was designed by the Institute of Energy Economics at the University of Cologne to provide long-term forecasts for the European power markets. For the purposes of this study, the model was redesigned for the three North African countries of Morocco, Algeria and Tunisia. DIME calculates the optimal dispatch as well as the investment pathway of commissioning and retirements of the conventional power generation system by minimizing the total discounted costs. Simulations are conducted over representative periods, ranging from 2010 to 2025 in 5-year intervals. For every period, the optimization is carried out under the boundary condition that electricity generation meets demand at any time during the sequence of representative days and throughout all simulation periods. There are 12 representative days consisting of four different seasons (winter, spring, summer, and autumn) and three days of the week (Wednesday, Saturday, and Sunday).

Renewable energies are introduced exogenously in the model: In the RES-E scenario, the installed capacities follow through the pathways given by the countries' renewable electricity goals. RES-E generation has prioritized access to electricity generation. This is realized in the model by deducting the feed-in of solar and wind power from the countries' specific electricity load curves. The remaining, residual load is then covered by the daily dispatch of conventional power plants. For each of the 12 days the dispatch is computed by DIME on an hourly basis in one hour intervals. The optimization also takes into account physical exchanges between the neighboring regions.

3.1. Model regions and interconnectors

In order to provide a complete picture and optimize the model's accuracy, we include several adjacent European countries that interact with the Maghreb electricity market (see Figure II-1). The Iberian Peninsula (Spain and Portugal) is included because it holds an already operational alternating current (AC) interlink to Morocco via the Strait of Gibraltar. Italy will, with its ELMED line have an HVDC interconnection link from Sicily to Tunisia (ELMED, 2010). This project is already in an advanced planning stage, as are other interconnectors between Morocco and Spain, and between Tunisia and Algeria. Furthermore, on a longer time horizon, two Algerian projects are being planned for interconnectors to the Spanish mainland and to Sardinia, Italy (Benabid, 2009).

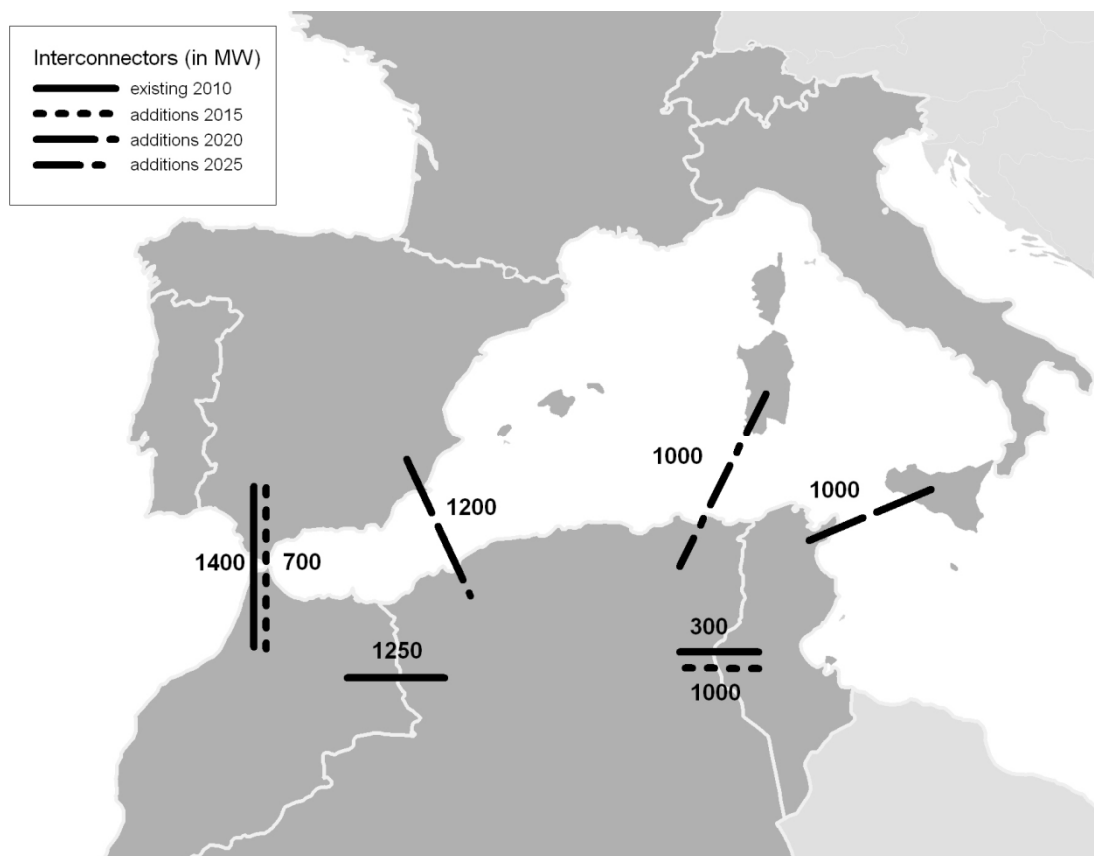


Figure II-1. Model regions and thermal transfer capacities, including expected construction pathways. Transmission lines between the European countries are not pictured in this figure.

The gross (thermal) transfer capacities between the Maghreb countries, as well as the trans-mediterranean interconnectors, are pictured in Figure II-1. For the net transfer capacities (NTC) required by our model as input parameters, either published data is used (ENTSO-E, 2009), or NTC values are estimated at 60 percent of the gross transfer capacity. Power losses alongside the transmission lines are likewise taken into account. Within a model region, no power losses occur. While Spain and Italy are (or will be) directly connected to the Maghreb electricity market, France is included as a model region because it bridges the Spanish and Italian power markets. Switzerland, as an important electricity transit country, likewise takes part in the model. For all 5 European model regions, power plant data, interconnection capacity, and the individual RES-E development plans have been assessed in former studies (EWI, 2010) and are introduced in our simulations.

3.2. Renewable electricity goals and model parameters

As outlined in chapter 2.3, the Maghreb states consider three basic renewable technologies as future contributors for their electricity supply—wind power, CSP and PV. Hydroelectric plants only play a noteworthy role in Morocco, but due to geographically limited expansion potentials, no major capacity additions are expected. The same accounts for other potential RES-S technologies, such as geothermal and biomass electricity, which do not currently appear on the RES-E development agenda of the Maghreb states.

Table II-1 summarizes how the renewable goals are translated into technical input parameters for the DIME model. Trends are extrapolated to provide a match to the type years required by the model. For Algeria, which expresses its renewable goals in percentages of overall electricity supply, we calculate the capacity values with the help of electricity demand projections (DLR, 2005) and the expected full-load hours of the technology.

Table II-1: RES-E scenario: cumulated installed generation capacity (MW) according to renewable goals and average and country-specific full load hours (FLH)

Year	Morocco (MW / FLH)			Algeria (MW / FLH)			Tunisia (MW / FLH)		
	Wind	CSP	PV	Wind	CSP	PV	Wind	CSP	PV
2010	250/ 3400	20/ 3300	-	50/ 2000	100/ 3500	-	19/ 2100	-	-
2015	1650/ 3400	800/ 3300	5 / 1650	525/ 2000	700/ 3500	-	175/ 2100	25/ 3300	10/ 1650
2020	2000/ 3400	2000/ 3300	100/ 1700	1100/ 2000	2000/ 3500	400/ 1700	1200/ 2100	100/ 3300	50/ 1700
2025	3000/ 3400	3000/ 3300	200/ 1800	2200/ 2000	4400/ 3500	1200/ 1800	1600/ 2100	200/ 3300	100/ 1800

The full-load hours in the table are estimations for typical RES-E power plants and reflect the geographical differences of the countries. Wind farms in Morocco feature higher performances than those in Algeria and Tunisia, while CSP plants in Algeria result in higher yields due to better sites with higher irradiation. The increase of PV full load hours over time reflects increases in module efficiency and the expected trend to large-scale PV plants in the future. The daily and seasonal feed-in profiles for PV plants are based on an exemplary 1 MW plant with crystalline cells on the 33° northern latitude. For the CSP feed-in, data from an exemplary 100 MW parabolic trough plant with a solar multiple of 1.5, including a thermal salt storage, is used. The intermittent character of wind is considered by implementing a random component in the feed-in profile.

3.3. Conventional Power Plants

3.3.1. Power plant data

The following conventional power plant technologies for the North African countries are incorporated into the model: hard coal power stations (only Morocco), liquid fuel (oil) power plants, open cycle gas turbines (OC), combined-cycle gas-fired power stations (CC), hydro-storage plants and pumped storage

hydropower plants (only Morocco). Our model uses a 2007 power station inventory by the Arab Union of Producers, Transporters and Distributors of Electricity (AUPTDE, 2007) as a database of existing power plants. This database is updated by more recent power plant projects following information published in the annual reports of the Maghreb utilities. For the three countries, the model's current power plant inventory encompasses 197 power generating units by the end of 2009. According to their construction years, the power plants are classified into different vintage classes, each having its specific parameters for efficiency, fuel consumption, ramp-up behaviour and operation and maintenance (O&M) costs (EWI, 2010).

3.3.2. Cost assumptions

Fuel costs are derived from market price assumptions for Europe that were carried out in 2009 (EWI 2010). Coal prices are estimated to be 9.9 €/MWh_{th} in 2010 and are expected to rise to 11.5 €/MWh_{th} by 2025. Gas prices start at a level of 20.1 €/MWh_{th} in 2010, reaching 26.8 €/MWh_{th} in 2025. These price levels are assumed to also be valid for the Maghreb power plants. For the gas-producing countries Algeria and Tunisia this might be bewildering at first glance, because it is well known that both countries supply their economies with cheap gas. In the logic of the model, however, it is necessary to consider the opportunity costs, as the countries could, in principle, sell their gas to Europe instead of using it in their own power plants¹¹. CO₂ costs are fixed and set exogenously, and they amount to 15€ in 2010, and increase by 5€ steps every 5 years. For reasons of fair competition on a common electricity market, an equal CO₂ price is assumed over all model regions.

The investment costs input parameters are based on an analysis of recently-completed plants, as well as costs of future power plant projects (EWI, 2010). The investment costs of conventional power plants are considered identical for all countries, and held constant until 2015. Afterwards, conventional power plant investment costs are expected to decrease, as Table II-2 indicates:

¹¹ Avoided transport costs for gas are nevertheless taken into account, as they reduce the opportunity cost of gas in the North African countries.

Table II-2. Investment costs of conventional power plants (source EWI 2010)

	Investment costs [€/kW]	Lifetime years	Net efficiency
Coal before 2015	1350	40	46 %
Coal after 2015	1200	40	50 %
CC gas before 2015	550	30	58 %
CC gas after 2015	550	30	61 %
OC gas before 2015	350	25	35 %
OC gas after 2015	350	25	40 %
Oil after 2010	450	25	40 %

3.3.3. Capacity additions and decommissionings

The model endogenously incorporates investment decisions concerning the commissioning of power plants and their retirement over the different representative periods. To better reflect the actual situation in the North African energy markets, several considerations, primarily political, are implemented in the model. Tunisia and Algeria, for example, have prioritized the use of natural gas as a domestic energy resource, while Morocco pursues a more diversified approach which includes the use of imported coal in its conventional power mix. A restriction is made with regards to nuclear energy—although the nuclear option for North Africa is an increasingly discussed topic, it is unlikely that first plants will come online before 2030 (Prognos, 2009). Therefore, nuclear capacity additions are not considered in the model setup.

3.4. Demand

For each country, DIME requires long-term electricity demand projections, as well as daily load curves representing the countries' characteristic electricity demand fluctuations throughout representative days. With regards to long-term demand projections, particularly those beyond 2020, unfortunately no data from North African sources are available in the literature or databases. Therefore, our model input reverts to a scenario outlined by the German Aerospace Center (DLR) in its MED-CSP study (DLR, 2005). DLR's demand forecast for Morocco, Algeria and Tunisia is based on historical electricity demand data, as well as

on assumptions for population growth and future growth of GDP. With annual rates of 7 to 8 percent per year, the DLR scenario leads, as pictured in Figure II-2, left, to rather high, but not unrealistic, demand increases. In fact, historic electricity demand data shows that, over the past years, demand increases in all the three Maghreb states has been consistently above 5 percent per year. (ONE, 2008b; STEG, 2008; CREG, 2009).

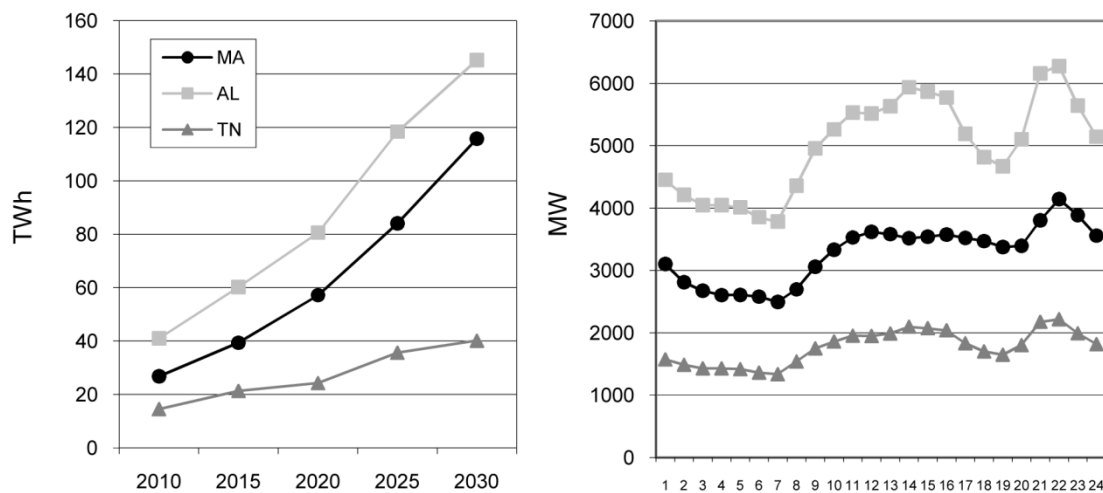


Figure II-2. Annual electricity demand growth of the Maghreb countries (left) and their daily electricity demand pattern (right) for a Wednesday in summer 2010.

Daily load profiles (see example of a summer day in Figure II-2, right) are retrieved from online databases and annual reports of the Maghreb national utilities or regulatory authorities. The load curves in North Africa show a relatively similar pattern with two characteristic maxima—one relatively broad-spread maximum at midday and a second, more distinct peak later in the evening hours. Extreme peak events usually occur either on hot summer days at the midday peak, due to extensive use of air conditioning, or at the evening peak in winter, due to electric heating. For simplicity, it is assumed that this typical shape of the load curves is not changing over the time periods covered by our simulation.

4. Results

4.1. Impact on the electricity mix

The first aspect the model examines is the change in the Maghreb's electricity mix over time and between the two scenarios. This is illustrated in Figure II-3. As a general finding, it can be seen that in both scenarios, the total electricity generation of the three Maghreb countries almost triples over the next 15 years - an obvious reaction to the anticipated massive power demand increase that the region faces in the coming years.

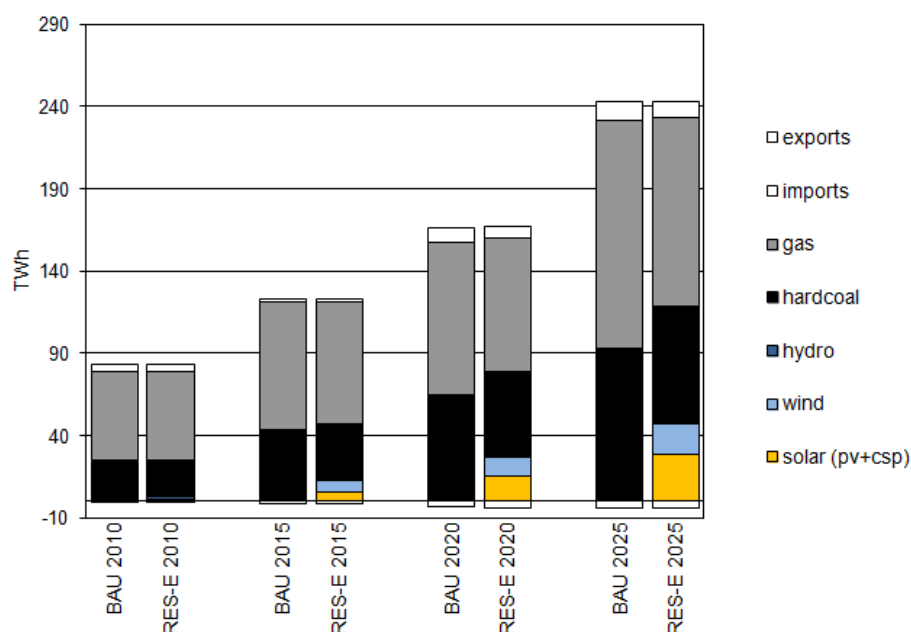


Figure II-3. Cumulated electricity generation of all three Maghreb countries (Algeria, Morocco and Tunisia) by technology in the BAU and the RES-E scenario.

Rising demand for electricity is also responsible for the observation that the RES-E goals, although ambitious from today's perspective, will in the future be superposed by the still much higher need for fossil fuel generation from gas and coal plants. Nevertheless, if comparing the generation shares in the RES-E sce-

nario with the BAU scenario, it can be seen that fossil generation cedes noticeable parts of its production to solar and wind electricity generation—if all three countries reach their respective renewable electricity targets, wind and solar electricity will replace approximately 20 percent of fossil fuel generation by 2025.

Table II-3. Renewable and fossil generation shares for Algeria, Morocco and Tunisia and for the three countries (NA-3) in the RES-E and the BAU scenario.

	Morocco 2025		Algeria 2025		Tunisia 2025		NA-3 2025	
	BAU	RES-E	BAU	RES-E	BAU	RES-E	BAU	RES-E
PV	0.0%	0.4%	0.0%	2.0%	0.0%	0.5%	0.0%	1.2%
CSP	0.0%	10.7%	0.0%	14.3%	0.0%	2.0%	0.0%	11.1%
Wind	0.9%	11.0%	0.0%	4.1%	0.0%	10.3%	0.4%	7.7%
Hardcoal	97.5%	75.7%	0.0%	0.0%	0.0%	0.0%	39.0%	30.0%
Liquid fuels	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Hydro	1.2%	1.3%	0.1%	0.1%	0.3%	0.3%	0.6%	0.6%
Nat. gas	0.4%	0.9%	99.9%	79.5%	99.7%	86.9%	60.0%	49.4%

Table II-3, which shows the detailed generation percentage differences between the two scenarios, allows a more thorough analysis of how RES-E generation replaces conventional generation in the RES-E scenario: 1) in Morocco, coal generation gives shares to wind and, to a minor extent, solar generation; 2) in Algeria, gas-generated electricity is partially replaced by solar and, to a minor extent, wind generation; 3) in Tunisia, gas-generated electricity cedes generation shares to wind and, to a minor extent, solar electricity.

As pictured in Figure II-3, the model also computes power exchanges between the North African countries and Europe. It can be seen that the Maghreb region remains in both scenarios a net importer of electricity. This result can be explained by exports from Spain, which incorporates nuclear power plants in its generation portfolio and is, according to our market model, able to provide a cheaper generation cost structure than the North African countries. Major differences between the RES-E and BAU scenarios with regards to the total exchanged amounts of electricity were not observed.

4.2. Impact on the power plant system

The interesting parameters for an analysis of the generation system are the installed capacities (see Figure II-4), as well as the full-load hours. The latter reflect the intensity of utilization of the technology. As in the previous chapter, we compare the modeled capacities of the RES-E scenario with those resulting from the BAU scenario for the year 2025.

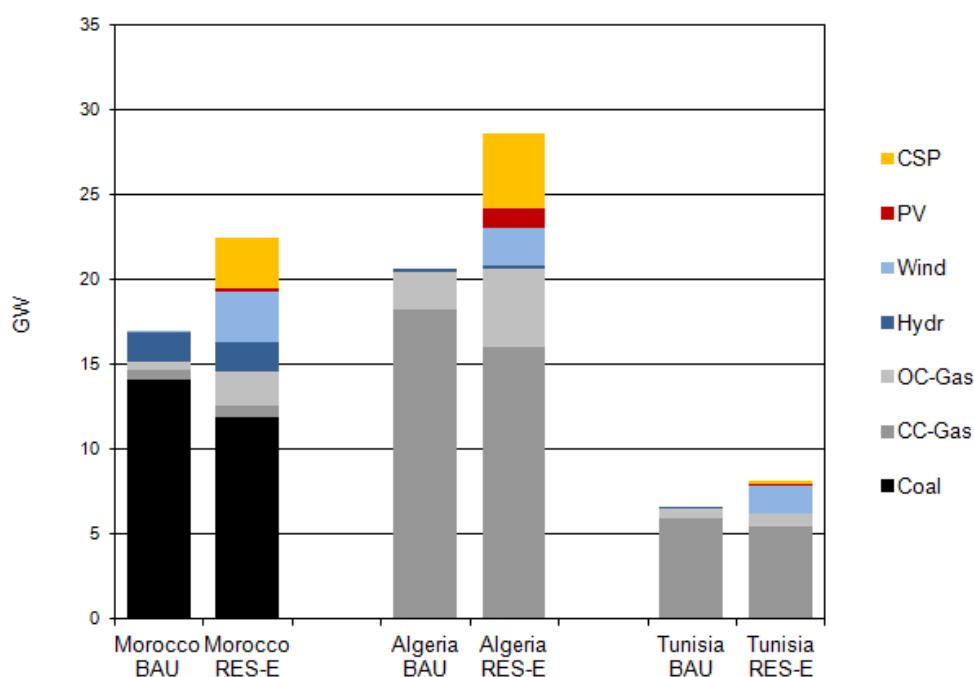


Figure II-4. Installed capacities (in GW) for the RES-E scenario and the business as usual (BAU) scenario for the year 2025.

For conventional power plants, the realization of renewable goals has the following consequences.

a) In Morocco, the RES-E scenario leads to a significantly reduced need for coal plant capacity. Around 2.2 GW (18 percent) less coal power stations need to be installed. The utilization of coal plants also decreases—while in the BAU scenario, they run at 6400 full load hours, this number drops to 5900 under RES-E penetration. Alternatively, Morocco's demand for gas-driven electricity plants increases from approximately 1.1 GW in the baseline scenario to 2.7 GW in the

RES-E scenario. This is almost entirely due to open cycle (OC) gas turbines. Although they operate only scarcely (our simulations partially show full load hours of below 100h), their spinning reserve is needed to cover potential peak demand in times of very low renewable feed-in. Combined cycle (CC) gas power plants are not added under the competitive market environment of the model. Existing CC plants, which have been built before 2010 still remain online, but their utilization decreases over the years. In 2025 they only run 800 full load hours in the RES-E scenario and 500 full load hours in the BAU case.

b) Algeria and Tunisia show a different pattern. Here, in the absence of other base-load technologies, CC gas power stations can be considered to be providers of base-load capacity. In both countries, the increased penetration of renewables leads to a reduction in installed CC gas capacity of 14 percent in Algeria and 8 percent in Tunisia with its lower RES-E penetration. Likewise, the average utilization of CC gas plants decreases from 5700 to 5300 full load hours in Algeria and 5600 to 5200 in Tunisia. In both countries, a significant increase in the OC gas power plant population is required to maintain peak reserve capacity; in Tunisia, OC gas capacity must be increased by more than 20 percent, while in Algeria, the RES-E scenario requires a doubling of the OC gas capacity.

From this it can be concluded that a high penetration of renewable generation capacity entails a significantly reduced need for new base-load coal plants in Morocco. Tunisia and Algeria, which currently have no coal power strategy on their short-term political agenda, show a reduced need for the addition of CC gas power plants. All three countries face a significant increase in OC gas capacity, which is the most cost-efficient solution for peak load coverage.

4.3. Impact on power plant dispatch

DIME allows the output of daily dispatching profiles for every reference day for each country. In order to illustrate the behavior of the power plant dispatch, the cases of Algeria and Morocco – exemplarily on summer week-days for the years 2010 and 2025 - are compared (see Figure II-5 and Figure II-6). Besides showing the RES-E penetration, the figures also illustrate the enormous challenges

which the Maghreb countries will likely face under the aspect of increasing capacity to meet projected demand within only a 15 year time period.

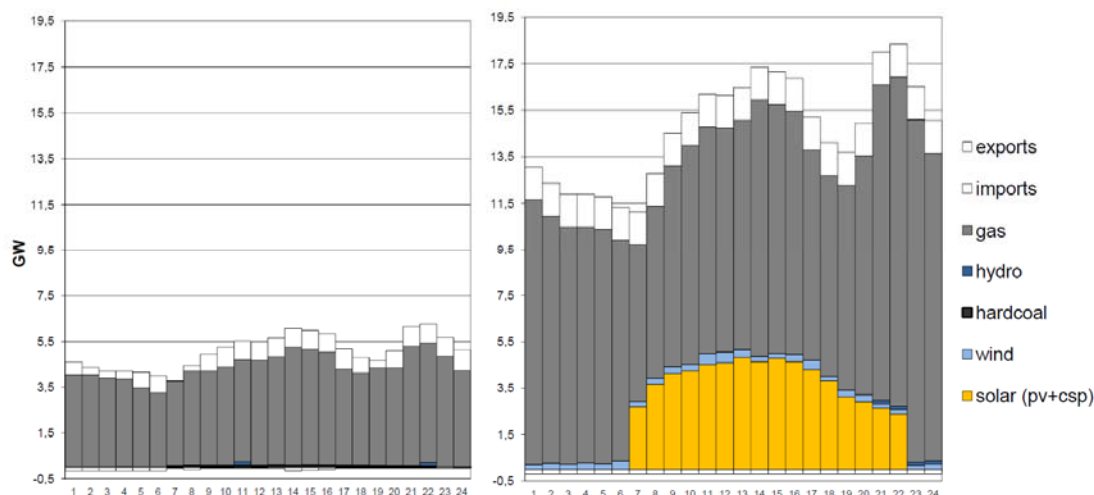


Figure II-5. Dispatch in Algeria 2010 (left) and 2025 (right) on a Wednesday in summer for the RES-E scenario.

In the Algerian case, it can be clearly seen how solar power plants start to generate electricity during daytime (there is an extension of solar generation into the evening hours, which is related to the use of thermal storage of the CSP plants). Conventional gas power plants (CC and OC are aggregated) react to this RES-E feed-in and cover most of the residual load. The small, constant band of imported electricity, visible on top of the domestic generation, completes Algeria's dispatch. A closer look reveals that these imports originate from Morocco, which, due to its coal-dominated power generation system, has cost advantages allowing electricity sales to Algeria. Morocco's exports to Algeria can also be identified in Figure II-6 as a 'negative' generation band on the bottom of the dispatch curve.

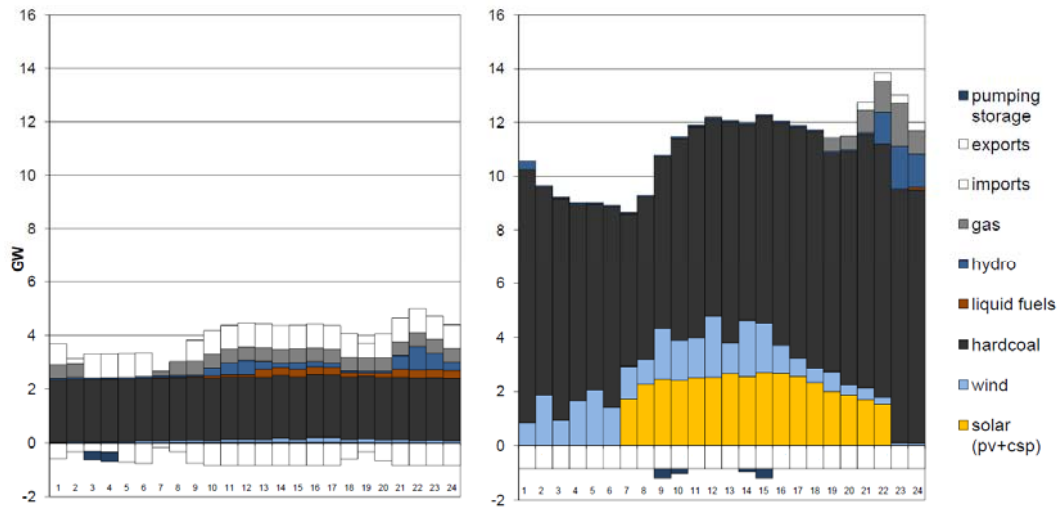


Figure II-6. Dispatch Morocco 2010 (left) and 2025 (right) on a Wednesday in summer for the RES-E scenario.

Morocco's dispatch pattern on a week-day in summer 2025 shows that, besides solar generation, an important amount of wind power also pushes into the daily generation of electricity. The residual load is mainly covered by coal power stations, but also by imports from Spain and the dispatch of hydro and gas plants. The example shows that during the evening load peak, especially when it coincides with a low wind feed-in, Morocco must launch gas and hydro plant generation and allow imports, in this case from Spain. A conclusion which can be drawn is that CSP plants with a higher storage capacity might be more advantageous for the Moroccan power system, as they would further extend solar production into the evening hours and consequently reduce the need for electricity imports and the costly dispatch of gas plants. A quantitative analysis of whether the related cost savings justify the investments in CSP plants with a higher storage capacity is currently under way.

4.4. Impact on system costs

In this chapter, we attempt to quantify the extent to which the integration of renewable energies leads to costs savings inside the conventional power genera-

tion systems of the Maghreb states. These savings will then be mirrored against the investment and operation costs of the RES-E plants, which the national economies must bear if the renewable electricity goals are to be achieved. In order to provide an accurate comparison, all considered costs are annualized and aggregated to a discounted net value in 2010 (€²⁰¹⁰).

The first value, the savings of the conventional power system, can be derived from the output results of the DIME simulations. Here, the differences between the BAU and RES-E scenarios regarding investment costs, O&M costs, fuel costs and other variable costs come into calculation. The second value, the aggregated RES-E system costs, is calculated separately on the basis of the renewable capacity installation pathway outlined in Table II-1 and under consideration of technology-specific cost assumptions, which are summarized below in Table II-4. The calculation of the annualized and aggregated costs is carried out by using a 25-year depreciation period for all renewable technologies at a real discount rate of 5 percent. It is worth to mention that the cost input parameters do not include risk primes or safety and security expenses related to the risk of political instability and terrorism. In particular, RES-E infrastructure in the desert regions of Algeria can be considered vulnerable with regards to terrorist attacks. A monetary assessment of this issue, however, cannot be given at the moment.

Table II-4. Cost input parameters for RES-E technologies

	Investment costs (€/kW)				O&M-costs
	2010	2015	2020	2025	(% of inv. costs)
CSP (SM 1.5)	4500	4000	3300	3200	2.3
Wind	1050	900	850	825	5.0
PV	3000	2300	1900	1300	1.0

The result of the cost calculations is presented in Table II-5. As expected, the realization of the renewable targets leads to substantial additional costs for all three countries (I). Until 2025, Morocco and Algeria have encountered total aggregated RES-E system costs of approximately € 7.5 billion and € 8.2 billion respectively, whereas Tunisia need only spend € 1.0 billion for the aggregated

RES-E investments and O&M costs. In all countries, the higher costs related to the setup of the RES-E system (I) are partially compensated by savings in the conventional power system (line II in Table II-5). For Morocco and Algeria, these savings amount to approximately € 1.3 billion and € 1.2 billion respectively, Tunisia realizes savings of € 0.3 billion.

Table II-5. Net present value of the aggregated costs until 2025

Discounted costs (€ ²⁰¹⁰ million)	Morocco	Algeria	Tunisia
CSP	5640	6670	290
Wind	1780	950	690
PV	110	540	60
Total costs RES-E system (I)	7530	8160	1040
Total savings conventional system (II)	1260	1230	280
Net costs to fulfill renewable targets (I-II)	6270	6930	750
RES-E system-efficiency (II/I)	0.17	0.15	0.27

A qualitative look at the origins of the RES-E induced savings reveals that the main contributors to the savings are not the avoided investments in conventional power plants, but rather the avoided fossil fuel consumption. In Algeria and Tunisia, avoided fuel costs contribute to around 90 percent of the total savings; in Morocco they amount to 80 percent. This difference between the countries can be explained by the fact that renewable plants in Tunisia and Algeria substitute electricity generated by gas, which is a relatively expensive fuel. In Morocco, where coal-fired power stations dominate the generation market, RES-E plants can only reduce the utilization of the less-costly hard coal. Therefore, fuel savings in Morocco are comparatively low.

Alternatively, renewable energies have a stronger impact on avoided investment costs in Morocco's conventional power generation system: By substituting expensive hard coal plants, more investment savings can be achieved in Morocco compared to in Algeria or Tunisia, where only the additions of less capital-intensive gas facilities are avoided.

Another important parameter –particularly relevant for political decision-makers in North Africa - is the resulting net costs which must be covered by the countries in the RES-E scenario. As outlined in Table II-5, these costs are the difference (I-II) between the aggregated costs for the RES-E system (I) and the resulting savings within the conventional power system (II). Until the 2025 period, these net costs amount to approximately € 6.3 billion for Morocco, € 6.9 billion for Algeria and €0.75 billion for Tunisia.

If the cost reductions of the conventional system (II) are put in comparison with the corresponding costs of the RES-E system (I), a further interesting aspect arises. The ratio between the two values (II/I) gives an indication how efficiently the countries' renewable energy targets match the conventional power system. In the RES-E scenario, the Tunisian RES-E mix substitutes for conventional electricity costs nearly twice as efficiently as the Algerian and the Moroccan RES-E mixes. For each Euro (€) spent for Tunisia's RES-E goals, a cost reduction of € 0.27 in the conventional power system can be achieved, whereas the corresponding savings in Algeria and Morocco are only € 0.15 and € 0.17, respectively. The explanation of these differences goes in line with the above-described cost effects of RES-E integration into the conventional power system: Tunisia, with its strong dominance of gas power plants, profits significantly from RES-E integration, because it substitutes for expensive gas fuel. Additionally, Tunisia takes advantage of a second effect—its high wind share compared to CSP and PV decreases the overall RES-E system costs. Algeria also profits from avoided expenses for gas fuel, but has more extensively invested in CSP and PV capacity. Therefore, the integration of RES-E sources into the Algerian power system is significantly less cost-efficient than in Tunisia. Morocco, alternatively, shows a mixed picture. Its renewable generation system contains relatively expensive solar power plants, as well as very cost-efficient wind farms, which, due to excellent conditions, contribute to more than 70 percent of the RES-E generation in 2025. On the other hand, the monetary fuel substitution effect in Morocco is low, because only relatively cheap coal-generated electricity is replaced. Both effects lead to a modest RES-E system efficiency in Morocco, which remains at the same level as Algeria.

From a purely economic perspective, setting aside technical aspects related to grid integration and system stability, this poses the question as to why Morocco

and Algeria currently put so much emphasis on solar power plants in their renewable goals, instead of focusing on wind power as the cheaper, ‘low hanging fruit.’ While in Algeria this might be due to a geographically-limited availability of wind sites (Himri, 2009), Morocco could very likely enhance the cost efficiency of its RES-E goals by increasing the presence of wind power, as suitable wind sites are considered abundant in the country (CDER, 2007).

5. Summary and Discussion

In this study, we used a linear power market optimization model to analyze the impact of renewable energy integration into the power systems of three North African countries—Morocco, Algeria and Tunisia. For that purpose it was assumed that the countries fulfill their self-imposed renewable electricity targets until 2025 and form a competitive regional electricity market that includes adjacent European countries. By comparing a renewable energy scenario (RES-E) to a baseline (BAU) scenario with no renewable electricity generation, we characterized and quantified some principal effects that accompany an increased RES-E penetration in the countries’ electricity systems. The model results show that for all countries, renewable energies are able to replace an important part of fossil generation. This leads to noticeable effects in the conventional generation system—the utilization of base load plants is reduced, while there is a stronger need for investments in flexible OC gas power plants.

Additionally, fluctuating renewable energy generation influences the hourly dispatch of conventional power plants which, under peak load conditions and weak RES-E feed-in, reacts with an increased dispatch of gas power plants. In a subsequent cost analysis, we compared the surplus costs incurred from achieving the RES-E goals with the savings resulting from avoided use of fossil fuels and investments in conventional power plants. For each Euro (€) spent on the RES-E goals, savings in the conventional power system of € 0.15 (Algeria), € 0.16 (Morocco) and € 0.27 (Tunisia) can be achieved. These relatively strong disparities between the countries’ specific RES-E savings raise the question of whether there is still room for improvement in the national renewable electricity targets of the Maghreb countries. Further work in this field will be required, for example by providing an analysis of how RES-E power plants could - in conjunction

with the conventional generation system - better reflect the specific renewable potentials of the Maghreb states. Another topic to be discussed is whether RES-E integration could be additionally optimized by more trans-national coordination and, perhaps, even a future harmonization of the renewable energy policies of the Maghreb states.

Acknowledgements

We would like to thank the company Fichtner GmbH, Stuttgart for providing technical parameters and feed-in profiles of CSP plants used in our simulations. Furthermore we thank Mr. Fawzi Kharbat (NEPCO, Jordan) for clarifications and background information on the AUPTDE power plant database. Special thanks likewise go to our colleague Heike Wetzel for her valuable discussions and scientific support.

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III. The value of dispatchability of CSP plants in the electricity systems of Morocco and Algeria¹²

Abstract

This paper examines the effects of an increased integration of concentrated solar power (CSP) into the conventional electricity systems of Morocco and Algeria. A cost-minimizing linear optimization tool was used to calculate the best CSP plant configuration for Morocco's coal-dominated power system as well as for Algeria, where flexible gas-fired power plants prevail. The results demonstrate that in both North African countries, storage-based CSP plants offer significant economic advantages over non-storage, low-dispatchable CSP configurations. However, in a generalized renewable integration scenario, where CSP has to compete with other renewable generation technologies, like wind or photovoltaic (PV) power, it was found that the cost advantages of dispatchability only justify CSP investments when a relatively high renewable penetration is targeted in the electricity mix.

1. Introduction

In recent years, concentrated solar thermal power (CSP) technologies have received increased attention not only from the electricity sector, but also from the media and research. One focal point of interest, and a strong contributor to the current popularity of CSP, concerns the desert regions of North Africa. Here, where solar irradiation is high and unused land abundant, many observers see significant opportunities for large-scale deployment of the technology. Promoters of green electricity export scenarios, like Desertec (Dii, 2011) or the Medi-

¹² This chapter is based on a peer-reviewed article published in the journal *Energy Policy*: Brand, B., Stambouli, A., Zeijli, D., The value of dispatchability of CSP plants in the electricity systems of Morocco and Algeria. *Energy Policy* 47 (2012) 321-331, doi: 10.1016/j.enpol.2012.04.073.

terranean Solar Plan (UfM, 2011), give CSP a paramount role in North Africa, particularly with regard to its geographical proximity to Europe as a solar electricity export destination.

Instead of examining such renewable export scenarios, the present work examines the effects of CSP integration into local electricity markets. Many North African governments already have their own policy goals for renewable energy in their future electricity mixes (Brand and Zingerle, 2010). For local planners and decision-makers, it is important to know to what extent CSP technology can be integrated into their countries' electricity systems and how this can be carried out in the most cost-efficient manner. Very often, economic considerations about renewable energy costs are limited to a mere comparison of the levelized costs of electricity (LCOE). If compared on the basis of this metric, the results show that wind – and by now, even photovoltaic (PV) power – features lower electricity generation costs than CSP (Kost and Schlegel, 2010). But choosing technologies on the basis of levelized cost comparisons alone can be misleading. Joskow (2010) points to the shortcomings of such an approach, stating that an LCOE-based comparison would generally overvalue intermittent generation technologies compared with dispatchable power plants.

CSP plants become dispatchable when they are outfitted with thermal storage systems, enabling them to store solar energy during daylight hours and transform it into electricity at a later time. On electricity markets, this ability to control the power output brings about two sources of economic value that are not reflected in a mere LCOE-based analysis.¹³ First, there is the “time of delivery value” (Sioshansi and Denholm, 2010), representing the cost savings within the daily dispatch of the power system. In CSP plants, the electricity production can be shifted to periods of high demand (Morocco and Algeria, for instance, feature pronounced demand peaks in the evening hours), thereby reducing the need for expensive peak load plants. The second source of economic value originates from the CSP plants' ability to provide firm capacity to the power system. If properly designed with thermal storage systems, and potentially also with auxil-

¹³ Sioshansi and Denholm (2010) mention a third source of economic value, “ancillary services,” such as the provision of regulation and spinning reserves. However, an analysis of this value is not included in the present work.

iary fossil fuel-fired heaters, CSP plants can guarantee full capacity at any time of the year. For the steadily expanding North African power generation systems, this means that CSP plants can avert future investments into conventional power plants – and reduce overall system costs.

The assessment of these economic advantages, in the literature often referred to as the “value of CSP dispatchability” or the “value of CSP storage,” is becoming an important research topic, as demonstrated by a number of recent studies. Sioshansi and Denholm (2010) calculate the potential operating profits of CSP plants with thermal storage in the electricity wholesale markets of the U.S. states of California, Texas, Arizona, and New Mexico. An analysis by Aga et al. (2011) focuses on the power markets of California and Spain, applying a numerical model to determine the revenue-maximizing storage configuration of CSP. Nagl et al. (2011) examine the value of thermal storage by applying a dynamic dispatch and investment model to the power system of the Iberian Peninsula (Spain and Portugal), while Poullikas et al. (2010) focus on the integration effects of solar thermal power plants in Cyprus. To date, the North African electricity systems have not been addressed by any model-based CSP integration studies.

For the present study, Morocco and Algeria were selected as case studies because of their promising solar potential, but also because of their distinct electricity systems, which provide an interesting comparison: Algeria features a flexible generation system with a large number of combined cycle (CC) and open cycle (OC) power plants, while Morocco relies on less flexible coal-fired steam power plants. This paper examines the interaction of storage-based CSP plants within these different electricity system environments. Additionally, we analyze what implications a higher share of other fluctuating renewable energy technologies like PV or wind power would have for the integration of solar thermal power plants in Morocco and Algeria.

We performed our analysis using an electricity market model which simulates CSP operation on the short-term level (daily dispatch) as well as in the long run, considering the different CSP penetration scenarios in the countries’ future electricity systems. The study begins with a brief analysis of the power systems of Morocco and Algeria and the political strategies for renewable and solar elec-

tricity in these countries, followed by an overview of CSP technologies. The modeling approach itself and the assumptions for the scenarios analyzed are described in the subsequent sections. In the concluding section, the model results are analyzed and discussed.

2. The power generation systems of Algeria and Morocco

2.1. The power system of Algeria

Algeria is one of the world's largest gas producers, and it is not surprising that the country uses this primary energy for domestic electricity generation. Currently, natural gas contributes 98% of the country's electricity production of around 45 TWh. The national power plant inventory consists of about 3.8 GW of open cycle (OC) gas turbines, 2.3 GW of combined cycle (CC) natural gas power stations, and 2.7 GW of gas-fired steam power plants (CREG, 2010). Since the Algerian government prioritizes the use of domestic fuels, it can be expected that gas power plants will continue to dominate the conventional part of Algeria's electricity system in the future. Renewable energies, likewise a domestic energy source, represent a further diversification option for Algeria.

A recent renewable energy development program published by the Algerian Ministry of Energy (MEM, 2011) identifies CSP and PV as the pillars of future renewable electricity generation. By 2030, it is intended that these technologies together supply about 37% of the national electricity production. An indicator that Algeria actually has taken steps toward a solar-based renewable energy strategy is the construction of the first integrated solar combined cycle (ISCC) plant in Hassi R'Mel (Central Algeria), as well as the government-supported set-up of a local PV module manufacturing industry to supply future domestic PV power plants projects (Photon, 2011). Wind energy, in contrast to the immense solar energy potential, has fewer development options and is given only a minor 3% share in the Algerian power mix by 2030. Likewise, few expansion options exist for hydropower, which today totals 160 MW of installed capacity.

2.2. The power system of Morocco

As the only North African country without any noteworthy fossil fuel energy reserves, Morocco must import almost all of its primary energy. Coal power plants, fueled by imported hard coal, form the backbone of the country's generation system. With 1.8 GW of installed capacity (Morocco's total capacity is 6.4 GW), they contribute about 45% of the country's forecasted electricity production of 26.5 TWh (ONE, 2010). It can be expected that at least for the next decade, this technology will continue to dominate the Moroccan electricity mix, as additions of approximately 2.0 GW of new coal-fired power stations are envisaged in the near future to meet the country's steadily growing electricity demand (Hajroun, 2010).

About 12% of Morocco's electricity demand is currently supplied by OC and CC gas power generation, and this share is expected to rise with additional CC power plants coming online in the near future. Liquid fuels (used in diesel- or oil-fired steam plants) still provide 13% of total demand, but the use of these energy resources will decrease in the long run due to high oil prices. With limited expansion possibilities, the share of hydropower, currently at 13%, is also set to decline in the future. A special feature of Morocco's power supply is the direct import of electric energy from Spain via a submarine interconnector through the Strait of Gibraltar. In 2010, this amounted to a substantial 14% of Morocco's electricity consumption. Wind energy currently covers only 2% of national electricity demand, although this share is likely to increase by 2020 if Morocco's electricity utility, Office National d'Electricité (ONE), rolls out a 2.0 GW wind development project, as planned. Solar energy, the country's second renewable energy asset, is also receiving strong government support. The Moroccan government recently embarked on a solar roadmap called the Moroccan Solar Plan, which targets the implementation of 2.0 GW of solar power plant projects through the end of this decade. Initially intended solely for solar thermal technologies, part of the 2.0 GW portfolio shall now also include PV technologies, although the exact proportions within the program are not yet known. A recently created government agency, the Moroccan Agency for Solar Energy (MASEN), is in charge of steering the roll-out of the Moroccan Solar Plan (Masen, 2010).

3. CSP technology

Concentrated solar thermal power plants can be categorized into four different technology types: parabolic trough systems, linear Fresnel mirrors, central receiver systems, and dish stirling systems. In this study, parabolic trough CSP plants have been chosen as the technology of reference.¹⁴ Through their success in commercial operations in a multitude of projects worldwide, they now represent today's most widespread CSP technology. At least for the coming decade, these plants are expected to dominate the CSP market (Sargent & Lundy, 2009). Standard parabolic trough plants are made of the following basic components: (1) a parabolic mirror collector field that produces solar energy by heating a synthetic heat transfer fluid (HTF); (2) the power block, consisting of a heat exchanger, a steam generator, and a steam turbine; and (3) a two-tank molten-salt thermal storage system. The configuration of these engineering components is represented by the following main economic sizing parameters of parabolic trough plants in our model analysis:

- Nominal capacity
- Solar multiple (SM)
- Storage size

The nominal power block capacity is measured in MW, while the solar multiple, a dimensionless parameter, expresses the relative size between the collector field and the power block. An SM of 1.0, for instance, equates to a mirror field that is able to drive the power block at nominal capacity under design-point conditions. These conditions are usually defined as the direct normal irradiation (DNI) on a typical solstice day at noon (Montes et al., 2009). Higher solar multiples allow the plant to produce surplus thermal energy, which can be stored for later use. The plant's storage capacity is rated by the number of hours (h) during which a fully charged storage system can supply thermal energy to feed the turbine at nominal power. Different combinations of SM and storage size change the properties of the plant's dispatchability, enabling the plant to be tailored to the flexi-

¹⁴ In the following sections, we use the terms "CSP" and "parabolic trough power plants" interchangeably.

bility requirements of the surrounding electricity system (Brand and Missaoui, 2011).

4. Model description

A bottom-up power supply model was used to analyze the integration of CSP into the electricity markets of Algeria and Morocco. The model is based on a linear optimization approach described by Bartels (2009), Swider et al. (2005), and Nagl et al. (2011), where both the cost-minimizing dispatch and the optimal investment pathway are calculated for each country's power generation system. For simplicity, the power systems of Morocco and Algeria are considered to constitute autonomous model regions that do not exchange electricity between themselves or with third neighbors. From this viewpoint, the model can be seen as a vertically integrated state-owned power company that acts as a cost-minimizing agent on its national electricity system, operates all existing power plants, and makes decisions about investments into new capacity. We opted for this model approach because it appears to fit very well with the current situation in Morocco and Algeria, where the state still rules over the power markets as a single agent. In the model, investments are carried out over 5-year periods, beginning with the year 2010. Each period consists of 32 representative days, with two typical load days (a working day and a weekend day) for each of the four seasons, and four typical meteorological situations per season. The limitation to 32 typical days, instead of the 365 days of a full year, was necessary to keep the calculation time within reasonable limits. Provided that the technical characteristics and operational constraints are respected, the agent has full control over the dispatch of the power generation system in 1-hour time steps within each of the representative days. By optimizing dispatch and investment decisions, the total discounted system costs, expressed by the following objective function, are minimized:¹⁵

¹⁵ All parameters in upper case represent variables being subject to optimization, whereas the parameters in lower case are fixed and set exogenously.

$$TOTAL = \sum_y disc_y \cdot (INV_y + OM_y + VAR_y + FUEL_y)$$

The four main cost constituents are investment costs, INV_y , fixed operation and maintenance costs, OM_y , variable operation costs, VAR_y , and fuel costs, $FUEL_y$. All cost components are aggregated to 5-year periods ($y = 2010, 2015, 2020, \dots$). The discount factor, $disc_y$, brings the costs to a net present value of total costs ($TOTAL$) in the basis year 2010 (€^{2010}). The overall real discount rate is 5%.

A set of equations links the cost components with technical variables. The investment costs, for instance, depend on additions of new plants to the total installed capacity, $CAP_{tech,y}$, (in MW) of the specific technology class ($tech = coal, gas, csp, wind, \dots$). Variable and fuel costs, are a function of the model variable $GEN_{tech,y,d,h}$, representing the actual electricity generation (in MWh) at any hour ($h = 1, \dots, 24$) and typical day ($d = 1, \dots, 32$) of the period y .

The optimization is carried out under the boundary condition that total electricity generation meets the demand at any hour of each representative day (energy balance constraint). Demand is exogenously determined by the typical electricity load pattern of Morocco or Algeria ($load_{y,d,h}$) plus an endogenous load referring to electricity drawn by pumps of hydro storage systems. The pumping option plays a role in Morocco, where two hydroelectric pumping systems with an overall capacity of 437 MW are installed.

An additional constraint (peak capacity constraint) prescribes that the available part of the total installed capacity must always be equal to or greater than an exogenous parameter $peakload_y$, the load maximum over the period y (usually incorporating a security margin). In Morocco and Algeria, annual peak load projections are part of the electricity demand forecasts published by the national utilities or regulatory bodies. The peak capacity constraint includes a factor for the capacity credit, indicating for each generation technology its capability to provide firm capacity to the power system. For renewable generation technologies, the capacity credit can be understood as the fraction of intermittent capacity by which conventional power generation capacity can be reduced without affecting the loss of load probability of the system (Giebel, 2005). In this sense,

the capacity credit of conventional power plants is always 100%, while its value for intermittent renewable technologies is generally lower.

The model likewise incorporates a number of further technical constraints corresponding to the dispatch properties of the thermal generation technologies (ramp-up and shut-down behavior, efficiencies, part-load conditions, ...), which cannot be elaborated here in detail (see Bartels (2009) and Swider et al. (2005)).

The contribution of hydropower is generally small in arid countries. Only Morocco holds noteworthy hydro power stations, but the technology in this country has reached its expansion limits, such that no major capacity additions are expected in the future. Nevertheless, operation of the existing capacities still has to be considered in the simulations. The model distinguishes between conventional hydroelectric dams and pumped storage facilities.

For renewable power plants without inherent storage technology (photovoltaic plants and wind farms), operational bounds prescribe that electricity production must always follow the daily variations of irradiation and wind speed. The limiting parameters, *p_v_profile* and *wind_profile*, are set exogenously, ranging from 0 to 1. They represent a deterministic 24-hour output pattern of a typical photovoltaic plant or wind farm and are derived from irradiation and wind speed time series. The model allows a “curtailment” of photovoltaic and wind power, which means that solar or wind resources can be left unused if this is beneficial for overall system costs.

CSP plants require a relatively extensive set of equations to describe their operation and dispatch constraints, which shall be outlined in the following in more detail. The equations differ from the proposal of Nagl et al. (2011) in that the operational constraints more precisely mirror the technical operation of the three main technical constituents of the plant: the mirror field, the power block and storage (see Figure III-1). Dumping of solar energy, for instance, is part of the optimization process. Moreover, the equations allow for easy variation of the solar multiple and the storage capacity, which is needed for the later optimization of the CSP configuration (see Section 6.1).

For parabolic trough power plants with a salt storage system, the central task of the linear optimization program is to decide which part of the hourly available

solar energy input (CSP_SUPPLY) is directly used to generate electricity and which part is routed to the thermal storage for a later dispatch. The variable CSP_SUPPLY can be regarded as equivalent to the solar thermal energy produced by the mirror field of a CSP plant:

$$CSP_SUPPLY_{csp,y,d,h} = CAP_OP_{csp,y,d} \cdot csp_profile_{d,h} \cdot csp_multiple_{tech}$$

In this equation, $CAP_OP_{csp,y,d}$ is the nominal online capacity of the CSP plant's power block. The exogenous factor $csp_profile$ varies with the hourly changes of direct solar irradiation; $csp_multiple$ represents the solar multiple (SM) of the plant.

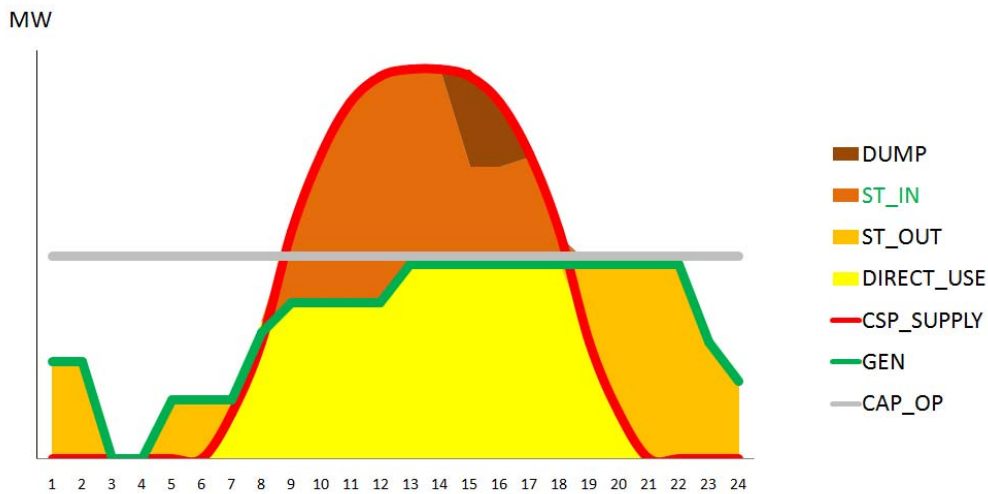


Figure III-1. Example operation profile of an SM 2/6h storage CSP plant on a cloudless summer day

Our approach considers a “rolling” 1-day cycle strategy for operation of the parabolic trough plants, as proposed by Sioshansi and Denholm (2010) and Giuliano et al. (2010). The model is simplified by assuming the plant operator has complete foresight and can store energy in anticipation of the next day's

demand (consecutive days are, as a further simplification, considered identical). Figure III-1 illustrates the interplay between SM, storage size, and the other model variables. The energy generated by the turbine (GEN) is the sum of the energy drawn from the storage system ($STORAGE_OUT$) and the directly used part of the thermal output of the mirror field ($DIRECT_USE$):

$$GEN_{csp,y,d,h} = ST_OUT_{csp,y,d,h} + DIRECT_USE_{csp,y,d,h}$$

The electricity production must always remain lower than the turbine's online capacity:

$$GEN_{csp,y,d,h} \leq CAP_OP_{csp,y,d,h}$$

The part of the mirror field's output that is used for direct electricity production must not exceed the energy supplied by the solar field (CSP_SUPPLY), and additionally, it must remain below the online capacity:

$$DIRECT_USE_{csp,y,d,h} \leq CSP_SUPPLY_{csp,y,d,h}$$

$$DIRECT_USE_{csp,y,d,h} \leq CAP_OP_{csp,y,d,h}$$

All solar energy supplied by the mirror field (CSP_SUPPLY) must be used directly by the power block, sent to the storage tank, or dumped:

$$CSP_SUPPLY_{csp,y,d,h} = DIRECT_USE_{csp,y,d,h} + ST_IN_{csp,y,d,h} + DUMP_{csp,y,d,h}$$

Dumping occurs when the storage volume is saturated and there is still more irradiation available than the plant's turbine can directly process. Technically, in this case, the solar collectors would be defocused. Power generated from the storage system cannot be higher than the online turbine capacity:

$$ST_OUT_{csp,y,d,h} \leq CAP_OP_{csp,y,d,h}$$

Within a one-day storage cycle (24 hours), the amount of energy taken from the storage system must equal the stored energy reduced by a loss factor $loss_{csp}$.

$$\sum_h ST_OUT_{csp,y,d,h} = loss_{csp} \cdot \sum_h ST_IN_{csp,y,d,h}$$

The factor $loss_{csp}$ represents thermal and heat exchanger losses during cycling of the storage system. For standard HTF/molten-salt systems, this factor is set at 0.985 (Sioshansi and Denholm, 2010). An additional constraint takes account of the limited capacity of the thermal energy storage. Rated in hours ($storage_hours$) of nominal plant capacity, it represents the upper limit of the energy discharged from the storage system each day:

$$\sum_h ST_OUT_{csp,y,d,h} \leq storage_hours_{csp} \cdot av_{csp,d} \cdot CAP_{csp,y}$$

Further operational constraints, which are not listed here in detail, exclude events of simultaneous charge and discharge of the storage system, and ensure that the storage system can only be discharged at times when the solar supply is insufficient to run the turbine at nominal capacity.

With the market costs of renewable energies still comparatively higher than conventional power, the model would omit renewable capacity additions in its optimization process, preferring a conventional investment pathway as the least-cost solution. Fixed renewable penetration targets were introduced to bring renewable power into play. The model receives an exogenous constraint in the form of a minimum energy production amount (in MWh), which must be supplied by renewable sources. This minimum renewable production is expressed as a percentage share of the total electricity demand and forces the model to actively install as much renewable capacity as is necessary to produce a certain “renewable” percentage (*res_share*) of the overall power demand.

5. Assumptions and input parameters

5.1. Demand

The model requires the following demand input parameters: the total annual electricity demand (*demand_y*) in TWh for each model period, the projected load maximum (*peakload_y*) in MW, and the typical hourly load pattern for each representative day (*load_{y,d,h}*). For the demand and peak load data, we utilized demand forecasts published by the Moroccan Ministry of Energy (Hajroun, 2010) and the Algerian electricity regulation authority (CREG, 2010).

Table III-1. Demand and peak load developments according to Hajroun (2010) and CREG (2010).

	2010	2015	2020	2025	2030
Morocco total demand (TWh)	26.5	40.0	57.0	85.0	116.0
Morocco peak load (MW)	4600	7400	9300	13900	18500
Algeria total demand (TWh)	45.2	60.0	79.0	105.0*	140.0*
Algeria peak load (MW)	8000	11000	14500	19000*	24700*

*Extrapolated values

Daily load profiles were retrieved from historical load data available in online databases of the Moroccan electricity utility ONE, and the Algerian transmission system operator Opérateur du Système Electrique (OSE). The load profiles were normalized and linearly scaled up with the respective expected annual demand in future periods.¹⁶

5.2. Generation

Eight generation technologies can contribute to electricity generation in the model: hard coal steam plants (Morocco only), gas-powered steam plants, CC gas plants, OC gas turbines, liquid fuel power plants (oil and diesel), hydropower (including pumped storage for Morocco), wind, photovoltaics, and CSP.

5.2.1. Conventional power plants

The model incorporates data from a power plant inventory of Morocco and Algeria (AUPTE, 2007); the data were updated with publicly available information about recently installed plants and future power plant projects. Technical plant parameters (e.g. plant efficiency, minimum load conditions, and availability) were derived from a recent study of the Tunisian generation system (GIZ, 2011), which we considered a reasonable estimate for Algerian and Moroccan conditions. The same publication provided financial parameters, such as investment costs, operation and maintenance (O&M) costs, and variable operation costs, as well as price developments for coal and natural gas. Carbon emission costs are not applied for fossil power plants in North Africa.

¹⁶ Forecasts of future load profile transformations in Morocco and Algeria were not available. It should be noted that model results might change, if, for instance, the mid-day peak becomes more pronounced in the countries due to progressing industrialization and an increased use of air-conditioning.

5.2.2. Concentrated solar power plants

The electricity generation of parabolic trough power plants in the model follows the descriptions outlined in Section 4. One important input parameter is $csp_profile_{tech,d,h}$, representing the normalized hourly energy supply of the plant's collector field for each of the 32 representative days. For simplicity, and to distinguish the effects of CSP integration from other, irradiation-dependent effects, we used one irradiation data set for both countries. The input refers to a site in eastern Morocco, not far from the Algerian border (Ouarzazate, average DNI of 2500 kWh/m²), for which an online database (Helioclim, 2010) provided hourly DNI irradiation data. These were subsequently processed by the System Advisor Model software (NREL, 2010) to obtain the hourly collector field output values for $csp_profile_{tech,d,h}$. As for the arid climate of North Africa, it is assumed that all CSP plants are dry-cooled. Likewise, it is assumed that all CSP plants are equipped with auxiliary fossil fuel-fired HTF heaters. The heaters' contribution to net electricity generation is negligible, but their presence guarantees plant capacity at peak load, even without sunshine and when the thermal storage is empty. Electricity system operators often require plants to hold such back-up heaters available (Goebel, 2010), as doing so increases the firm capacity of the power system. As a result, a relatively high capacity credit of 90% is assumed in our model for all parabolic trough configurations.

The financial parameters of parabolic trough power plants mainly depend on the sizing of the three main subunits (mirror field, power block, and storage system). Therefore, the specific investment costs (€/kW) are a function of the plant's solar multiple and storage capacity. With the help of the component-specific costs database of NREL's System Advisor Model (NREL, 2010), we performed a breakdown of the plant's total specific capital costs for each examined combination of SM and storage capacity, arriving at three cost portions: costs of the collector field (C_C), costs of the power block (C_P), and costs of the storage system (C_S). These costs are considered current equipment costs in the basis year 2010. To obtain costs for the subsequent model periods, y (2015, 2020, ...), we applied a standard learning curve approach (e.g. as described in Junginger et al., 2010):

$$C_y = C_{2010} \cdot \left(\frac{I_y}{I_0}\right)^{\frac{\lg(PR)}{\lg 2}}$$

This calculation was carried out for each cost constituent, C_C , C_P , and C_S , assuming progress ratios (PR) of $PR_C = 90\%$, $PR_P = 94\%$, and $PR_S = 88\%$ for the collectors, the power block, and the storage systems, respectively (Junginger et al., 2010). The increase in the worldwide cumulative installed CSP capacity from 2010 (I_0) to future years (I_y) is based on the assumptions of an “optimistic-realistic” global CSP market development scenario, as outlined by Viebahn et al. (2010). The described method can be used to calculate the specific investment costs for the future periods of any plant configuration with different solar multiples and storage capacities. Table III-2 displays the cost parameters of one exemplary configuration of an SM2/6h parabolic trough plant until 2030. Fixed O&M costs are set at 53.8 €²⁰¹⁰/kW_a and variable costs at 2.3 €²⁰¹⁰/MWh (NREL, 2010).

Table III-2 Specific investment costs for a parabolic trough plant with SM2 and 6h storage.

		Specific investment costs (€ ²⁰¹⁰ /kW)		
		2010	2020	2030
SM 2 / 6h	Collector field	2800	1870	1470
	Power block	1850	1460	1270
	Storage system	1430	880	660
	Total	6090	4200	3400

5.2.3. Wind and PV

The hourly time series of photovoltaic generation (*pv_profile*) is based on an exemplary PV plant with crystalline cell technology at Ouarzazate, while *wind_profile* is derived from wind speed time data from representative wind sites (Tetouan and Essaouira in Morocco and Adrar in Algeria). The specific

investment costs for wind and PV plants are shown in Table III-3 (EWEA, 2009; IEA, 2009; IEA, 2010; and EPIA, 2010). O&M costs are estimated at 20.0 €/2010/kWa for PV and 40.0 €/2010/kWa for wind power.

Table III-3 Specific investment costs for onshore wind and PV power plants.

	Specific investment costs (€/2010/MW)		
	2010	2020	2030
PV power plants	2500	1380	1000
Wind farms	1150	980	930

With both technologies providing intermittent feed-in, our estimates for the capacity credits remain conservative, at 5% for wind power and 0% for photovoltaic plants.

6. CSP integration into the conventional power systems of Morocco and Algeria

In this section, we examine whether the characteristics of Algeria's and Morocco's conventional electricity systems have an effect on the integration efficiency of parabolic trough plants. To facilitate understanding of the basic effects of CSP integration, we assume that until 2025 Morocco will continue to base its conventional generation system on a combination of coal and gas plants, and Algeria will maintain its strategy of prioritizing gas power generation. Furthermore, it is assumed that, apart from hydropower and the already existing wind parks in Morocco, no renewable technology other than CSP will take part in future electricity generation. This restriction was required to discern the influence of the conventional power system on CSP integration from the effects of other intermittent renewable sources. (An analysis of the integration of CSP plants in a renewable power mix with wind and PV is provided in Section 7).

6.1. Impact of CSP configurations on total system costs

The first analysis to be performed by the model was to determine the optimal CSP configuration for different solar penetration scenarios in Morocco and Algeria. Two scenarios were examined: (1) a “low penetration” scenario, where the CSP share in both countries increases in a stepwise manner from 0% in 2010 to 5% in 2025; and (2) a “high penetration” scenario, where the CSP share climbs to 20% by 2025. Several model runs were performed to calculate the total costs ($TOTAL_{SMx/y}$) for different combinations of solar multiples and storage capacities. Solar multiple values were set at $x = 1, 1.5, 2, 2.5$, and 3, while the corresponding storage sizes ranged from $y = 0\text{h}$ (no storage) to 10h in 1-hour increments¹⁷. It was found that for each penetration scenario, there is one ideal combination of SM and storage capacity that reduces the total costs to an absolute minimum ($TOTAL_{min}$). These optimum combinations are shown in Table III-4.

Table III-4 Cost-minimizing combinations of solar multiple (SM) and storage capacity for the integration of parabolic trough plants into the conventional power systems of Morocco and Algeria until 2025.

	Morocco	Algeria
5% CSP penetration target	SM 1.75 / 4h	SM 1.75 / 3h
20% CSP penetration target	SM 2 / 6h	SM 2.25 / 8h

Figure III-2 depicts contour charts of the surplus costs that incur when SM/storage combinations other than the optimized configurations shown in Table III-4 are simulated. The x-axis of each graph refers to the solar multiple and the y-axis to the capacity of the storage system in hours. The surplus costs, or more precisely, the relative deviations from the total cost minimum $TOTAL_{min}$,

¹⁷ To improve precision, cost minima were additionally determined with SM 0.25 resolution

$$\frac{TOTAL_{SMx/y}}{TOTAL_{min}} - 1$$

are displayed in the colored areas. It can be seen that in the case of the 5% CSP penetration scenario (Figure III-2a and 2c), the cost deviation from the minimum configuration remains relatively low over large areas of SM/storage combinations (see blue area, where the cost deviation is smaller than 0.5%). However, if a goal of 20% CSP penetration is targeted (Figure III-2b and 2d), this area begins to narrow. Particularly in the case of Morocco (Figure III-2b), shifting away from the optimum configuration means a relatively strong penalization by increased total system costs. In the case of Algeria (Figure III-2d), this effect appears to be less pronounced. Here, the area of moderate additional costs is larger compared to that of Morocco and extends to CSP configurations with smaller solar multiple and storage sizes. Thus, it can be stated that the dispatchability of CSP plants matters more in the Moroccan power system than in the Algerian one, and that this effect becomes more significant with increasing CSP penetration.

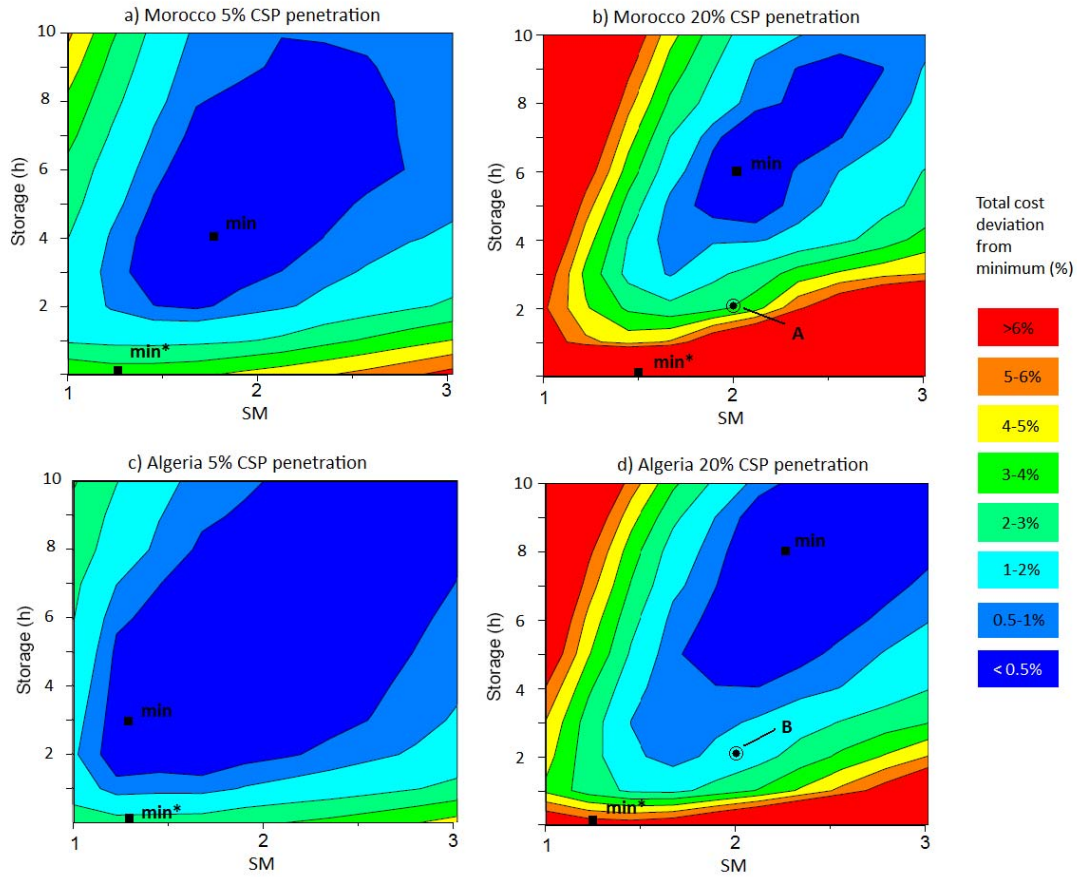


Figure III-2. CSP integration costs charts. Deviation from the cost minimum, *min*, in percent for 5% and 20% solar penetration in Morocco and Algeria. The configuration at *min** represents the cost minimum for non-dispatchable CSP plants

An explanation of this behavior is qualitatively given by some typical parabolic trough plant operation patterns. Figure III-3 (left, bottom) depicts the hourly dispatch of the best-choice CSP configuration (SM2 and 6h storage, *min* in Figure III-2b) in Morocco on a summer day in 2025 in the 20% CSP penetration scenario. The model optimizes daily electricity production by releasing stored solar energy at hours of high demand, between 19h and 23h, and to a minor extent, also in the morning hours. A look at the details of the plant's operation (Figure III-3, left, top) reveals that most of the available solar energy is actually utilized and converted into electricity; only a small amount has to be dumped. Nevertheless, dumping is part of the optimal model solution because there is a

trade-off between the avoided investments into a larger thermal storage system and the value of the unused solar energy.

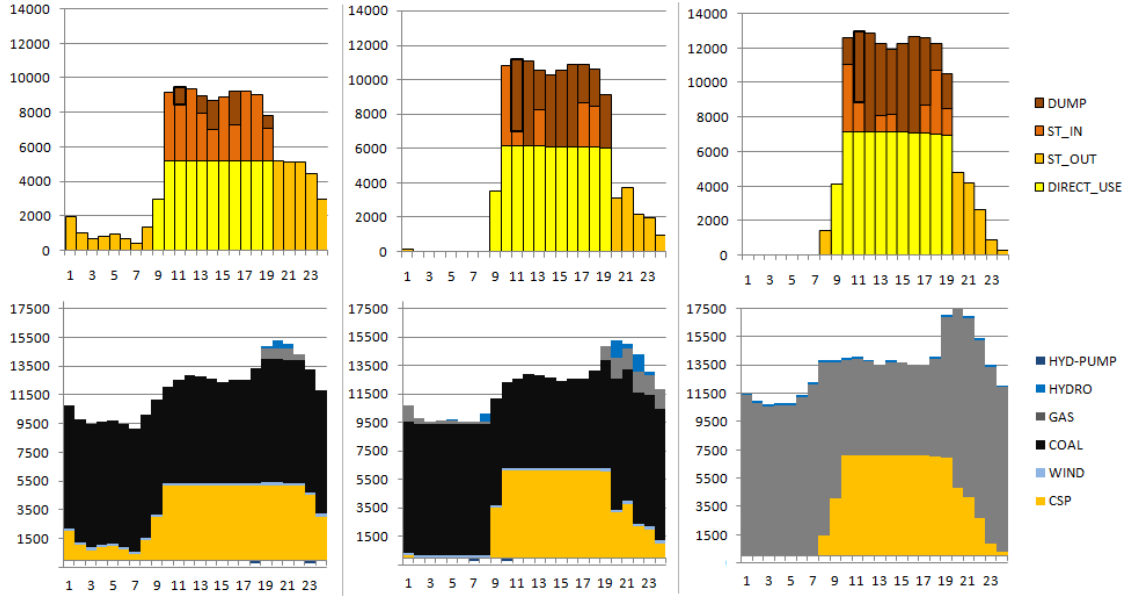


Figure III-3. Left: Dispatch (in MW) for 20% CSP penetration in Morocco, SM2, 6h storage. Center: 20% penetration, SM2, 2h storage (see A in Figure III-2a). Right: 20% CSP penetration in Algeria, SM2, 2h storage (see B in Figure III-2d).

Dumping becomes detrimental when the thermal storage capacity is significantly undersized in relation to the collector field. Figure III-3 (center) shows the dispatch of an SM2 configuration with only 2 hours of storage capacity. The integration of such inappropriately sized CSP plants leads to an economically suboptimal solution for the entire power system (see point A in Figure III-2b). The first and obvious reason for this outcome is that significant amounts of solar energy are wasted because the storage system is undersized. The model is constrained to compensate these losses by installing more CSP plant capacity to achieve the 20% solar share in the overall electricity mix; the related investments ultimately increase the total costs. The second consequence of an undersized storage system is that the plant's ability to shift solar generation to other periods of the day is reduced: the power system becomes less flexible, and in periods of high demand, hydro and gas power has to be deployed. Another drawback of the limited ability to shift generation is the strong CSP feed-in dur-

ing the daylight hours from 9h to 19h. This is particularly disadvantageous in the Moroccan case (Figure III-3, center, bottom), as coal power plants have to be ramped down or operated in an efficiency-reduced partial load mode. Algeria, alternatively, has fewer problems with large amounts of solar power pushing into generation (Figure III-3, right) because flexible gas power plants can more easily (and, comparatively, more cost-efficiently) follow the variations of solar electricity production. As a result, the cost disadvantages are smaller than in the Moroccan case (see point B in Figure III-2d).

6.2. Monetary value of CSP dispatchability

The previous section has demonstrated how outfitting CSP plants with storage systems (and, thus, making them dispatchable) has a perceptible impact on the total costs of the entire power system. It has also been demonstrated that the degree of this impact depends on the intended CSP penetration as well as on the characteristics of the electricity system where the plants are to be integrated. In this section, we want to provide a more quantitative picture of the monetary advantages of integrating storage-equipped CSP plants into the electricity systems of Morocco and Algeria. A possible metric for the “value of dispatchability” can be the cost difference between a power system with storage-optimized CSP plants and a system where non-dispatchable CSP plants are integrated. The least-cost configuration of these zero-storage CSP plants is given at a solar multiple of SM1.5 or SM1.25 (depending on country and penetration, see min^* in Figure III-2). Therefore, $TOTAL_{min^*}$ are the total system costs with SM1.5/0h or SM1.25 plants, while $TOTAL_{min}$ refers to the total costs with the optimized SM/storage combinations found in the previous chapter (see min in Figure III-2). Following the general approach of calculating levelized generation costs, we put the cost difference into relation with the amount of CSP electricity produced,

$$\left(TOTAL_{min^*} - TOTAL_{min} \right) / \left(\sum_y disc_y \cdot \sum_{csp,d,h} GEN_{csp,y,d,h} \right)$$

and obtained a per-unit value (in €²⁰¹⁰/MWh), which can be regarded as a proxy for the value of dispatchability. Table III-5 lists the results for Algeria and Morocco. To provide a more complete picture, we analyzed not only the penetration cases of 5% and 20%, but also those of 10% and 30%.

Table III-5. Value of dispatchability. Achievable cost savings per MWh (right) by integrating CSP plants with optimized configurations (left) compared to non-dispatchable configurations under different solar penetration quota.

CSP penetration by 2025	Optimum configuration		Value of dispatchability (€ ²⁰¹⁰ /MWh)	
	Morocco	Algeria	Morocco	Algeria
5 %	SM 1.75 / 4h	SM 1.75 / 3h	39	29
10 %	SM 2 / 6h	SM 1.75 / 6h	38	30
20 %	SM 2 / 6h	SM 2.25 / 8h	44	28
30%	SM 2.5 / 9h	SM 3 / 13h	55	35

An initial finding is that in Morocco the value of dispatchability is, in all penetration cases, significantly higher than in Algeria. This can be seen as an indicator that CSP storage does indeed have a higher economic value in the Moroccan power system than in the Algerian one. We also observe that in Algeria the value of dispatchability remains relatively insensitive to CSP penetration, while, in contrast, the value increases in Morocco, especially at the very high CSP penetration of 30%. This confirms our previous conjecture that flexible, storage-based CSP plants provide a comparatively better match for Morocco, with its inflexible coal-based power plant environment, than for Algeria, whose large number of gas power stations already offer a flexible power system environment.

7. CSP integration in a renewable power mix with wind and PV power plants

While the analysis of the previous section shed a light on the origins and dimensions of the economic advantages of CSP in conventional power systems, it did

not discuss the interplay of CSP with other renewable power plant technologies. In Morocco and Algeria, wind and PV are likewise promising renewable diversification options; but unlike CSP, both can only deliver intermittent power, have no storage capacity, and are largely incapable of backing the power system with firm capacity. Nevertheless, both wind and PV power challenge parabolic trough plants with their potentially lower levelized costs of electricity generation (Kost and Schlegel, 2010). Therefore, from the viewpoint of a decision-maker, it would be important to know whether (and if yes, under which conditions) the economic advantages of dispatchability can outweigh the disadvantages of CSP's higher LCOE. To answer these questions, we enabled the model to simulate competition among the three renewable technologies: wind, PV, and CSP. To provide a realistic picture, we decided to orient the analysis toward the "official" renewable energy goals made public by Morocco's and Algeria's governments and planning authorities.

It must be premised that simulations with dispatch and investment models over a long time horizon are generally subject to high sensitivities with regards to the input parameter assumptions. Alterations of cost parameters, for example, the CSP, PV, and wind investment costs, can entail significant changes in the results. Furthermore, it is possible that technology innovations in the CSP sector will lead to a future market preference for other, perhaps less costly, CSP technologies (e.g. central receiver systems) instead of the parabolic trough power plants that this study focused on. Because the scope and the space for this article did not permit a sensitivity analysis of such different CSP technology pathways or cost development scenarios, the following findings reflect only the general trends related to one specific cost and technology scenario.

7.1. Optimized renewable power mix in Algeria

Algeria has abundant potential for solar power plants, but offers only few promising sites for wind farms. The orientation of the Algerian Ministry of Energy and Mines reflects this reality. In its renewable development program, *Programme des Energies Renouvelables et de l'Efficacité Energétique* (MEM, 2011), published in 2011, wind is given only a 3% share of total electricity consumption by 2030, while the goal for photovoltaic and CSP together is set at

37%. We parameterized the model such that wind generation in the overall mix is exogenously constrained to a quota of 3%, while the renewable scenario constraint (res_share_y , see Section 4) forces PV and CSP generation together to reach 37% by 2030. The outcome of the simulation thus provides an optimized breakdown between these two solar technologies. This should, from a planner's perspective, allow conclusions about which combination of Algeria's two renewable "pillar" technologies (PV and CSP) is the most cost-efficient solution in terms of system integration.

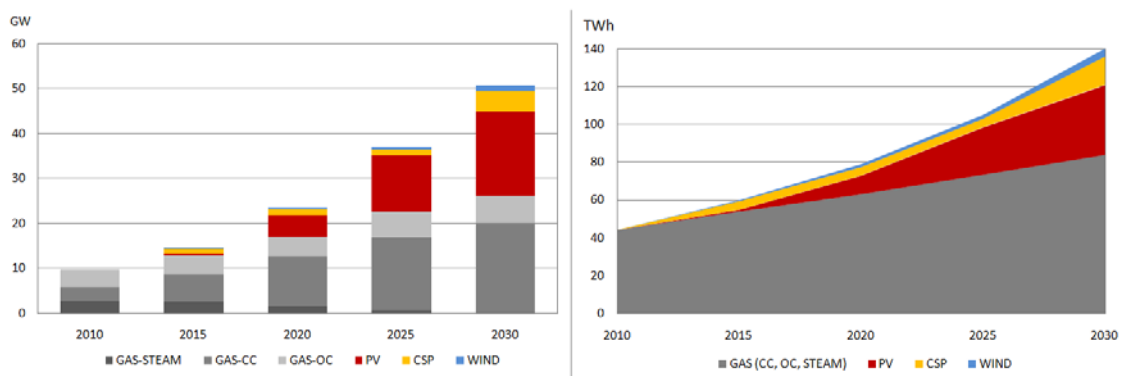


Figure III-4. Capacities (left) and generation (right) of the Algerian electricity system until 2030.

Figure III-4 depicts the results for the Algerian electricity mix until 2030. An initial observation is that both CSP and PV power plants take part in solar generation, although PV to a much greater extent. The dominance of PV can be explained by its lower electricity generation costs, which significantly undercut the costs of CSP. This triggers PV capacity additions, while CSP stagnates, at least until 2025. Nevertheless, in 2030 this picture begins to change with a noticeable recovery of CSP. We explain this finding with the previously outlined concept of the value of dispatchability: from a certain PV penetration level on, further photovoltaic feed-in becomes disadvantageous because PV electricity production peaks at the height of the day and cannot be shifted to later hours. More PV capacity leads to a suboptimal system configuration during daily operation, because it either forces PV production curtailment or requires the conventional power plants to reduce their output. Only by introducing renewable technologies with a high value of dispatchability – in other words, CSP plants – can the relat-

ed cost disadvantages for the overall system be avoided. In our example, simulations lead to an optimal CSP configuration of SM2/6h storage, with an installed capacity of 4.6 GW in 2030, contributing to 11% of the 2030 generation mix. The installed PV capacity in 2030 reaches 18.7 GW, with a production share of 26%.

7.2. Optimized renewable power mix in Morocco

Unlike Algeria, Morocco doesn't express its renewable goals in electricity production shares, but instead in installed capacity. For wind power, a capacity target of 2.0 GW is foreseen by 2020 (ONE, 2008), which translates into an annual production of around 7.3 TWh, or approximately 13% of the projected national electricity demand. As the proportions of CSP and PV plants in Morocco are not known, the 2.0 GW solar capacity goal of the Moroccan Solar Plan (Masen, 2010) does not allow any conclusions to be drawn about future solar electricity production. We therefore decided to base our analysis on an estimated production goal of 5.6 TWh by 2020, representing a 10% share in the national mix. Assuming that this 10% solar share will be maintained until 2030, we examined, as in the previously described Algerian case, which combination of PV and CSP capacities would prove the most cost-efficient for the Moroccan generation system.

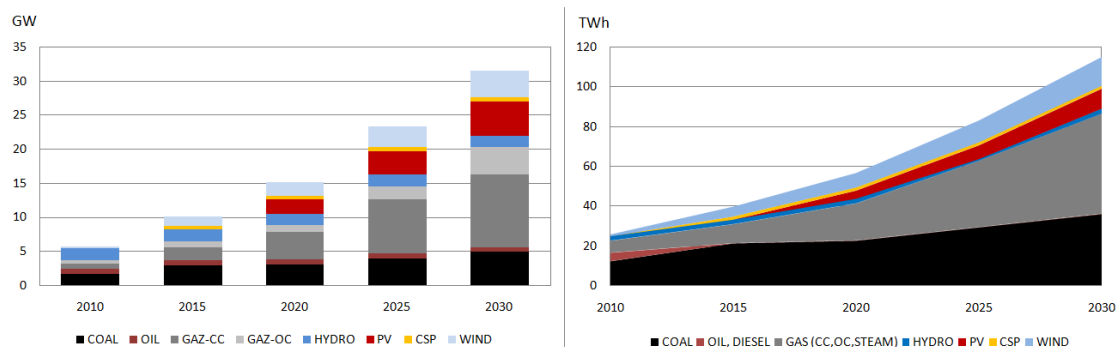


Figure III-5. Capacities (left) and generation (right) of the Moroccan electricity system until 2030.

The optimization results in Figure III-5 show that for Morocco, a PV-biased solar mix is also preferred until 2030; CSP power plants would only cover a small part of the generation. We carried out several model runs with different combinations of solar multiple and storage capacity and found that the least-cost configuration for CSP is given at SM1.5/3h storage. Its entire production in 2030, however, would only cover 1.4% of the total demand (with a capacity of 570 MW), while the PV share reaches 8.6% (with around 5 GW capacity). The low requirement for CSP indicates that its value of dispatchability remains low when only a small renewable quota (10% solar and 13% wind) is targeted. Even the 4 GW intermittent wind capacity can apparently still be handled by the conventional electricity system without the need for additional CSP plants. From a purely economic viewpoint, it can be stated that Morocco, with its current solar development plan, should – at least in the short run – rather emphasize projects with “low-hanging renewable fruits” like wind and PV instead of investing into the currently more costly CSP technologies. On the other hand, CSP could very well constitute an appropriate technology if Morocco revised its renewable goals and opted, for example, for more wind power in the future. For instance, the current 2 GW wind target appears relatively small compared to the country’s overall wind potential of 25 GW (Benkhadra, 2011). Under high wind penetration, dispatchable CSP plants with high storage capacity will very likely prove a better-fitting renewable complement to the intermittent wind feed-in.

8. Conclusion

The ability to store and dispatch electricity distinguishes CSP power from other, intermittent renewable generation technologies and promises general cost advantages when it comes to integration into electricity systems. In this study, we used a linear power market optimization model to examine these advantages – the “value of dispatchability” – and quantified the consequences for CSP in different power system environments. Morocco and Algeria, both countries with high ambitions for renewable energy, were chosen as case studies. It was shown that a value of CSP dispatchability exists in both electricity systems, and that this value is influenced by three main factors: (1) the properties of the surrounding conventional power system (e.g. coal- or gas-based power systems), (2) the

technical configuration of the CSP plants themselves, and (3) the targeted renewable energy penetration. We observed, however, that the value of dispatchability does not necessarily guarantee the competitiveness of CSP in comparison with other renewable generation technologies. When the targeted amount of renewable generation in an electricity system is low, the value of dispatchability is often not sufficient to justify the investments into CSP capacities. In Morocco, which in our examined scenario targets only a small solar quota, photovoltaic power appears to be the more cost-efficient means of solar electricity generation. Algeria, on the other hand, has subscribed to a solar electricity goal of almost 40%, which can only be reached by a major participation of CSP plants. Nevertheless, drawing final conclusions about the general competitiveness of CSP is still premature, since renewable cost developments and learning rates, especially for CSP, are currently not very well explored. Further uncertainties might likewise emerge from the hardly predictable PV market, which is currently seeing dramatic price drops on a worldwide level. In the not unlikely event that PV power plant costs decrease more quickly than in our assumptions, CSP deployment in Morocco and Algeria would be pushed even further into the future. Another aspect is the future development of alternative storage devices, such as electrochemical storage systems; innovations and lower costs in this area could further challenge the competitiveness of CSP plants in electricity systems. On the other hand, it should also be mentioned that our model did not consider the aspect of grid infrastructure expansion, where CSP might provide certain cost advantages compared to other intermittent renewable technologies. Moreover, there is a political dimension: CSP is considered to have a high potential for creating economic value and jobs in North Africa (Fraunhofer and Ernst & Young, 2011), and certain governments might become inclined to prioritize this technology in their national renewable expansion plans. In conclusion, it can be stated that any long-term planning to incorporate CSP plants into power systems must consider not only the technical and economic aspects of system integration, but also the potential future welfare effects on local economies.

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IV. Multi-criteria analysis of electricity generation mix scenarios in Tunisia¹⁸

Abstract

The diversification of the national electricity generation mix has risen to the top of Tunisia's energy planning agenda. Presently, natural gas provides 96% of the primary energy for electric power generation, but declining domestic gas reserves and a soaring electricity demand are urgently calling for alternative fuel strategies. Currently discussed diversification options include the introduction of coal and nuclear power plants and/or an increased use of renewable energies. This article presents a methodology to assess different electricity system transformation strategies. By combining an electricity market model with a subsequent multi-criteria decision analysis (MCDA), we evaluate five power mix scenarios regarding power generation costs as well as non-economic dimensions such as energy security, environmental impact and social welfare effects. Based on criteria valuations obtained during consultations with Tunisian stakeholders, a final, best-ranking electricity scenario was selected, consisting of 15% wind, 15% solar and 70% natural gas-generated electricity in the national power mix by 2030.

1. Introduction

The Tunisian electricity sector is faced with a multitude of challenges, pressing for decisions about a new national power supply strategy. As for many other North African countries, one particular concern is the ever-growing electricity demand, which, despite successful efforts to reduce energy intensity in the

¹⁸ This chapter is based on a peer-reviewed article published in the journal *Renewable and Sustainable Energy Reviews*: Brand, B., Missaoui, R., Multi-criteria analysis of electricity generation mix scenarios in Tunisia. *Renewable and Sustainable Energy Reviews* 39 (2014) 251-261, doi: 10.1016/j.rser.2014.07.069.

past¹⁹, is still on a constant surge in Tunisia. Between 1990 and 2010, domestic power consumption almost tripled, from 4.9 TWh to 14 TWh; for the next two decades, the demand is expected to further grow, reaching 25-33 TWh by 2030 (STEG, 2010). The challenge to cope with this soaring demand is joined by another issue: Tunisia's unbalanced primary energy supply of power generation. Currently, the national electricity system is almost entirely fueled by natural gas, making up around 96% of the generation mix. This dependency will become particularly problematic in light of an imminent natural gas deficit that is forecasted for 2018 or 2020 (DGE, 2013), putting Tunisia's electricity supply security at risk. So far, the following options to render the Tunisian electricity mix more diverse have been proposed:

(1) *Coal Power*. The use of imported hard-coal for power generation is already extensively practiced in another North African country, Morocco, and decision makers frequently discuss whether Tunisia should likewise adopt this model. A study by the national electricity and gas utility STEG (2010) evaluated the option to install up to three coal-fired steam power plants in the country, each with a capacity of 600 MW.

(2) *Nuclear Power*. This option is also discussed in Tunisia, since in 2006, the former government signed a bilateral agreement with France on the civilian use of nuclear power (Haddad, 2011). Under exploration is the installation of one pressurized water reactor with a tentative capacity of 900-1000 MW, slated to become operational by 2027 (STEG, 2010).

(3) *Renewable Energies*. Tunisia is endowed with excellent renewable resources. The country's wind potential is valued at 8 GW, with three main regions being particularly suitable for large wind farm projects: the North-East, the Central-West and the South-West of Tunisia. Solar radiation conditions are the best in Southern Tunisia, but generally very favorable across the entire territory. Gross estimations for Tunisia's solar potential reach 844 GW for PV and 65 GW for CSP technology (ANME, 2010) although these values must be assessed with care, as they usually do not include considera-

¹⁹ Tunisia was one of the first North African countries to subscribe to a national energy efficiency strategy, which it has pursued since the 1980s.

tions about the actual availability of the land for solar power projects. The “Tunisian Solar Plan,” a renewable energy roadmap drafted by the Tunisian energy conservation agency ANME, acknowledges the high renewable potential and points particularly to wind, photovoltaic and concentrated solar power (CSP) as diversification options to replace natural gas in the electricity mix (ANME, 2013a).

The extent to which these three diversification options - or a combination of them - can become part of a Tunisian electricity strategy is a matter of ongoing deliberations. The political context in which these discussions are taking place is marked by the Tunisian revolution of January 2011, which brought a new way of political decision making to the country along with considerable consequences for the culture of energy system planning. In the past, Tunisia’s electricity strategies were discussed in closed circles accessed only by a small group of public decision makers, many of them with strong links to the national gas and electricity sector. The decision process itself was biased by the view of the incumbent national gas and electricity utility STEG, whose maxim for power system optimization was a least-cost planning approach, mostly disregarding social and environmental aspects. Nowadays, after the upheavals in early 2011, the interests of the civil society and other stakeholder groups can no longer be ignored. One aspect frequently raised in the discussions about power system choices are socio-economic benefits, most prominently the aspect of domestic added value, local manufacturing opportunities for the Tunisian industry, and its potential effects on job and income creation. Tunisia is suffering from a lack of jobs; the country’s youth is plagued by a high unemployment rate, which, according to a study of the World Economic Forum (2012), is currently around 30%. Aspirations for job creation and concerns of the local population (social acceptance) need to be given much more attention in the new democratic context. Also environmental aspects and the question of ecological sustainability of the national energy supply are increasingly discussed topics in the Tunisian society. Moreover, as a signatory of the Kyoto protocol, Tunisia has taken international responsibility to combat climate change by reducing its carbon emissions. A survey carried out in 2012 at a multi-stakeholder workshop within the framework of a project commissioned by the Tunisian energy conservation agency ANME and the German development agency GIZ (Wuppertal Institut/Alcor,

2012) revealed four major groups of criteria that are pertinent to electricity strategy development in Tunisia:

- Economic costs;
- Security of supply;
- Ecological sustainability; and
- Socio-economic benefits.

It is obvious that the above-listed goals are partially conflicting. Therefore, any decision on future electricity system transformation pathways can only be a compromise, which, at best, maximizes the satisfaction of the majority of the stakeholders, taking into account their varied preferences and objectives.

An analysis framework that includes these objectives must consider other criteria than just the economic costs of electricity generation. One possible approach would be to use external costs as guidance for decision-making on electricity strategy options (Mirasgedis and Diakoulaki, 1997). The valuation of externalities, however, is a difficult undertaking as it requires attributing quantified ‘market’ prices to the different societal and environmental impacts of electricity production. Although there have been attempts to quantify such externalities, for instance for public health and environmental impacts in Europe (Krewitt, 2002), estimations are generally associated with high uncertainties. Moreover, it is not clear whether the externality values obtained from a European project can properly reflect the North African conditions: for the Tunisian context, for instance, no valid quantifications of the external costs (or benefits) of social acceptance, job creation and energy security are available at the moment.

An alternative - and for our purpose a more suitable approach - is multi-criteria decision analysis (MCDA). Dealing likewise with strategy optimization problems, MCDA has been widely applied to social, economic, agricultural, industrial, ecological and biological systems, and likewise to energy supply systems (Wang et al. 2009). In the present study, we use MCDA in conjunction with an electricity generation system model to calculate and evaluate different Tunisian power system scenarios. This combined approach is presented in Section 2,

where the basic features of the model as well as the MCDA method are described. Section 3 proceeds with an outline of the simulated scenarios and gives details about the assumptions and parameters used in the study. Section 4 presents the results, followed by a concluding discussion in the last chapter (Section 5).

2. Methodology

As pointed out in a literature review by Pohekar and Ramachandran (2004), the inclusion of multi-criteria techniques in energy system planning has gained increasing popularity in academic research since the 1990s. As for the area of electric energy supply (power generation), one can distinguish between two types of studies:

First, there are approaches that analyze and compare individual generation technologies based on their inherent economical, social and ecological properties. Recent examples of such studies can be found in Streimikiene et al. (2012), Pappas et al. (2012) and Boran et al. (2013), where renewable, fossil and nuclear power plant technologies are compared (or ranked) within a multi-criteria framework.

A second type of studies rather evaluates combined power systems instead of single technologies. Here, the multi-criteria analysis focuses on the performance of the overall power supply system, usually by examining different electricity mixes for a specific region or country. Finland, for instance, was the subject of a scenario study of Häyhä et al. (2011), where three alternative power system transformation pathways until 2050 with different levels of renewable, fossil and nuclear penetration underwent a multi-criteria analysis. A similar analysis has been carried out by Diakoulaki and Karangelis (2007) for the Greek power system regarding economic, environmental, and energy security aspects. The development of the Spanish electricity system until 2030 was the focus of a study by Del Riego et al. (2012), while Ribeiro et al. (2012) analyzed electricity supply scenarios for Portugal until 2020 by combining a multi-criteria decision analysis with a linear power system optimization model.

In the present study, we opt for a two-step methodology, i.e. linking an electricity system model with a multi-criteria analysis, to analyze different Tunisian electricity mix scenarios with a time horizon of 2030. Figure IV-1 gives an overview of the methodological sequences of the analysis.

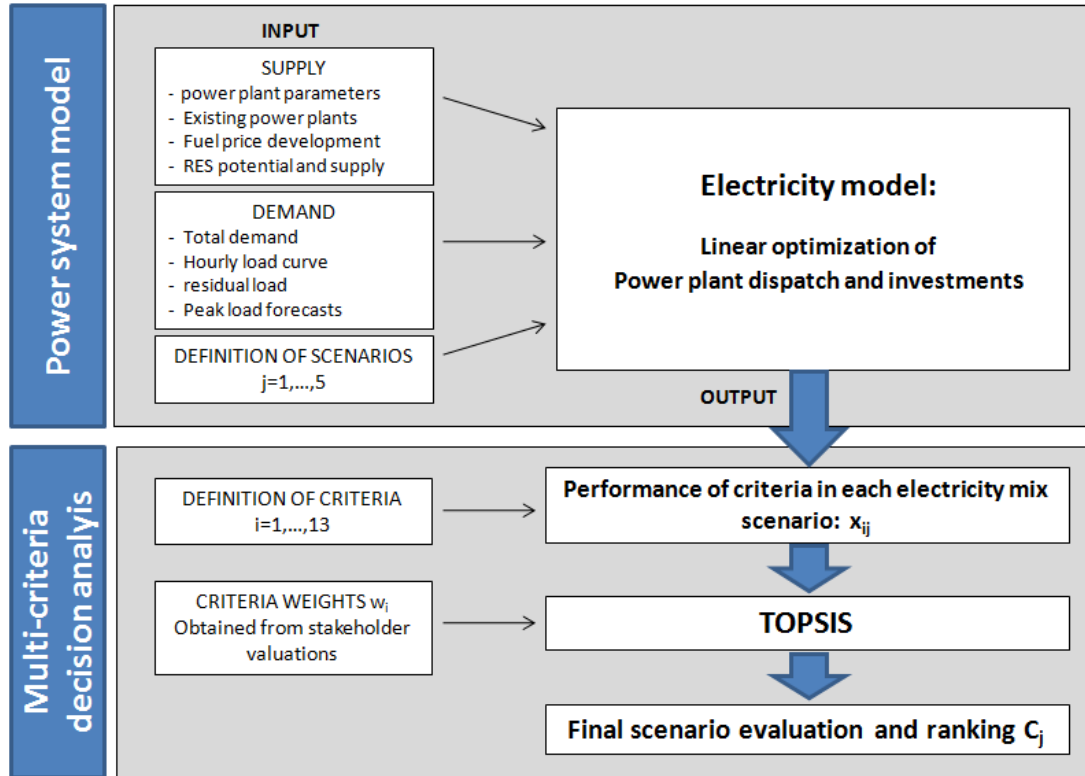


Figure IV-1. Process structure of the analysis

2.1. Electricity Model

There are several reasons why an electricity system modeling step has been placed prior to the multi-criteria analysis. First, the model was needed to calculate and quantify the most relevant values of the criteria: total system costs, primary energy consumption, emissions, etc. (for an overview, see Table IV-4). Second, the use of the model ensures that only technically viable and economically optimized power system configurations will be evaluated in the MCDA analysis. This is why both the daily power plant dispatch as well as the system transformation in the long-run - regarding commissioning and decommissioning of power plants - have been calculated under consideration of technical and economic constraints. Finally, the purpose of using an electricity system model was

to provide the interested stakeholders with other relevant information concerning the impact of the electricity mix scenarios on the Tunisian economy, e.g. in terms of investments needed, infrastructure requirements, primary energy inputs and costs.

We use a bottom-up linear optimization model, which has already been applied and described in more detail in a previous generation system study of North African power systems (Brand et al., 2012). By minimizing the total discounted system costs,²⁰

$$TOTAL = \sum_{y=2010,2015,...} disc_y \cdot (INV_y + OM_y + VAR_y + FUEL_y)$$

the model calculates the optimal dispatch as well as the least-cost pathway for investments and the retirement of power plants in the Tunisian electricity system. Starting with the existing power plant stock of the year 2010, the simulation forecasts the configuration of the Tunisian power system until 2030 in 5-year periods. Each forecast period consists of 32 characteristic days, representing different electricity load profiles (working days and weekend days) as well as hourly wind-speed and solar irradiation patterns for 4 different seasons. The optimization process is carried out under the condition that the electricity supply meets the demand at any time:

$$load_{y,d,h} = \sum_{tech} GEN_{tech,y,d,h}$$

²⁰

The total system costs (the objective function) are the sum of investment costs (INV), fuel costs (FUEL) and variable costs (VAR), as well as costs for operation and maintenance (OM). The model does neither consider carbon emissions costs, nor revenues from avoided carbon emissions. The discount factor (disc) was calculated on the basis of an interest rate of 8%.

The model variable $GEN_{tech,y,d,h}$ represents the actual electricity generation (in MWh) for the specific generation technology $tech$ at any hour ($h = 1, \dots, 24$) and typical day ($d = 1, \dots, 32$) of the period y , while $load_{y,d,h}$ is the exogenously determined electricity load pattern of Tunisia, derived from historic load data of the Tunisian electricity. Further boundary conditions are the technical constraints of the involved generation technologies. Thermal power plants need to respect constraints concerning part-load conditions and other dispatch properties like ramp-up and shut-down behavior, while renewable power generation is limited by the availability of the wind or solar resource. For wind speed and solar irradiation, we used hourly, historic data of the year 2010 matching to the date and hour of the electricity load of the corresponding representative day. Cost of transmission system expansion, as well as transmission losses were neglected in this simplified modeling approach.

The model delivers a number of *physical parameters* related directly to the future power system configuration as well as a set of *non-physical, qualitative parameters* relevant for the multi-criteria analysis:

- **Physical parameters:** For each forecast period, the model provides information about the installed generation capacities and power generation (per technology and per fuel) and primary energy consumption (overall and per energy carrier). Furthermore, the model calculates the quantities of gaseous emissions like carbon dioxide, but also nuclear waste.
- **Non-physical and qualitative parameters:** The most prominent non-physical parameters, directly available from the model output, are the economic costs of the power system scenarios. In addition, according to the proportions of each generation technology in the electricity mix, the model delivers a number of other parameters of qualitative or semi-qualitative nature, like the job creation potential, indicators for supply security, and the social acceptance of the different electricity mix scenarios.

Together with Tunisian stakeholders, a set of the 13 most relevant parameters, summarized in Table IV-4, were chosen as key evaluation criteria of the study's power system analysis.

2.2. Multi-criteria decision analysis

The problem of decision-making on energy strategies can be principally addressed by a variety of multi-criteria methods. An overview by Pohekar and Ramachandran (2004) presents a method inventory summarizing the most popular MCDA energy planning techniques: the analytical hierarchy process (AHP), the preference ranking organization method for enrichment evaluation (PROMETHEE), the elimination and choice translating reality (ELECTRE), the technique for order preference by similarity to ideal solution (TOPSIS), the weighted sum method (WSM), the weighted product method (WPM), compromise programming (CP), and the multi-attribute utility theory (MAUT).

In the present study, we decided to use the TOPSIS method developed by Hwang and Yoon (1981). TOPSIS was chosen primarily due to practical considerations, because it has “a sound logic, representing the rationale of human choice; a scalar value that accounts for the best and worst alternatives, and a simple computation process” (Kim et al., 1997) making it a suitable tool that can easily be understood and reproduced with reasonable effort also by users outside of the scientific sphere.

The scope of the TOPSIS method is to determine the “best scenario” among a set of scenarios, that is, the one closest as possible to an hypothetical ideal alternative and at the greatest distance to a negative-ideal alternative. This is realized by a geometrical mapping of each scenario's performance calculating its relative (Euclidean) distance to the ideal and negative-ideal solution. In detail, the procedure of the TOPSIS method consists of the following steps:

- (1) Formulate a decision matrix which is formed by $j=1, \dots, M$ columns for the scenarios and $i=1, \dots, N$ lines for the criteria. The matrix entries x_{ij} are the raw results of the simulations with the Tunisian electricity system model (see Section 2.1.) and represent the performance of the criterion i with regard to scenario j (see Table IV-6).

(2) Calculate a normalized decision matrix. The TOPSIS method uses vector normalization to eliminate the units of the criteria, resulting in comparable matrix entries r_{ij} normalized to $|r_{ij}| \leq 1$:

$$r_{ij} = x_{ij} / \sqrt{\sum_{j=1}^M x_{ij}^2}$$

(3) Construct the weighted normalized decision matrix.

$$v_{ij} = w_i \cdot r_{ij}$$

The weighting parameters w_i for each criterion are derived from subjective preferences expressed by Tunisia experts within a voting process (see Section 3.3).

(4) Calculate the ideal and the negative-ideal solution. The ideal solution A^* is a set of elements v_i^* selected from those elements of the normalized decision matrix v_{ij} , that provide the maximum benefit with respect to criteria i .

$$A^* = \{v_1^*, \dots, v_N^*\}$$

Note that v_i^* must be selected from v_{ij} by respecting the benefit attribute of each criterion. For example, regarding the criterion “job creation,” the highest value $v_i^* = \max_j v_{ij}$ must be chosen (benefit attribute is positive: “more jobs are better”); for CO₂ emissions (negative benefit attribute: “less CO₂

emissions are better”), the lowest possible value $v_i^* = \min_j v_{ij}$ must be selected. Reciprocally, the negative ideal solution

$$A^- = \{v_1^-, \dots, v_N^-\}$$

is a selected set of v_i^- associated with the lowest benefit among all v_{ij} .

(5) For each scenario $j=1, \dots, M$, calculate the Euclidean distance to the ideal alternative A^* ,

$$D_j^* = \sqrt{\sum_{i=1}^N (v_{ij} - v_i^*)^2}$$

as well as the negative ideal alternative A^-

$$D_j^- = \sqrt{\sum_{i=1}^N (v_{ij} - v_i^-)^2}$$

(6) Compare the separation values D_j^* and D_j^- by calculating the relative closeness to the ideal solution:

$$C_j = \frac{D_j^-}{D_j^* + D_j^-}$$

(7) Finally, establish a ranking list, with the highest C_j indicating the best-ranking scenario j .

The input parameters needed for the TOPSIS method are the performance x_{ij} of the different criteria i , and their respective weights w_i in each scenario j . While x_{ij} are obtained from the results of the power system model (see Section 4.2), the weights w_i of the criteria are derived from Tunisian stakeholders valuations (Section 3.3).

3. Scenarios, input parameters, and criteria

In a project preceding this study²¹, five different scenarios for the Tunisian electricity mix until 2030 were defined by the key stakeholders of the Tunisian power sector (Ministry of Energy, Ministry of Industry, STEG, ANME, etc.). These scenarios reflect the discussed policy options for power system transformation along the main diversification axes: nuclear energy, coal power and/or the increased use of renewable energies for electricity generation. A business as usual (BAU) case represents the scenario where no action is taken, and the current electricity mix (i.e. natural gas-generated power with a minor contribution of wind power) is maintained until 2030 (see Table IV-1).

²¹ The previous study was commissioned by the German development agency Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) in 2011.

Table IV-1. Scenarios.

Scenario	<i>j</i>	Description
BAU (Business as usual)	1	Reference scenario, current power mix carried forward to 2030: 95% gas, 5% wind
DivCoal	2	Diversification with an important contribution of coal power plants to be build starting from 2020. Targeted electricity mix by 2030: 60% coal, 35% gas, 5% wind.
DivNuc	3	Nuclear power for diversification, ~1000 MW reactor to be built in 2025. Targeted electricity mix by 2030: gas 75%, nuclear 25%, wind 5%.
DivCoalRes	4	Combined diversification with coal and renewable power. Targeted electricity mix by 2030: 50% coal, 10% wind, 5% solar (3% PV, 2% CSP), 35% gas.
DivRes	5	Diversification with 30% renewable power. Targeted electricity mix by 2030: gas: 70%, wind 15%, solar 15% (10% PV, 5% CSP).

3.1. Input parameters

3.1.1. Demand

It is obvious that a more efficient use of electricity is important for Tunisia to alleviate energy dependency, save fuel costs, and mitigate the need for new generation capacity and grid expansion. Tunisia traditionally puts high efforts into promoting energy efficiency. Despite certain concerns about the ‘rebound effect’ of energy efficiency measures (Wang et al., 2014), the Tunisian government pursues an active policy to reduce electricity consumption in households as well as in the industry (GIZ/ANME, 2012). In our study, we use a demand forecast by the Tunisian Energy conservation agency ANME (2013b). This demand scenario assumes an energy efficiency pathway with a decoupling of the Tunisian electricity demand from economic growth, translating into an average decrease of the electricity intensity of 2% per year between 2013 and 2030. The resulting demand and load input parameters for the electricity model are displayed in Table IV-2.

Table IV-2. Assumptions for annual gross demand growth and peak load development of the Tunisian electricity system.

	2010	2020	2030
Annual demand (TWh)	14.8	20.7	28.9
Peak load (MW)	3010	4900	6840

Regarding the daily variations of the demand, hourly historic load profiles of the year 2010, provided by the Tunisian electricity utility STEG, were used. These profiles were linearly scaled up according to the expected increases of the annual demand in the future model periods until 2030.

3.1.2. Generation

As mentioned, the model assumes that power supply is exclusively realized by domestic power plants; electricity exchanges with neighboring countries (power imports or exports) are neglected. For simplicity we also consider that the model can only deploy a limited set of different generation technologies: gas power plants (open cycle and combined cycle power plants²²), coal steam power plants, nuclear power plants, photovoltaic (PV), concentrating solar power (CSP) and wind turbines. The technical and economic modeling parameters of these power plants are displayed in Table IV-3. Data originated from a previous report about Tunisian power system strategies (Wuppertal Institut/Alcor, 2012) and were agreed upon in a workshop with Tunisian power system experts. The same applied for the fuel price assumptions (for natural gas, steam coal and uranium), which were derived from an analysis of market price forecasts from different organizations, such as the International Energy Agency (IEA), the US Department of Energy (US-DOE), and the World Bank. An average reference scenario representing moderate price increases until 2030 (see Table IV-3) was chosen and agreed upon by the Tunisian stakeholders.

²² Additionally, the model takes account of the existing stock of older, gas-powered steam plants in the Tunisian power system.

Additional supply input parameters are solar irradiation and wind speed. Solar output patterns were calculated with the System Advisor Model of the US National Renewable Energy Laboratory (NREL) based on global horizontal irradiation (GHI) for PV or direct normal irradiance (DNI) for CSP power plants (NREL, 2012). Irradiation time series were provided by Helioclim (2012) for an exemplary site near the city of Tataouine, Southern Tunisia in the year 2010. It is assumed that PV power plants are composed of crystalline cell modules and installed ground-mounted, while CSP power plants are standard dry-cooled parabolic trough technology, with 6 hours thermal storage and a solar multiple of 2. The methodology of CSP dispatch modeling is described in Brand et al. (2012). Hourly wind power production patterns were calculated with the MINT software (Sander+Partner, 2012), based on MERRA wind speed time series of NASA (2012) for typical commercial wind turbines (Vestas V80, 2 MW) at three typical wind power sites in Tunisia: Bizerte, Haouaria and Kebili.

Table IV-3. Power plant parameters assumptions

	Generation technology						
	Coal	Nuclear	Gas OC	Gas CC	Wind	PV	CSP
Invest. (€/kW) 2010 → 2030	2130	4170	675	900	1150 → 930	2500 → 1000	6090 → 3400
Fix. O&M (€/kW _a)	22.8	53.2	8.2	9.7	40.0	20.0	53.8
Var. O&M (€/MWh _d)	3.1	0.4	2.5	1.6	0.0	0.0	0.0
Fuel costs (€/GJ) 2010 → 2030	2.79 → 2.97	0.47 → 0.64	6.38 → 8.69	6.38 → 8.69	-	-	-
Technical availability (%)	85	85	85	97	99	96	97
Efficiency (%)	40	33	35	53	-	-	-
Capacity credit (%)	100	100	100	100	3	13	90
Minimum load (%)	50	65	20	40	-	-	20
Specific CO ₂ emissions (kg/MWh)	815	-	575	400	-	-	-
Specific SO ₂ emissions (g/MWh)	150	-	-	-	-	-	-
Specific NO _x emission (g/MWh)	250	-	50	50	-	-	-
Specific Nuclear waste (g/MWh)	-	2.7	-	-	-	-	-
Specific fine dust emissions (g/MWh)	15	-	-	-	-	-	-
Social acceptance [-2,+2]	-2	-2	1	1	0	2	2
Local production share (%) 2010 → 2030	25	5	15	25	43	41 → 71	36 → 57
Contribution to energy independency [-2,+2]	-1	-2	-2	-2	2	2	2
Apptitude to respond to peak load events [-2,+2]	0	0	2	2	-1	-1	1

3.2. Criteria

Our analysis covers 13 different criteria pertaining to four main groups: economic costs, supply security, socio-economic criteria and ecological criteria (see Table IV-4).

Table IV-4. Criteria and criteria groups

Criteria group	<i>i</i>	Criteria	Unit	Benefit attribute*
A. Economic costs	1	Net present value of total costs	billion €	NEG
	2	Specific generation costs	€/MWh	NEG
B. Supply security	3	Aptitude to respond to peak load events	point [-2...2]	POS
	4	Total natural gas consumption	PJ	NEG
	5	Contribution to energy independency	point [-2...2]	POS
C. Socio-economic criteria	6	Local manufacturing share	%	POS
	7	Social acceptance	point [-2...2]	POS
	8	Average jobs created	number	POS
D. Ecological criteria	9	CO ₂ emissions	Mt CO ₂	NEG
	10	SO ₂ emissions	t SO ₂	NEG
	11	NO _x emissions	t NO _x	NEG
	12	Nuclear waste	t nuc. waste	NEG
	13	Fine dust emissions	t fine dust	NEG

*) benefit attribute for TOPSIS analysis: NEG: 'less is better'; POS: 'more is better'

3.2.1. Economic costs

Total system costs (1)

The total system costs are the discounted sum of investment annuities, fuel costs, and operation and maintenance (O&M) costs accumulated over the entire model period until 2030. From the standpoint of a national planner, these net capital costs are the most relevant economic indicator, as they allow a direct assessment of how much the different scenarios would cost the Tunisian economy.

The electricity model calculates the total costs as net present value for the year 2010 (€²⁰¹⁰), using a discount rate of 8%.

Specific generation costs (2)

Another important cost indicator is the specific electricity generation costs, a per-unit value expressed in €/MWh. They can be regarded as a proxy of the electricity costs to be incurred by the Tunisian electricity consumer. The criteria input used in this study is the average of the annual specific cost between the years 2010 and 2030.

3.2.2. Supply security

Supply security is a prominent objective in energy planning in Tunisia. For simplicity, we subsume under this criteria group short-term aspects, as the technical ability of the power system to provide capacity at any time (peak load response), as well as the following long-term energy independency aspects.

Aptitude to respond to peak-load events (3)

For this criterion, we give each generation technology a qualitative attribute (scores between -2 and +2) for its aptitude to respond to peak load events. Gas power plants, with their high capability to follow load variations, receive a score of +2, while less flexible base-load technologies (coal, nuclear) are given 0 points. The score of intermittent renewable power technologies is -2, because they cannot fully guarantee reserve capacity at any time. Based on these numbers, we calculate for each scenario an aggregated indicator for each scenario by averaging the different technology scores according to their proportions of installed capacity in the generation system until 2030.

Total natural gas consumption (4)

As outlined in Section 1, it is a key objective of Tunisia's energy policy to reduce the consumption of natural gas in order to delay dwindling of domestic gas reserves and to alleviate long-term dependency on gas imports (from Algeria). Therefore, the total gas consumption in terms of primary energy for electricity

generation is a characteristic criterion for energy security. The indicator used in the TOPSIS analysis is the overall gas consumption (in PJ) until the year 2030.

Contribution to energy independency (5)

Substituting natural gas with other energy carriers does not increase the security of supply if these carriers expose to Tunisia to new dependencies. The criterion “contribution to energy independency” provides for each primary energy source a subjective ranking between -2 and +2. Renewable energies receive the highest score (+2), as they diversify with domestic energy carriers. As the reduction of natural gas consumption is the key target of Tunisia’s diversification efforts, generation technologies using natural gas are given a score of -2. Coal has the inconvenience of being imported, but the risk of becoming energy-dependent is attenuated by the fact that steam coal is available from numerous exporting countries worldwide (score -1). In contrast, uranium, the fuel for nuclear technology, is only available from rather monopolized markets, which raises the proneness toward energy dependency (score -2). The overall score is calculated by building an average of the individual scores, weighted by the fuel consumption proportions for each scenario in the overall electricity production until 2030.

3.2.3. Socio-economic criteria

This criteria group, consisting of three sub-criteria, addresses socio-economic welfare incurred by the different electricity scenarios.

Local production share (6)

The local production share (or local content) indicates how much of the investment into new generation capacity would flow back to the Tunisian manufacturing industry. The estimates (expressed in percentage points) are based on ratings provided by Tunisian power and renewable energy experts (see Table IV-3). For solar technologies, a gradually increasing local manufacturing share is assumed from today’s levels until 2030 (Fraunhofer/Ernst & Young, 2011). The final local production share for each power mix scenario is calculated by averaging the

individual technology scores according to their proportions of the installed capacities until 2030.

Social acceptance (7)

Social acceptance is given a rating between -2 and +2 in an effort to reflect the population's expected attitude vis-à-vis the occurrence of new power plant technologies in the country. The worst value (-2) is received by nuclear power plants, whose implementation is presumed to be disapproved both by the local population as well as the Tunisian society in general (Haddad, 2011). A zero score (0) is given to technologies which are regarded indifferently or where reservations by the local population (noise, visual disturbance) are outweighed by a generally positive reputation of the technology on the national level, for example, wind power. Solar technologies are ranked highest with a rating of +2. The total social acceptance rating per scenario is calculated as an average of the annual acceptance ratings until 2030.

Job creation (8)

The estimation of the job creation potential for the different scenarios is based on the method used by Rutovitz and Harris (2012). For each scenario, the total number of jobs is calculated as the sum of jobs in the power plant construction sector, the component manufacturing industry, and the labor needed for operation and maintenance (O&M) of the already installed power plants. Jobs in the construction sector are obtained by multiplying the capacity (in MW) of new power installations with a technology-specific employment multiplier, corrected by a regional adjustment factor listed in the above-mentioned study of Rutovitz and Harris (for Tunisia, the correction factor of the Middle East region was applied). The jobs created in the power plant component manufacturing industry receive an additional adjustment considering the local production share in Tunisia (see criteria 6 in Table IV-6). Jobs in the O&M sector are calculated by applying an O&M employment factor to the existing, operational power plant capacities. Finally, for each scenario, a final overall criteria score x_{ij} is established by calculating the average number of jobs in the electricity sector until 2030.

3.2.4. Ecological criteria

This criteria group encompasses the most relevant emissions of the different power plant technologies: *CO₂ emissions (9)*, *SO₂ emissions (10)*, *NO_x emissions (11)*, *nuclear waste (12)*, and *fine dust (13)*. For each category and scenario, total emissions until 2030 are calculated based on the technologies' specific emissions per MWh (see Table IV-3).

3.3. Criteria weights

Assigning weighting parameters for the criteria is a difficult undertaking, as it requires knowledge about the decision-makers' preferences on energy policy issues. Unfortunately, looking at the ongoing political transition process in Tunisia, it was (at the moment of this writing) impossible to discern the orientations of a yet-to-be elected Tunisian government. Therefore, our study was constrained to fall back to a relatively coarse and simplified method to elicit the weighting parameters. The method consisted of carrying out a voting among 37 Tunisian experts during a stakeholder workshop held in Tunis. As illustrated in Figure IV-2 the involved stakeholders were representatives of different institutions and organization: the national electricity utility STEG, its renewable branch STEG Energies Renouvelables, the Ministry of Industry, the Ministry of Environment, the national agency of energy conservation (ANME), the Ministry of Planning and Regional Development, the Ministry of Finance, as well as individual consultants and industry representatives.

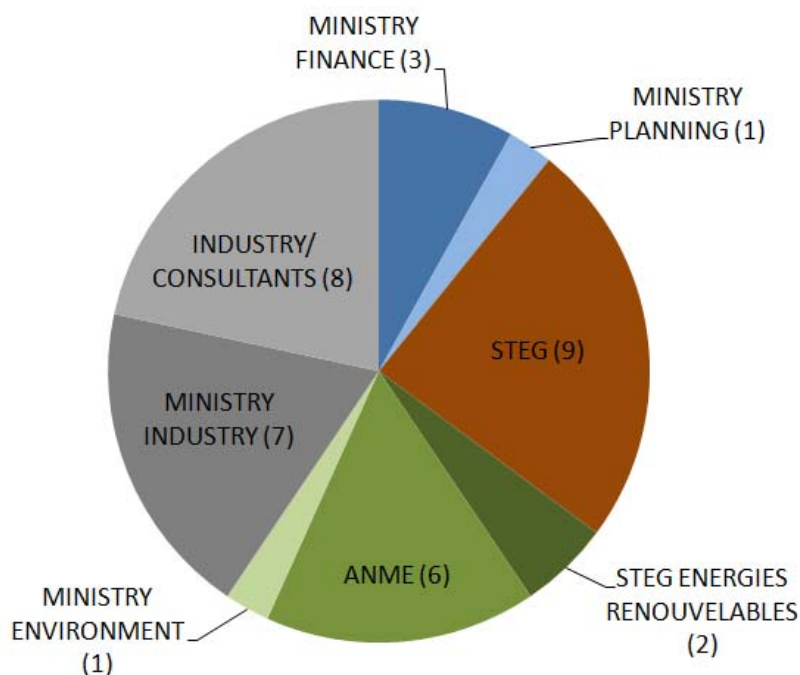


Figure IV-2. Participants of expert workshop by institution

Although it is difficult to establish a direct link between institutional affiliations and the participants' individual preferences, it can be presumed that, for example representatives of the renewable energy sector, the Ministry of Environment or ANME rather tend to favor environmental criteria, while members of the electricity utility STEG might be more concerned by aspects of security of supply. Industry representatives might also emphasis supply security criteria, but should be also concerned by cost aspects and socio-economic criteria. Cost criteria are likely to be the priority for members of the finance and planning ministries.

In order to avoid that the votes follow too closely the orientations of the involved institutions, each expert was asked to designate not only one, but two favorite criteria groups he deems most important for a future Tunisian energy strategy. A final count of these votes revealed the following ranking among the criteria groups: Minimizing economic costs was given the highest priority (34%), followed by concerns about supply security (27%); socio-economic considerations (20%) and environmental aspects (19%) were valued significantly lower. Subsequently, the weights for the criteria groups were apportioned equal-

ly to the different sub-criteria in order to obtain the 13 individual weighing parameter w_i used in the TOPSIS analysis.

Table IV-5. Weighting parameters for criteria groups

	Reference	Sensitivity cases			
	Expert voting ^{*)}	Holistic approach	Technocratic approach	Mercantilist approach	Eco-social approach
A. Economic costs	34%	25%	40%	10%	10%
B. Supply security	27%	25%	40%	40%	10%
C. Socio-econ. benefits	20%	25%	10%	40%	40%
D. Ecological criteria	19%	25%	10%	10%	40%

^{*)} obtained at a stakeholder workshop in Tunis.

Due to the above-mentioned uncertainties about the actual Tunisian policy preferences, we decided to repeat the multi-criteria analysis with distinct assumptions regarding the weighting parameters for criteria groups. Four different sensitivity cases were analyzed: (1) a ‘holistic approach’, where each criteria group receives the same weight, (2) a ‘technocratic approach’ characterized by a strong emphasis on economic costs and security of supply, (3) a ‘mercantilist’ approach where decision making is oriented towards social-economic benefits and the security of Tunisia’s energy supply, and (4) an “eco-social” approach where ecological and social criteria dominate (see Table IV-5).

4. Results

4.1. Power system transformation pathways

Figure IV-3 depicts the outcome of the linear optimization process, giving an overview of the system configuration (installed capacities) as well as the generated electricity by power plant and fuel type from 2010 until 2030. Looking at the installed capacities (Figure IV-3, left), it becomes obvious that the need for

new power plant capacity in Tunisia is substantial. Not surprisingly, the scenario with the highest capacity requirements is the renewable diversification scenario (DivRes). Due to their low capacity factor, a comparatively high number of wind and solar power plants needs to be installed in order to meet the 30% renewable electricity share by 2030. Furthermore, as these technologies also feature a low capacity credit²³ (see Table IV-3) more conventional back-up power capacity (mostly OC gas turbines) is required to ensure the reliability of the system in case of peak load events.

The development of the power generation mix (Figure IV-3, right) indicates that natural gas still remains the dominant fuel if the cumulated gas-based generation between 2010 and 2030 is regarded. This means that Tunisia will still require a substantial amount of natural gas during the transition period until the diversification goals are reached in 2030. This accounts even for those scenarios that portray a strong substitution of natural gas with coal by 2030 (DivCoal and DivCoalRes). Tunisian decision makers should be aware that due to the long lead times for coal power plant construction (the first plants are unlikely to become operational before 2018), the provision of natural gas still remains a crucial issue for the near future.

²³ The capacity credit is the ability of a generation technology to provide firm capacity to the power system.

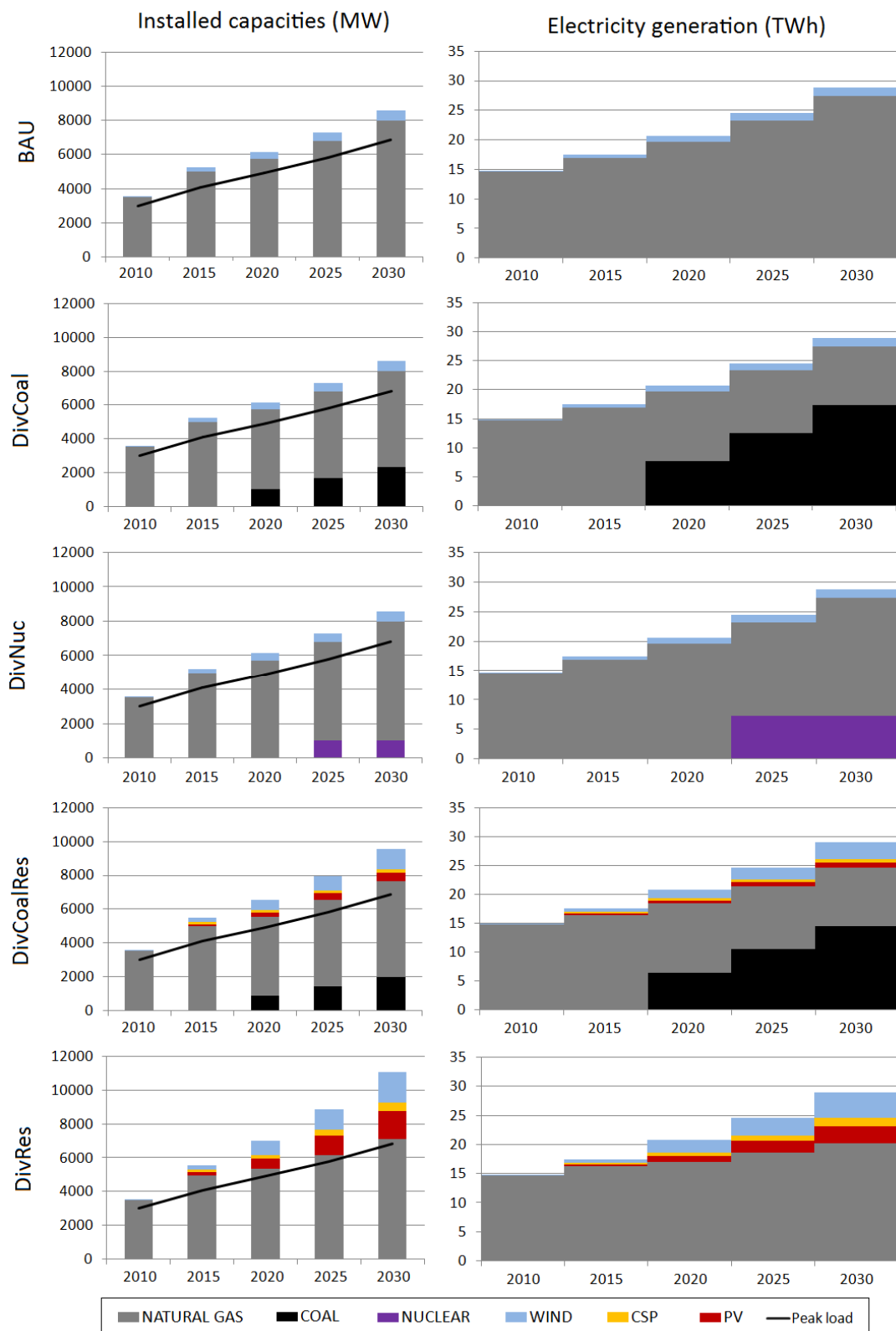


Figure IV-3. Installed capacities (left) and generation (right) for the different scenarios of the Tunisian electricity system until 2030.

4.2. Scenario performance by criteria

Table IV-6 summarizes the model results for the different criteria. Each element of the table represents the performance value x_{ij} of the criterion i with respect to scenario j . It is obvious that due to the multitude of the criteria, their different dimensions, and resulting performances scores, no easy conclusion about the best performing scenario can be possible – at least not by a simple look at the table. This underscores the need for a multi-criteria method to evaluate the different scenarios in a more analytical fashion.

Table IV-6. Results of the electricity system calculations: Performance of the criteria for each scenario.

Criteria i	Scenario j					Unit
	1	2	3	4	5	
	BAU	DivCoal	DivNuc	DivCoalRes	DivRes	
	Performance x_{ij} of criteria i in scenario j					
1	13.9	13.1	13.7	13.6	14.4	billion €
2	69.9	64.4	68.1	67.0	72.4	€/MWh
3	1.63	1.36	1.53	1.20	1.13	point [-2...2]
4	2822	1846	2447	1831	2424	PJ
5	-1.85	-1.55	-1.85	-1.39	-1.36	point [-2...2]
6	21.7	21.4	18.9	24.7	29.5	%
7	0.75	0.42	0.53	0.81	1.08	point [-2...2]
8	3650	5690	5460	7360	8890	jobs
9	167	226	144	206	143	Mt CO ₂
10	0	22	0	18	0	kt SO ₂
11	19	48	16	42	16	kt NO _x
12	0	0	149	0	0	t nuc. waste
13	0	2150	0	1800	0	t fine dust

4.3. Results of the Multi-criteria analysis

By applying the TOPSIS method to the results of Table IV-6 and considering the weighting parameters w_i obtained at the expert workshop (Table IV-5), we calculate the final scenario scores in Table IV-7. The highest ranking is the diversification scenario with renewable energies (DivRes), followed by the business as usual scenario (BAU) and the diversification scenario combining renewable energies with coal power contribution (DivCoalRes). At the bottom of the ranking with relatively close scores are the nuclear (DivNuc) and coal (DivCoal) strategies.

Assuming that our criteria framework and the applied weighting parameters actually match with the rationale of the Tunisian decision-makers, it can be stated that the electricity mix diversification strategy with renewable energies should actually receive high emphasis in deliberations about future Tunisian energy policies.

Table IV-7. Scenario score for TOPSIS (weighting parameters obtained from expert voting)

Scenario	j	Scenario score	Rank
BAU (Business as usual)	1	0.633	2
DivCoal	2	0.444	5
DivNuc	3	0.484	4
DivCoalRes	4	0.555	3
DivRes	5	0.773	1

4.4. Sensitivities: Ranking for different policy preferences

Although the analysis in the previous section reveals a clear preference in favor of the renewable diversification scenario (DivRes), one could argue that this outcome is distorted by the criteria weight selection, which features a certain

bias of the stakeholders in favor of ecological and social criteria; thus, it may have put the renewable scenario in an overly optimistic light. What would be the outcome of the TOPSIS analysis if other policy preferences, such as those described in Section 3.3 (Table IV-5), were used to determine the weighting parameters? In order to answer this question, additional rankings for different sensitivity cases are calculated.

Table IV-8. Ranking of the different sensitivity cases

Rank	Reference	Sensitivity cases			
	Expert voting	Holistic approach	Technocratic approach	Mercantilist approach	Eco-social approach
1	DivRes	DivRes	DivCoalRes	DivRes	DivRes
2	BAU	BAU	DivRes	DivCoalRes	BAU
3	DivCoalRes	DivCoalRes	DivCoal	BAU	DivCoalRes
4	DivNuc	DivNuc	BAU	DivCoal	DivNuc
5	DivCoal	DivCoal	DivNuc	DivNuc	DivCoal

The results in Table IV-8 demonstrate a relatively high robustness, at least with regards to the first rank: except for the ‘technocratic’ approach, which prioritizes the coal-RES diversification scenario (DivCoalRes) because of its favorable supply security and energy independency attributes, all other sensitivity cases exhibit a clear preference for the renewable diversification scenario (DivRes). Regarding the second and third rank, both are predominantly occupied by the business as usual (BAU) and the coal-renewable diversification scenario (DivCoalRes). The two last ranks are shared between the DivCoal and the DivNuc scenario.

In summary, the overall analysis framework indicates that the renewable diversification scenario (DivRes) represents a favorable strategy for the majority of all possible decision-making environments. The DivRes scenario is followed by the business-as-usual (BAU) baseline case and the combined diversification scenario with coal and renewable energies (DivCoalRes). The nuclear power and the exclusively coal-based diversification (DivCoal) scenarios appear to be the least

advantageous strategies for the transformation of the Tunisian electricity system until 2030.

5. Conclusion

The methodology presented in this article aims to support the complex decision making process that is currently ongoing concerning Tunisia's future energy strategies. After the Tunisian revolution in 2011, the country is standing not only at a crossroad in the political sense but also in terms of energy supply; thus, important decisions need to be made. Regarding the electricity sector, prime concerns are the dependency on natural gas, cost aspects, and the social aspects of future electricity supply schemes, in particular with regard to job creation. Environmental sustainability issues are also increasingly promoted by Tunisia's civil society.

In this study, we subjected five different electricity system transformation scenarios to a two-stage analysis method which consisted of a cost-minimizing electricity market model used in conjunction with a multi-criteria analysis. In order to address the multiple aspects of the decision making context in Tunisia, it was necessary to establish a mapping of the Tunisian stakeholder preferences prior to the analysis. This was carried out within a participative stakeholder process, organized by the Tunisian energy conservation agency ANME during expert consultations about a new national renewable energy roadmap: the Tunisian Solar Plan. The criteria preferences (obtained by expert voting) served as weighting parameters for the above-mentioned multi-criteria analysis, and allowed us to establish an indicative ranking of the scenarios. The scenario providing the maximum utility for the examined group of stakeholders was the "Renewable Diversification Scenario," targeting a renewable contribution of 30% in the overall power generation mix in 2030.

As our multi-criteria analysis rests on criteria valuations of a relatively limited set of experts, there remain uncertainties as to whether the obtained ranking result really mirrors the intentions of the Tunisian society as a whole. In a sensitivity analysis, we calculated hypothetical cases where the criteria preferences are very asymmetrically biased toward mercantilist, eco-social and technocratic criteria. The results show a relatively high robustness of the ranking, placing the

renewable diversification scenario in a superior position for most of the analyzed alternative decision environments. One exception is the “technocratic” decision environment, where the coal-renewable diversification (DivCoalRes) is preferred to the renewable diversification (DivRes). Interestingly, the MCDA study of Ribeiro et al. (2012) about the Portuguese power system comes to a similar result: here, the “maximum renewable scenario” is challenged by an equally high-ranking “coal scenario”. The studies featuring Finland (Häyhä et al., 2011) and Greece (Diakoulaki and Karagelis, 2007), however, both exhibit a clear preference of the RES-E scenarios compared to other alternatives of the multi-criteria decision framework. In the Spanish study (Del Riego et al., 2012), where a system development model is combined with a dynamic TOPSIS-based decision making tool, renewable technologies also clearly increase their weight in electricity generation. Our results, together with the above-mentioned findings from similar studies in other countries, should be seen a strong indicator that renewable energy mix visions actually merit a stronger consideration in the discussions about future Tunisian energy policies.

To conclude, it must be noted that our analysis has the drawback of being a purely national approach. The effects of integrating the Tunisian electricity system into a larger regional or even trans-Mediterranean electricity supply scheme were not within the scope of this study. Additional investigations are desirable with this respect, e.g. to assess the effects of a stronger embedding of the Tunisian power system in an international electricity supply scheme with more cross-border power trade (imports and exports of electricity). In this case, interactions of the different national policy strategies also need to be taken into account, e.g. between the Tunisian energy roadmap and those of its neighboring countries Algeria and Libya. Further work is also required to assess the impact of common trans-Mediterranean power system roadmaps, like Desertec (Dii, 2013) or the Mediterranean Solar Plan of the Union for the Mediterranean (UfM, 2013) on the sustainability of the electricity generation mix in Tunisia.

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V. Transmission topologies for the integration of renewable power into the electricity systems of North Africa²⁴

Abstract

A cost-minimizing electricity market model was used to explore optimized infrastructures for the integration of renewable energies in interconnected North African power systems until 2030. The results show that the five countries Morocco, Algeria, Tunisia, Libya and Egypt could together achieve significant economic benefits, reaching up to €3.4 billion, if they increase power system integration, build interconnectors and cooperate on joint utilization of their generation assets. Net electricity exports out of North Africa to Europe or Eastern Mediterranean regions, however, were not observed in the regime of integrated electricity markets until 2030, and could only be realized by much higher levels of renewable energy penetration than currently foreseen by North African governments.

1. Introduction

The tremendous growth in demand for power is one of the principal challenges faced by the electricity sector in North African countries. By 2030, the entire region bordering the southern shore of the Mediterranean Sea – from Morocco to Egypt – will be consuming 2.0-2.4 times more electric energy than it is presently (OME, 2011), and only massive investments into new generation capacity, paralleled by reinforcements of the transmission infrastructure, can avert supply shortages and subsequent damages to the countries' economic development. Undoubtedly, the pressure to keep up with the soaring demand for electricity is

²⁴ This chapter is based on a peer-reviewed article published in the journal *Energy Policy*: Brand, B., Transmission topologies for the integration of renewable power into the electricity systems of North Africa. *Energy Policy* 60 (2013) 155-166, doi: 10.1016/j.enpol.2013.04.071.

placing great strain on the North African economies, especially in times of political upheaval in the aftermath of the Arab Spring and an imminent worldwide economic crisis. However, an accelerated upgrade of electricity infrastructures also offers outstanding opportunities: North African countries can step out of their accustomed national planning strategies and reshape their electricity supply and transmission systems in a more sustainable and coordinated manner.

1.1. Opportunities for power system integration and renewable energies

One major opportunity arising from the pressing need to upgrade North African electricity systems is the creation of a coordinated strategy among the different national power sectors and the setup of strengthened transnational interconnectors for electricity exchange. Until recently, electricity system planning in the southern Mediterranean region has focused on national power demands; this is why today only weak cross-border capacities connect the North African countries. The few existing transmission lines are only sporadically utilized, serving mainly for unscheduled energy transfers ('mutual aid') to counter temporary supply shortages – rather than for systematic, market-based electricity trade (MED-EMIP, 2010). Looking back at the historical development in Europe, the current status of North Africa's power systems can be compared with the European situation in the 1960s and 1970s, when the development of interconnectors was still driven by system security and reliability requirements. Larger cross-border electricity transmission only started in later decades, before finally reaching today's levels on liberalized European power markets (ECN, 2006). A similar evolution of power system integration also would have multiple advantages for North Africa: power exchanges over large distances can level out structural differences between the countries' electricity markets due to different fuel prices.

es, power plant operation costs, and load patterns²⁵. Regions with low electricity production costs benefit from export revenues, while importing countries avoid additions of expensive generation capacity. Furthermore, interconnected power systems are more robust against network failures, unforeseen load increases and tripping of power plants, and hence require a lower reserve capacity, rendering the utilization of installed power plants more efficient. As a further advantage, cross-border integration allows access to a more diverse mix of primary energy sources for electricity generation, which consequently increases energy supply security in the interconnected regions.

A second opportunity, closely linked to the issue of transmission infrastructure, is the increased utilization of renewable energy sources (RES-E). As is commonly known, the entire southern Mediterranean area is endowed with excellent solar and wind resources. Particularly in North Africa, renewable energies promise important strategic benefits given their potential to reduce energy dependency and to slow down the extraction of oil or gas reserves. Interconnected electricity systems facilitate the integration of RES-E by bridging regional heterogeneities of intermittent renewable power. Transmission system operators (TSO) could jointly forecast renewable feed-in and organize the dispatching of conventional power plants to balance out variations of solar and wind generation across larger geographical areas. With RES-E technology costs constantly decreasing, renewable power is garnering greater attention across the entire Middle East and North African (MENA) region. The recent RES-E expansion goals, and even detailed roadmaps, that many governments have embraced are clear indications of this trend (see section 4.5.).

²⁵ Due to their specific industry structure and household behavior, the typical load profiles of North African countries show significant differences on a daily and seasonal basis. Likewise, worth mention are the distinct weekend concepts applied by the countries (Saturday/Sunday weekends vs. Friday/Saturday weekends) and the time shift of electricity consumption across the Southern Mediterranean rim, which stretches over three time zones.

1.2. Transcontinental context and institutional developments

In addition to the above-mentioned opportunities for North African power systems there is the aspect of pan-Mediterranean cooperation. Already in 1995, with the Barcelona Process, a framework was created for economic collaboration between the European Union (EU) and its southern neighbors, targeting improved integration of southern Mediterranean energy markets with those of the EU. These activities received a fresh impetus in 2008 with the founding of the Union for the Mediterranean (UfM), a more formal political partnership between the EU members and 16 southern Mediterranean countries. UfM's principal institutional vehicle pertaining to energy issues is the Mediterranean Solar Plan (MSP), a project dedicated to improving the political and regulatory conditions for a future trans-Mediterranean electricity system based on renewable energies (MSP, 2011). The concept, which is being advanced by private sector initiatives like the Desertec Industrial Initiative (Dii, 2012) and the Med-Grid consortium (MedGrid, 2012), is indeed unique in the history of power systems. This is not only because of the technical challenges, but also because of the ambition to reunite two structurally, completely disparate electricity systems: in the North, the already well-integrated and fairly liberalized European power system, characterized by relatively stable electricity demand and the political will to reduce greenhouse gas emissions; in the South, the fragmented national electricity markets, often ruled by state monopolies whose main occupation is to cope with the ever-increasing electricity demand.

Despite these apparent differences, a number of recent developments indicate that the convergence of Euro-Mediterranean electricity systems is being taken seriously on the institutional level. In 2007, national regulatory agencies from both sides of the Mediterranean founded MEDREG, the Association of Mediterranean Regulators for Electricity and Gas (MEDREG, 2012). The year 2012 brought the creation of a similar multilateral body, the umbrella organization of the Mediterranean Transmission System Operators (MED-TSO), with the objective to coordinate the development and operation of Mediterranean electricity grids (MED-TSO, 2012).

On a more regional level, attempts to spur electricity system integration are visible in the North African Maghreb region: the electricity ministers of the Arab

Maghreb Union²⁶ have cooperated since 1989 within the Maghreb electricity committee Comité Maghrebin de l'Electricité (COMELEC). Three COMELEC members –Morocco, Algeria, and Tunisia– already operate a synchronized and interconnected electricity grid (Brand and Zingerle, 2010) and are striving for a progressive integration of their electricity markets and the implementation of joint transmission projects (Algiers Declaration, 2010). Similar activities have been established in the Mashreq, where an eight-country electricity agreement between Egypt, Iraq, Jordan, Lebanon, Libya, Palestine, Syria, and Turkey was initiated in 1988 (ADB, 2012). The Arab Union of Electricity (AUPTDE, 2012), founded under the auspices of the League of Arab States, has likewise subscribed to the idea of fostering the coordination of their members' electricity systems, including the strengthening of transnational transmission capacities.

1.3. Barriers

Despite such multiple activities, the development of integrated MENA electricity systems still faces a number of challenges and barriers. One particular obstacle is the high investment cost of transnational transmission assets, and the question how these costs can be allocated among the concerned parties. Financing transnational transmission projects still requires – directly or indirectly – negotiations between the involved governments, which is one important reason why projects between North African countries are constantly plagued by delays and why interconnector capacities are still insufficient today. Moreover, though not often openly discussed, many governments in North Africa have political reservations against too much interdependence with their neighboring energy sectors (Ghiles, 2010). Energy is a sensitive area in MENA politics: Many governments view increased dependence on neighboring nations or diminished control, even partly, of domestic resources –for instance, through electricity imports– as a serious loss of national sovereignty. General reservation regarding cooperation can also be observed when looking at the renewable electricity goals of the North African countries; so far, none of the national roadmaps envisages RES-E synergies with its neighbors, although the different geographical and temporal

²⁶ The Arab Maghreb Union (AMU) was founded by the governments of Algeria, Libya, Mauritania, Morocco, and Tunisia.

availability of wind, solar, and hydro power would speak strongly for a joint and coordinated utilization of these resources.

1.4. Research questions

The present paper sets out to unfold and quantify some of the major effects of increased cooperation and integrated power system planning among North African countries. In particular, the following questions are addressed: (1) What are the economic benefits of an increased multilateral cooperation with regard to electricity exchanges via transnational interconnectors? (2) How would increased RES-E integration influence the electricity generation scheme and the transmission topologies between the North African countries? (3) Under which circumstances could rising renewable energy penetration in North Africa facilitate electricity exports to neighboring regions; e.g., to Europe?

The analysis was performed using a multi-regional electricity market model that simultaneously optimizes investments in generation and transmission capacities, as well as power plant dispatch and electricity exchanges between different countries. The study begins, in section 2, with a description of the article's scope within the research context of power system modeling of interconnected North African electricity markets. This is followed by a model description (section 3), and a definition of the parameters and scenarios (in section 4). Finally, sections 5 and 6 provide the analysis of the results and a concluding discussion.

2. Scope of the study

Modeling renewable energy integration into interconnected North African electricity systems is a relatively young research area that has been addressed by only a few publications. So far, many studies have been largely motivated by the vision of cross-continental power networks, considering Europe and the MENA region (EU-MENA) as one large interconnected system, substantially –or even fully– supplied with renewable power. A drawback of these (normative) approaches is that they often omit analysis of the transformation pathways from the current fossil-based power supply infrastructures into the future. Additional-

ly, the needs and particularities of North African electricity markets are often not sufficiently considered in the research.

Czisch (2005) used a linear optimization model to simulate a cost-minimized, transcontinental 100% renewable electricity supply scheme on the basis of HVDC transmission. In his study, however, the North African electricity systems were only represented in a relatively coarse manner, with several countries merged into simplified model regions. A more clear-cut regional analysis is provided by Dii (2012), simulating a 90% RES-E supply in 2050 for the entire EU-MENA region, made up of 38 different countries, including the five North African countries as distinct model regions. Unlike Czisch's work, Dii's study features detailed transmission corridors, including both AC as well as HVDC transmission technology. Both studies, however, remain idealized, as they simulate a hypothetical, newly built generation and transmission system (Greenfield approach), neglecting the existing generation and transmission infrastructures (power plant inventory and transmission lines) as boundary conditions of the analysis. Haller et al. (2012) also explored the temporal transition of the EU-MENA electricity systems over several decades. Here, the investment pathways for generation and transmission assets were optimized under the assumption of strong carbon emission reduction targets until 2050. Power system data of the North African countries, however, was only approximated. A study providing greater insight into North African power infrastructures and their individual electricity generation preferences is provided by Mercados (2011); but this work does not include transmission capacity optimization and is regionally limited to the Maghreb countries Morocco, Algeria, and Tunisia.

The analysis presented in this paper aims to provide a realistic picture of what can be done in terms of power system development in North Africa from today until 2030, taking into account the existing infrastructures and different national planning strategies in North Africa, but also considering technical and non-technical barriers to electricity system planning and the integration of renewable energies. The following key specifications were considered paramount for the model development:

- Capability to simulate different power system transformation scenarios from today until 2030.
- Inclusion of the status quo of power system infrastructures as well as the near- and long-term planning processes for transmission and generation assets in North Africa.
- Hourly representation of the countries' electricity load in order to explore interactions with the intermittent RES-E generation and the dispatching of conventional power plants.
- The ability to mirror different levels of regional cooperation.

Geographically, this study focuses on the five North African countries that are represented as distinct model regions. Additionally, in order to reflect electricity exchanges with adjacent electricity systems, Europe and eastern Mediterranean countries are included, but in a less refined manner as aggregated satellite regions (see Figure V-1).

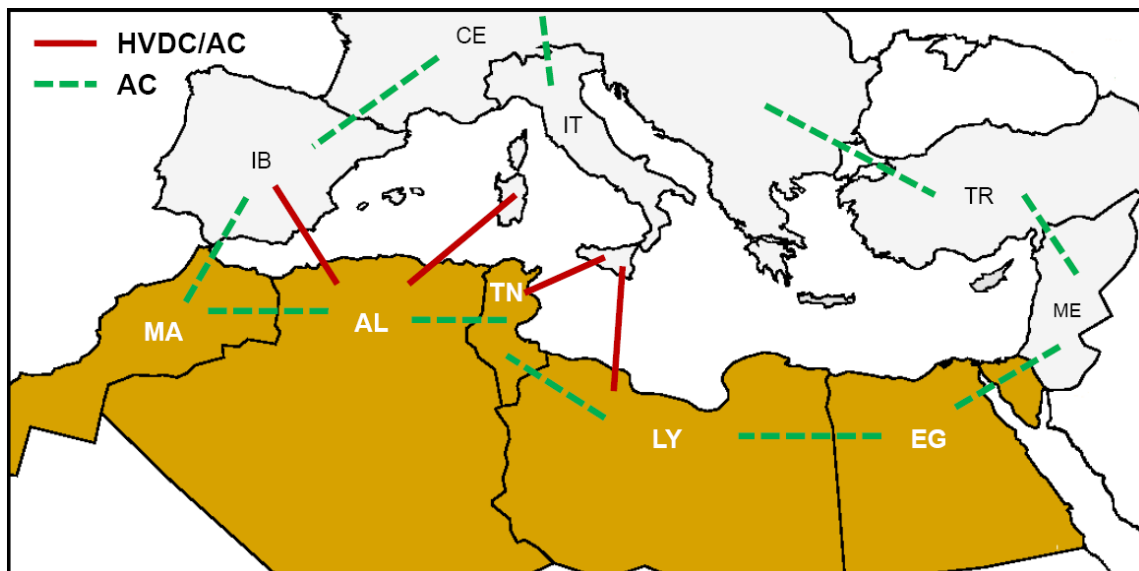


Figure V-1. Model regions: Morocco (MA), Algeria (AL), Tunisia (TN), Libya (LY) and Egypt (EG). Satellite regions: Iberian Peninsula (IB), ENTSO-E Continental Europe (CE), Italy (IT), Turkey (TR) and East Mediterranean countries (ME).

3. Model description

Analyzing the transformation of interconnected power systems typically is performed with linear models that combine the long-term optimization of investment planning with the short-term dynamics of system operation on a seasonal, daily and often hourly basis. The methodology in this study likewise follows this approach, as it is done by a number of similar dispatch and investment models, for instance applied by EWI/Energynautics (2011), Haller et al. (2012), or Bartels (2009). For reasons of space, and because the approach has been sufficiently described in the literature, the following paragraphs only sketch the most important model functions and particularities pertinent to the analysis of North African power markets.

3.1. Objective function and decision variables

The objective function, the central element of any linear program, is the total discounted system cost, the sum of investment annuities of generation capacity (INV_y), the costs of operation and maintenance, (OM_y), other variable costs, (VAR_y), and fuel costs ($FUEL_y$), as well as the investment associated with the set-up of net transfer capacity (NTC) for power transmission (INV_NTC_y).

$$TOTAL = \sum_y disc_y \cdot (INV_y + OM_y + VAR_y + FUEL_y + INV_NTC_y)$$

The different cost constituents are aggregated into 5-year periods ($y=2010, 2015, 2020$, etc.) and discounted with an overall real discount rate of 5% to a net present value in 2010 (€^{2010}). The model seeks to minimize the total costs $TOTAL$ by finding optimized values for the following decision variables:

- Investments in generation capacity $CAP_{i,r,y}$ (in MW) of each technology type $i=coal, gas, csp, wind, \dots$ in the period y and region r .

- The expansion of installed transmission capacity $NTC_{r,r^*,y}$ (in MW) between two model regions r and r^* .
- The hourly (h) electric energy generation $GEN_{i,r,y,d,h}$ (in MWh) for each representative day (d) in the period y .
- The hourly electric power transmission $TRANS_{r,r^*,y,d,h}$ (in MW) from region r to region r^* .

The indices associated with the variables characterize the main regional, temporal and technological features of the model and are listed in Table V-1.

Table V-1. Model definition

Index	Definition	Denotation
i	generation technology	coal, lignite, nuclear, gas_cc, gas_oc, gas_st, oil, pv, csp, wind_onsh, wind_offsh, biomass, hydro, hydro_pump.
r, r [*]	model region	MA, AL, TN, IB, IT, CE, TR, ME, EG, LY
y	model period	2010, 2015, ...
d	representative day	1,..., 24
h	hour	1,..., 24

3.2. Energy balance constraint

The energy balance constraint ensures that electricity supply meets the electricity demand in any region at any given moment in time.

$$\begin{aligned}
 & load_{r,y,d,h} + \sum_{r^*} TRANS_{r,r^*,y,d,h} + HYD_PUMP_{r,y,d,h} \\
 &= \sum_{tech} GEN_{i,y,d,h} + \sum_{r^*} (1 - loss_{r,r^*,y}) \cdot TRANS_{r^*,r,y,d,h}
 \end{aligned}$$

The left side of the equation, the total regional electricity demand, is the sum of the hourly electricity load ($load_{r,y,d,h}$), the energy transmitted from region r to neighboring regions r^* ($TRANS_{r,r^*,y,d,h}$), and the electricity used for pumping into hydro storage systems ($HYD_PUMP_{r,y,d,h}$). Electricity transmission between two regions must not exceed the net transfer capacity ($TRANS_{r,r^*,y,d,h} \leq NTC_{r,r^*,y}$).

3.2 Operational constraints for power plants

Modeling the operation of power plants requires a large set of constraints, describing the technical particularities of each power generation technology. For thermal power plants (fossil and nuclear power, as well as biomass-driven units), the dispatch properties are mainly ruled by the limitations of the steam or gas turbine: efficiencies, ramp-up behavior and part-load conditions. Hydroelectric power plants have only limited significance for North Africa, as only Egypt and Morocco exhibit noteworthy capacities. These are simulated by taking into account seasonal water inflows and the reservoir size of the water storage in the case of hydro dams or pumped storage facilities. Renewable power generation technologies like wind and photovoltaic (PV) plants are bound to the intermittent availability of irradiation and wind speed. The model reflects this by deterministic output patterns of standard wind turbines and PV plants that were calculated on basis of hourly historic meteorological data of representative sites in each region. Unlike wind and PV, concentrated solar power plants (CSP) can feature storage systems to transform (direct) solar irradiance into dispatchable electricity. For this study, we assumed a dry-cooled parabolic trough power plant, with 6 hours thermal storage and a solar multiple of 2. The simulation of storage-based CSP plant operation within linear models is described in Brand et al. (2012).

3.3 Constraints of supply security and interregional cooperation

Technical and security requirements, as well as aspects of interregional cooperation are mirrored by further equations. The peak load constraint ensures that in each period (y) sufficient generation capacity is available to meet the maximum

power demand. This criterion is of high importance for the North African countries, where the peak loads set new records every year. The parameter $peakload_y$ (in MW, including a security margin) represents the expected maximum load in each period y , corresponding to forecasts published by national TSOs or electricity regulators. In interconnected power systems, the capacity assigned to cover the peak load can be differentiated as follows (see Al Khal et al., 2006):

$$peakload_{r,y} \leq \sum_i CAP_{i,r,y} \cdot av_{i,r,y} \cdot cap_credit_{i,r,y} + \sum_{r^*} (1 - loss_{r^*,r,y}) \cdot CAP_{S_{r^*,r,y}} - \sum_{r^*} CAP_{S_{r,r^*,y}}$$

First, it can be provided domestically, by the sum of all installed power generation technologies $CAP_{i,r,y}$ in the model region r . The parameters $av_{tech,y,d}$ and $cap_credit_{tech,y}$ are percentage values referring to the typical plant availability and the capacity credit²⁷ of each technology. A second contributor to peak load coverage could be foreign capacity, if one region r^* guarantees another region r apart of its generation capacity for the event of peak loads. This ‘shared’ capacity, $CAP_{S_{r^*,r,y}}$, is reduced by a loss factor $(1 - loss_{r^*,r,y})$ to consider power drops along the transnational interconnector. Conversely, all capacity that region r reserves for other regions, $CAP_{S_{r,r^*,y}}$, needs to be subtracted as it is not available for domestic peak coverage. A further boundary condition ($CAP_{S_{r^*,r,y}} \leq NTC_{r,r^*,y}$) constrains the shared capacity to not exceed the interconnector’s NTC value.

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In this study, the capacity credit is understood as the capability of a specific generation technology i , to provide firm capacity to the power system. The capacity credit of conventional power plants is always 100%, while for intermittent renewable technologies it is significantly lower.

The technical possibility to counter peak loads with foreign generation capacity does not necessarily mean that all countries actually desire this option. Especially in North Africa, one could imagine political misgivings concerning dependency on other countries' generation assets. Following the approach by Al Khal et al. (2006) an additional equation was introduced to define the level of accepted reliance of a country from external capacity:

$$\sum_{r^*} (1 - loss_{r^*,r,y}) \cdot CAP_{S_{r^*,r,y}} \leq (1 - autonomy_{r,y}) \cdot peakload_{r,y}$$

By setting an individual autonomy factor ($autonomy_{r,y}$), each region can decide the extent to which it would tolerate other regions' capacity to cover its peak loads. If 100% autonomy is chosen ($autonomy_{r,y} = 1$), then $CAP_{S_{r^*,r,y}}$ becomes zero, meaning that the region wants to secure its peak load entirely with own capacity. Different autonomy factors thus allow for simulation of different regional cooperation scenarios regarding the joint utilization of generation capacity.

A second dimension of cooperation concerns electricity imports. Here, the key criteria is not capacity, but energy. In the model, the level of energy independency is defined by the share of the annual electric energy demand (in GWh) that must be generated domestically:

$$annual_demand_{y,r} \cdot self_supply_{r,y} \leq \sum_{i,d,h} GEN_{i,y,d,h}$$

With the parameter $self_supply_{r,y}$, the allowed level of import dependency can be adjusted: setting $self_supply_{r,y} = 100\%$ is equivalent to a situation where a region is constrained to fully satisfy its demand by own generation; lower values of $self_supply_{r,y}$ would represent a higher tolerance of electricity imports. Limiting import dependency can be considered a strategy for those countries

whose governments protect the local power generation industries and support the use of domestic energy resources.

4. Assumptions and scenarios

4.1. Demand and load

Demand and load projections in this study rest on historic electricity data for the year 2010 that were carried forward until 2030 on the basis of annual growth rate assumptions made by OME, 2011 (“conservative scenario”, assuming continuation of past demand trends into the future). Table V-2 lists the annual demand ($annual_demand_{y,r}$) of the North African model regions, as well as the corresponding forecasts for the maximum peak load ($peakload_{r,y}$) for the years 2010, 2020, and 2030.

Table V-2. Assumptions for annual demand and peak load developments.

Region	Annual demand (TWh) / Peak load (MW)		
	2010	2020	2030
Morocco (MA)	27.5 / 4790	46.1 / 8030	64.0 / 11140
Algeria (AL)	45.2 / 7720	75.8 / 12940	105.1 / 17950
Tunisia (TN)	14.9 / 3010	25.0 / 5050	34.7 / 7000
Libya (LY)	30.4 / 5280	51.0 / 8850	70.7 / 12280
Egypt (EG)	133.3 / 22750	223.4 / 38130	310.0 / 52900

For Turkey and the East Mediterranean model region, the demand scenarios likewise refer to OME’s growth forecasts, while the corresponding projections for the EU satellite regions are based on the “EU Reference scenario” data in Capros et al. (2009). The hourly electric load was calculated on basis of historical time series provided by TSOs or national electric utilities²⁸. To the extent it was possible, data was used from the basis year 2010, of which 24 representative days (d) - of each month one working day and one weekend day - were se-

²⁸

For the European model regions, 2010 data from ENTSO-E was used.

lected. These load patterns were normalized and scaled up with the annual demand in order to obtain the hourly load ($load_{r,y,d,h}$) for future periods until 2030.

4.2. Generation

The following generation technologies are covered by the model: nuclear power, lignite and hard coal, gas-powered steam plants (Gas ST), combined cycle (Gas CC) gas and open cycle (Gas OC) gas turbines, liquid fuel-fired power plants (Oil), hydropower (including pumped storage), biomass, wind (onshore and offshore), photovoltaic (PV), and CSP. The most important technical and financial model parameters are listed in Table V-3. Most parameters were taken from a previous study on the Tunisian power system (GIZ, 2011), which was considered reasonable for the other North African countries, as well.

Table V-3. Power plant parameters assumptions

	Invest. (€/kW) 2010/2020/2030	Fix. O&M (€/kW _a)	Var. O&M (€/MWh _{el})	Tech. avail. (%)	Effi- cienc y(%)	Cap. credit (%)	Min. load (%)	CO ₂ emission (tCO ₂ /MWh _{th})
Coal	2130	22.8	3.1	85	40	100	50	0.34
Lignite	1800	40.0	1.0	85	42	100	60	0.41
Nuclear	4170	53.2	0.4	85	33	100	65	-
Gas OC	675	8.2	2.5	85	34	100	20	0.20
Gas ST	900	15.0	1.6	85	39	100	45	0.20
Gas CC	900	9.7	1.6	85	53	100	40	0.20
Oil	775	8.2	2.5	85	32	100	20	0.28
Biomass	1500	60.0	2.8	85	32	100	20	-
Hydro	2000	20.0	0.0	99	-	100	-	-
Wind onsh.	1150/980/930	40.0	0.0	97	-	5	-	-
Wind offsh.	2,000	100.0	0.0	95	-	8	-	-
PV	2500/1380/1000	20.0	0.0	99	-	0-5*	-	-
CSP	6090/4200/3400	53.8	0.0	96	-	90	20	-

* the capacity credit for PV is considered 0% for European countries, 5% for MENA due to more constant irradiation and a trend to midday load peaks.

To reflect region-specific particularities, the model included several exogenous restrictions to the possible expansion of generation technologies. For political and technical reasons, the installation of nuclear power, lignite, biomass and off-shore wind technologies was precluded in North Africa until 2030. Coal steam power plants can only be built in Morocco –where the technology is already in use – and Tunisia, which is considering a coal diversification strategy starting in 2020 (GIZ, 2011). The 2010 power plant stock of all North African countries is summarized in Table V-4; data was taken from annual reports of North African utilities and AUPTDE publications. The power plant inventories of the European satellite regions are based on ENTSO-E (2011), TEİAŞ (2012) for Turkey and PWMSP (2011) for the East Mediterranean model region.

Table V-4. North African power plant inventory by the end of 2010.

	Installed generation capacities (MW) in 2010				
	Morocco	Algeria	Tunisia	Libya	Egypt
Coal	1785	-	-	-	-
Gas OC	915	6565	1512	2019	2841
Gas ST	-	2487	1080	-	10334
Gas CC	850	2042	784	2430	7137
Oil	803	245	-	2000	1124
Hydro	1770	228	66	-	2800
Wind	221	-	54	-	490
PV	-	-	-	-	-
CSP	-	-	-	-	-

4.3. Renewable and fossil resources

4.3.1 Meteorological data

The electricity generation of solar power plants and wind farms is ruled by irradiation conditions and wind speed. The deterministic operation patterns needed for CSP and PV modeling were calculated with the System Advisor Model (NREL, 2012) on basis of global horizontal irradiation (GHI) and direct normal irradiance (DNI) time series at representative sites in the different model regions. The time series -provided by Helioclim (2012) - were selected such that they correspond to the 24 calendar dates of the electricity load in 2010. Wind power production, likewise matching to these days, was calculated with the MINT software (Sander+Partner, 2012) on basis of MERRA wind speed time series (MERRA, 2012). For hydropower generation, historic seasonal water inflow profiles were used. It must be premised that the use of deterministic meteorological data (wind and solar radiation), represents a simplification. Therefore, for future studies, it is desirable to likewise consider the stochasticity of intermittent generation in the simulations. A further aspect worthy to be taken into account – especially when it comes to long-term electricity system forecasts – is climate change and its potential impact on the availability of the renewable resources in the MENA region over the coming decades.

4.3.2. Fuel costs

Table V-5 displays the price scenario used for this study. Prices are expressed in $\text{€}^{2010}/\text{MWh}_{\text{th}}$ (lower heating value) and represent each fuel's opportunity cost. It has to be noted that in North Africa, the assumption of uniform opportunity costs can be undermined by the fact that gas-producing countries, especially those with intertwined gas- and electricity state monopolies, often tend to supply the power system with gas at much lower costs. The associated distortions that such indirect subsidies could have on the model results are analyzed in a sensitivity analysis in section 5.4.1.

Table V-5. Fuel price pathways. Source: IEA, 2011 (current policies scenario), own assumptions.

	Fuel costs (€/MWh _{th})		
	2010	2020	2030
Nuclear	1.7	2.1	2.3
Coal	10.2	11.2	12.0
Oil	36.2	54.7	62.3
Lignite	1.5	1.5	1.5
Natural Gas	19.3	28.3	32.4
Biomass	15.0	15.0	15.0

4.3.3. Carbon emissions

Threatened by water scarcity and desertification, and with a densely populated coastal zone exposed to sea-level rise, the North African region is particularly vulnerable to climate change (AfdB, 2012). Although all North African countries have signed and ratified the Kyoto Protocol and are occasionally also involved in Clean Development Mechanism (CDM) projects within this framework, the decarbonization of power systems currently is only a minor motivation for the regional decision makers; energy security, industrial development and job creation, are seen as much more important drivers for RES-E projects in North Africa. Therefore, it remains uncertain whether the region will soon get involved in larger emission trading schemes, such as practiced in the EU. Nevertheless, the question of an equitable accounting of CO₂ emissions will be raised in North Africa as well if the power systems become embedded in liberalized, integrated electricity markets with strong interconnections with the EU. In this case, in order to provide for equal competition, and to prevent carbon leakage through electricity transmission (e.g. the setup of carbon-intensive generation technologies in the South), it can be expected that a level playing field with consistent costs for CO₂ emissions will be established. In this model, therefore, a scenario of homogenous CO₂ costs was applied, amounting to 15 €/tCO₂ in 2010, with a stepwise increase to 30 €/tCO₂ until 2030 (IEA, 2011, current policies scenario). A sensitivity analysis was carried out in section 5.4.2 to analyze the case of disparate CO₂ costs between different countries.

4.4. Transmission

In the model, upgrades of NTC capacity ($NTC_{r,r^*,y}$) can only take place along predefined transmission corridors (see Figure V-1) referring to technically and economically reasonable Mediterranean interconnection options until 2030 (see MED-EMIP, 2010). It is assumed that no HVDC network technology will be utilized for terrestrial applications until 2030. Therefore, overland transmission is realized strictly with AC technology. Interconnectors crossing the Mediterranean Sea, however, involve the utilization of HVDC cables, connecting DC/AC converter stations at the two sides of the shore²⁹. Undersea cable distances and landing points are taken from MED-EMIP (2010), while the length of the overland lines is estimated as distance between centers of high population density. The technical and cost parameters associated with the transmission assets are summarized in Table V-6.

Table V-6. Transmission technology parameters. Sources: Realisegrid (2010), MED-EMIP (2010). The investment cost assumptions include a security margin of 100% for AC and 20% for HVDC (Dii, 2012).

	Transmission lines		Converter terminal pair	
	Inv. costs (€/MW _{NTC} · km)	Line losses (%/1000 km)	Inv. costs (€/kW _{NTC})	Converter losses (%/1000 km)
AC overland (400kV)	800	10.0	-	-
HVDC submarine (400 kV)	1200	5.0	150	1.4

The existing NTC values in the base year 2010 are taken from MED-EMIP (2010), while data published by ENTSO-E (2012), was used for interconnectors between and within the European satellite regions. The model likewise considers

²⁹ Exceptions are Morocco-Spain connection, realized in AC, and the Libya-Tunisia corridor, for which an HVDC back-to-back converter station is foreseen to asynchronously link the two electricity systems. The corridors to Italy via the islands Sardinia and Sicily include additional AC and HVDC connections to mainland Italy, which are for clarity purposes not shown in Figure V-1.

endogenous NTC additions until 2015, if they are part of fixed planning programs, for example outlined in TYNDP (2012).

4.5. Renewable electricity goals

One particular objective of this study is to analyze the impacts of RES-E penetration on interconnected regional power markets in North Africa. All five countries have drafted roadmaps for RES-E deployment (see the synopsis in Table V-7). As these goals have been pronounced by official sources – mostly ministries or government agencies – they are considered as the baseline scenario for RES-E deployment in this study. Nevertheless, it is worth noting that North African RES-E targets underwent several modifications in the past, and it is not unlikely that they might become subject to further revision if, for instance, political or economic frameworks change.

Table V-7. North African RES-E goals for wind, PV and CSP capacity

Region	Wind (MW)	CSP (MW)	PV (MW)	Source
	2010 / 2020 / 2030	2010 / 2020 / 2030	2010 / 2020 / 2030	
MA	221 / 2000 / 4000*	0 / 1000** / 2000*	0 / 1000** / 2000*	ONE, 2008, Masen, 2012
AL	0 / 270 / 1750	0 / 1500 / 7725	0 / 830 / 2800	MEM, 2011, PWMSP, 2012
TN	54 / 500 / 1520	0 / 200 / 600	0 / 800 / 1930	GIZ, 2011; African Manager, 2012.
LY	0 / 1500 / 2500*	0 / 800 / 1600*	0 / 150 / 850*	REAOL, 2012
EG	490 / 7200 / 14800*	0 / 800** / 2550*	0 / 800** / 2550*	NREA, 2012; Younis, 2012

*) extrapolated **) solar capacity goals split into 50% PV and 50% CSP

The model likewise incorporates RES-E targets of the satellite regions: for the East Mediterranean region, these assumptions are based on PWMSP (2012); for European countries, the model relies on EU's National Renewable Energy Action Plans (NREAP) based on data from ECN (2011). Trends were extrapolated until 2030.

5. Results

5.1. Baseline scenario

The first scenario to be examined is a baseline scenario in which North African countries fulfill their RES-E capacity goals, but remain reluctant regarding upgrades of transnational interconnectors. Therefore, no endogenous transmission asset additions are considered in the baseline scenario. Moreover, it is assumed that all regions exhibit aversion to power system cooperation: the model parameters $self_supply_{r,y}$ and $autonomy_{r,y}$ stay at their current, high levels³⁰. Figure V-2 shows the simulated transmission capacities and power exchanges under these conditions for 2030. As expected, power transfers remain low, in particular in the eastern part of North Africa (Libya and Egypt), which continues to be disconnected from the western system (Morocco, Algeria, Tunisia) because the baseline scenario excludes the construction of a Libya-Tunisia interconnection. Electricity exchanges between North Africa and Europe only take place on the Spain-Morocco interconnector, with a net power flow of 240 GWh in the North-South direction in 2030. This is a significant decrease compared to 2010, for which the model calculated 5400 GWh of Moroccan imports from Spain³¹. The finding that Morocco can reduce its imports until 2030 can be attributed to lower electricity generation costs due to an increased domestic construction of RES-E and coal power plants, which render –in the long run– imports less attractive.

³⁰ For both parameters an average value of 97% is assumed, except for Morocco, where a self-supply rate of 83% reflects high electricity imports from Spain amounting to 17% of the national demand in 2010.

³¹ According to the utility's reports, the actual imports amounted to 3900 GWh in 2010 and 4500 GWh in 2011 (ONE, 2011).

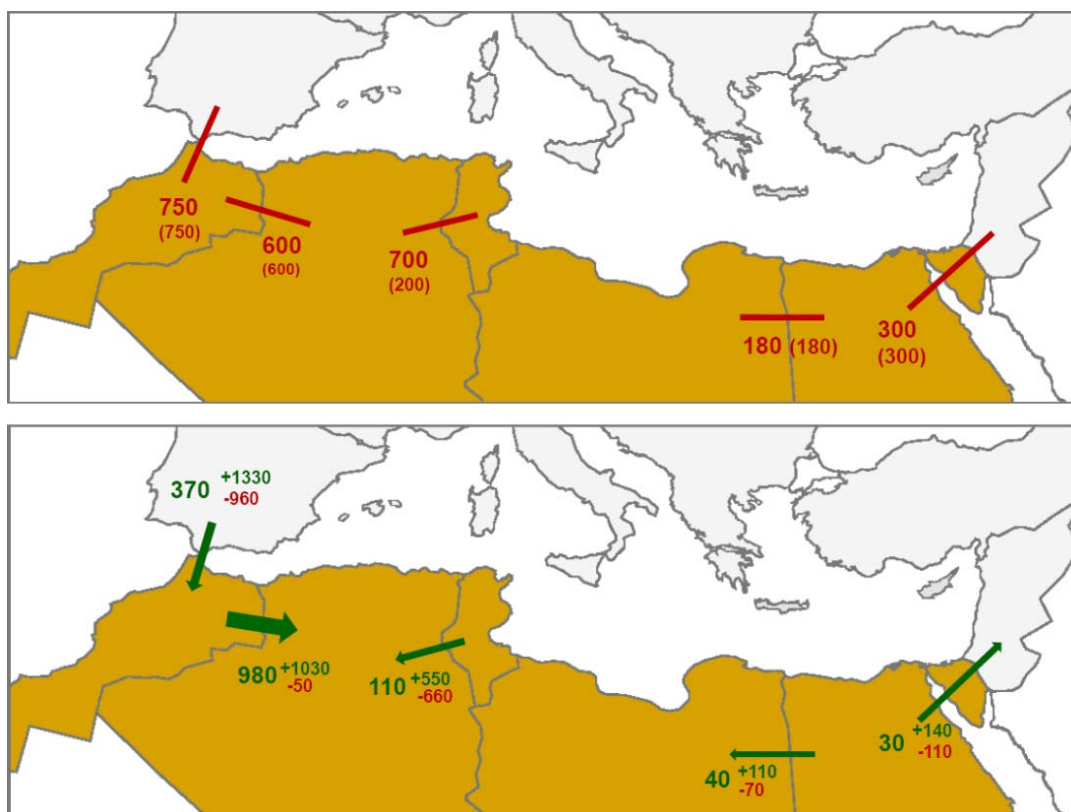


Figure V-2. Baseline scenario. Top: 2030 installed NTC capacities in MW (values in brackets: 2010 capacities). Bottom: Annual net energy flows in GWh in 2030 (+/- symbols indicate the gross exchanges between the model regions)

5.2. Power system cooperation scenarios

5.2.1. Economic benefits of cooperation

This section analyzes the extent to which transnational electricity interconnectors would be constructed if the North African countries were to fulfill their individual RES-E goals but accept higher reliance on neighboring generation capacity and electricity imports. As described in chapter 3.4., one can distinguish between two dimensions of reliance: first, the tolerance of electricity imports; and second, the willingness to jointly share generation capacity for peak-load coverage. In the model, both dimensions were simulated by adjusting the parameters $self_supply_{r,y}$ and $autonomy_{r,y}$, in 25% steps. Setting $self_supply_{r,y}=75\%$ and $autonomy_{r,y}=100\%$ corresponds to the case where a country is willing to

accept up to 25% of foreign electricity imports, but prefers full independency when it comes to safeguarding its power system against peak load events. Vice-versa, high self-supply requirements ($self_supply_{r,y} \approx 100\%$) are tantamount to electricity import aversion, for example, if governments pursue ‘mercantilist’ strategies by giving priority to domestic over foreign electricity generation.

Table V-8. Cost benefits of cooperation: Cumulated cost savings of all North African countries until 2030 compared to the baseline scenario.

	Cooperation scenarios		
	“Import reliance / capacity autonomy”	“Import aversion / capacity reliance”	“Mixed capacity and import reliance”
	($autonomy \approx 100\%$, $self_supply = 75\%$)	($autonomy = 75\%$, $self_supply \approx 100\%$)	($autonomy = 75\%$, $self_supply = 75\%$)
Savings in the generation system (€ ²⁰¹⁰ million)	2209	1282	3942
Surplus costs for the transmission system (€ ²⁰¹⁰ million)	-500	-306	-554
Total benefit (€²⁰¹⁰ million)	1709	976	3388

Table V-8 displays the discounted and aggregated cost savings between the baseline scenario and three different cooperation scenarios until 2030. It can be observed that considerable economic gains for the entire generation system (in terms of fuel savings, and avoided capacity additions) can be achieved with increased cooperation. These gains largely overcompensate for the surplus costs of the transmission capacity required for the different power systems - leaving an overall net benefit in all cooperation scenarios. It reaches a maximum of around €3.4 billion³² in the “mixed capacity and import reliance scenario,” in which the model regions simultaneously cooperate on interregional electricity exchange and peak capacity sharing ($self_supply_{r,y}$ and $autonomy_{r,y}$ both set at 75%). Additional model runs show that with a further increase of interregional reliance

³² This value represents approximately 1% of the total costs of the North African power system until 2030.

5.2.2. Interconnector capacities and power transfers

The figure consists of two maps of the Mediterranean region, showing the distribution of the number of languages spoken. The top map displays the number of languages spoken in each country, with values ranging from 180 to 790. The bottom map displays the number of languages spoken in each country, with values ranging from 30 to 1740. The maps are color-coded: yellow for the number of languages spoken and green for the number of languages spoken.

Top Map: Number of Languages Spoken (Yellow)

Country	Number of Languages Spoken
Spain	750
France	790
Italy	420
Greece	460
Turkey	180
Syria	300
Lebanon	180
Israel	180
Jordan	180
Libya	180
Algeria	180
Morocco	180
Tunisia	180
Malta	180
Cyprus	180
Slovenia	180
Croatia	180
Bosnia and Herzegovina	180
Montenegro	180
Albania	180
Macedonia	180
Serbia	180
Romania	180
Bulgaria	180
Greece	180
Turkey	180
Syria	180
Lebanon	180
Israel	180
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Serbia	180
Romania	180
Bulgaria	180
Greece	180
Turkey	180
Syria	180
Lebanon	180
Israel	180
Jordan	180
Libya	180
Algeria	180
Morocco	1

Power transmission in Libya and Egypt remain moderate, which can be attributed to prohibitive high investment costs and energy losses of AC transmission

crossing these large territorial states³³. A further finding is that Europe remains a net electric energy exporter in 2030, transmitting around 4.6 TWh to North Africa via three new power lines across the Mediterranean Sea. Only Morocco exports a minor amount of power – 0.3 TWh to Spain. Electricity generation cost differentials between the two continents explain why North Africa obviously still prefers imports from Europe: The European satellite regions operate a comparatively cheaper conventional power plant fleet (coal, nuclear power), which, together with the expected expansion of renewable power according to the NREAP goals, will -for the coming two decades- keep the (marginal) electricity generation costs in Europe still more competitive than in North Africa. The resulting North-South electricity transfers are further eased by the situation that demand in Europe stagnates while it increases in North Africa.

It is likewise remarkable that Algeria becomes a major import destination in 2030, receiving net electricity transfers from all adjacent model regions. This surprising result (typically, Algeria is known as an energy-exporting nation) can be explained by the price differentials of electricity generation between different model regions: Algeria's conventional power generation system is almost entirely fueled by natural gas, and the dominance of this expensive primary source keeps the Algerian generation costs high in comparison to other regions – despite the construction of an important number of solar power plants in the country by 2030. It is noted that this observation rests on the underlying assumption that opportunity fuel costs are being applied in the model. In practice, however, this economic rationale is often ignored: especially in oil- and gas-exporting countries, national electricity utilities receive fuels at significantly lower prices than the opportunity costs, or even enjoy fuel subsidies (for a discussion in which way such a behavior distorts the model results, see section 5.4.1).

³³

As a reminder, the present study assumes that no HVDC grid technology would be used for overland interconnection between North African model regions until 2030.

5.2.3. Capacities, generation and power plant dispatch

The power dispatch behavior in Figure V-4 shows an example of the generation systems' operation and power exchanges between the five North African model regions on a weekday in summer (July).

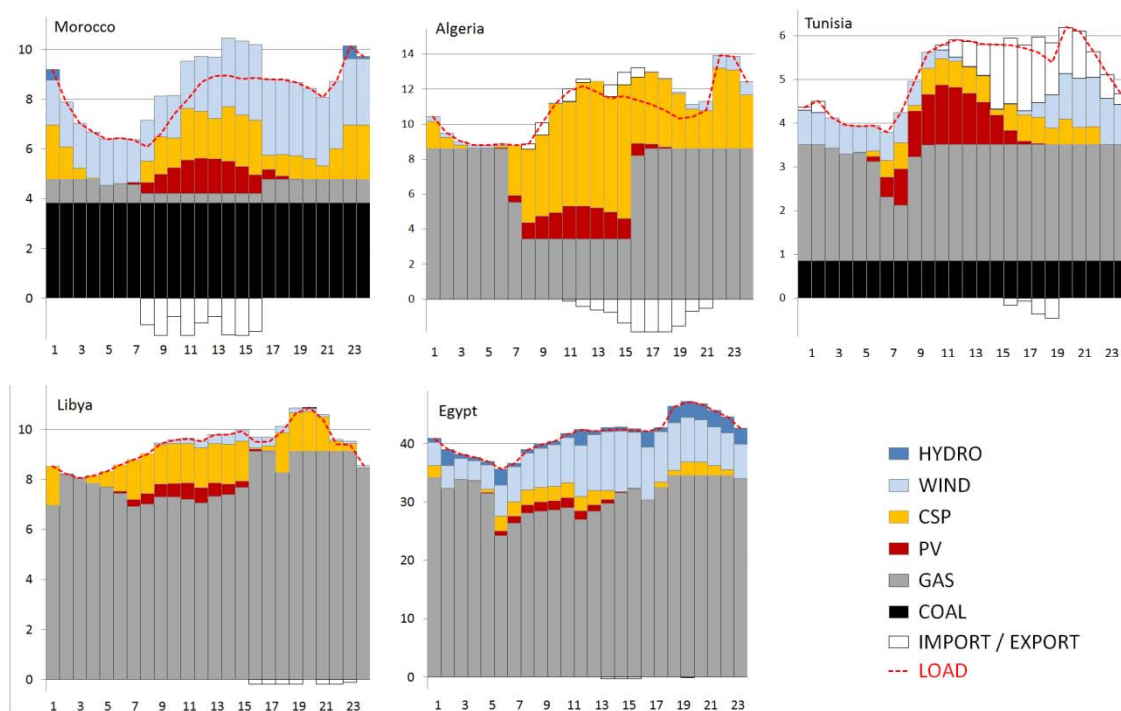


Figure V-4. Dispatch pattern (in MW) for the five North African model regions on a summer day in 2030. "Mixed capacity and import reliance" scenario.

As explained above, power export affinity increases if the marginal system costs of a model region are lower than in other regions. This is usually the case when only inexpensive intermediary or base-load power plants need to be dispatched, e.g. at times of low electricity demand (at night and in the early morning), or when RES-E power generation significantly reduces the residual load that needs to be covered by conventional power plants. In the present example (see Figure V-4, top), such an event can be observed in Morocco and Algeria, where a temporary high penetration of wind and solar power reduces the output of gas power

stations and enables electricity exports –even during the mid-day demand peak. In both cases, significant exports are the result. Conversely, the proneness to imports rises when high electricity demand meets low domestic renewable energy output. This is, in the given example, the case in Tunisia, where electricity imports are needed to cover the high residual demand in the afternoon and evening. A special feature –likewise observed in Tunisia, but also in Algeria – is simultaneous import and export of electricity. Such electricity transits can occur when a model region exhibits lower generation costs compared with one region, while they are higher in comparison with the other region.

Figure V-5 depicts the cumulative installed generating capacities (in MW) and power generation (in TWh) in all five North African model regions until 2030. It can be seen that despite the countries' renewable power goals, the bulk of the North African power supply in 2030 will still rely on natural gas, apart from a minor share of coal-fired power provided by steam coal power plants in Morocco and Tunisia. RES-E generation reaches 141.2 TWh, totaling approximately 24% of the five countries' total demand of 584.4 TWh in 2030.

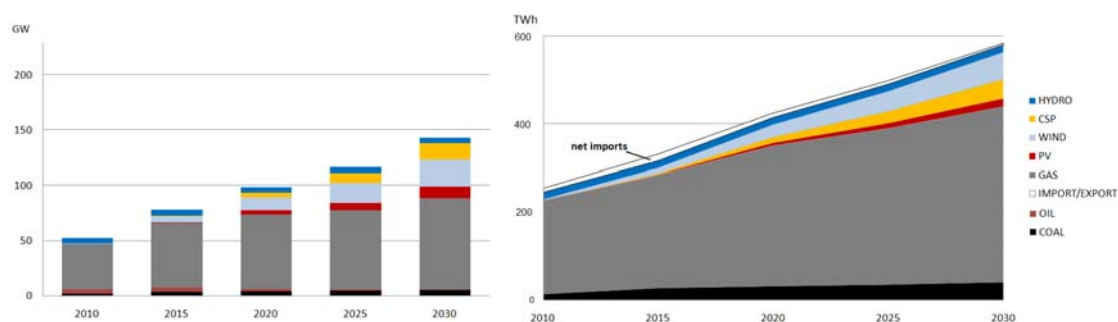


Figure V-5. Cumulative installed capacities (left) and generation (right) of the five North African countries in the “mixed capacity and import reliance” scenario.

Looking at the electricity exchanges of North Africa with its satellite regions, it is noteworthy that the construction of trans-Mediterranean interconnectors apparently constitutes a signal for electricity transfers from Europe to North Africa - and not the other way around. Although the net North-South electricity flows

of 4.3 TWh observed in the baseline scenario are comparatively small, this nevertheless points to a remarkable phenomenon that needs to be considered when designing trans-continental electricity interconnectors. That is, North African countries could be (at least temporarily) inclined to import rather than to export electricity from Europe on the transition pathway toward integrated EU-MENA power supply schemes.

5.3. Increased RES-E penetration

The outcome of the previous paragraph has shown that the power supply pattern in a North African integration scenario –in which the region reaches around 24% RES-E penetration by 2030– is not (yet) enough to trigger power exports from South to North. In this section, it shall be explored whether increased RES-E penetration could reverse this situation, and if so, what level of penetration would be required. To this end, additional model runs were carried out, simulating higher RES-E shares, reaching up to 50% of the overall North African demand by 2030. It is assumed that the additional capacity needed to fulfill these goals will be allocated according to a technology-neutral, least-cost integrated planning approach. It is additionally assumed that regardless of the North African RES-E capacity upgrades, the satellite regions' renewable expansion plans remain unaltered.

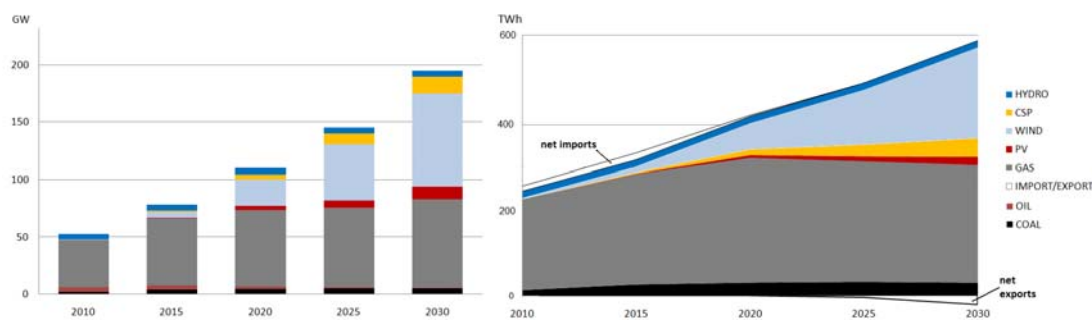


Figure V-6. Cumulative capacities (left) and generation (right) in the case of 50% RES-E penetration in North Africa until 2030.

The results for the 50% penetration case are shown in Figure V-6. Here, the least-cost approach significantly favors wind power (in comparison to CSP and

PV technology) to be installed mainly in Egypt, Morocco and, to a lesser extent, Tunisia. As a second result, a shift in the exchange pattern between North Africa and its satellite regions can be observed: North African countries actually become net exporters of electricity, but only from 2025 onward, and only if their RES-E generation exceeds >40% of their domestic demand. This result qualitatively confirms a previous finding by Haller et al. (2012), where net transmission flows from MENA to Europe were likewise observed only at relatively late stages of the transformation process – by 2030 – when high RES-E integration levels are reached.

Figure V-7 illustrates the extent to which high RES-E penetration is a prerequisite for enabling net electricity flows out of North Africa into the satellite regions. In addition to the 50% and the 24% penetration, the analysis also explores exports under a 40% penetration scenario as well as the case of total disregard of the national RES-E goals (no further additional RES-E capacity until 2030, 0%).

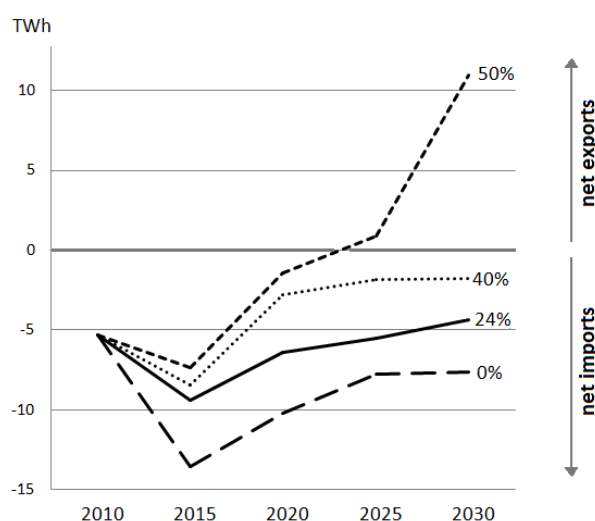


Figure V-7. Net electricity exchanges of North Africa with the satellite regions depending on different RES-E penetration ambitions by 2030. The 24% penetration represents the scenario in which the North African countries reach their current RES-E goals (“mixed capacity and import reliance scenario”).

In all RES-E expansion scenarios, the construction of new trans-Mediterranean interconnectors in 2015 would initially lead to increased net electricity flows from Europe into the North Africa model regions. These North-South flows par-

tially decrease in the following periods, as North Africa catches up with the construction of its own power plants – both renewable and non-renewable. But, in the overall picture, net imports only vanish in the very high 50%-RES-E penetration scenario, in which North Africa would be able to export around 10.9 TWh.

The finding that such great RES-E integration efforts are required before the North African region could realize first exports raises doubts about whether the paradigm of competitive, integrated EU-MENA electricity markets actually provides the right framework for enabling South-North electricity transfers. If this is an actual priority for Mediterranean politics, then it appears, in the short term, more conducive to improve the conditions for particular, ‘organized’ export projects (e.g. through bi- or multilateral off-take agreements for renewable power) than to wait for the market forces to leverage the comparatively better renewable energy resources in the South.

5.4. Sensitivities

It is known that simulations of dispatch and investment models covering long time periods are generally prone to sensitivities over technical and economic input parameter assumptions. An analysis of such sensitivities, therefore, deserves special attention - all the more so since the complexity of the model (which covers a high number of interconnected regions) increases the uncertainties. By carrying out test runs, the model results were checked for sensitivities regarding electricity demand, increased capital costs for power generation technologies, lower learning rates (and thus slower cost reductions) for renewable technologies and different fuel price scenarios. Generally, these sensitivities remain moderate, especially if the input parameter modifications are equally applied to all model regions. The strongest impact results from heterogeneities of cost input parameter between the model regions. The variable costs of electricity generation are first and foremost decisive: Even small interregional disparities in this area were able to significantly distort the outcome of the simulated generation infrastructures and inter-country transmission topologies. The following two examples briefly demonstrate that such cost disparities could very well constitute an issue in practice.

5.4.1. Impact of fuel costs disparities

Contrary to the widely applied approach of using opportunity costs for fossil fuels in power system modeling, it could be argued that due to strategic behavior (for political or social reasons) certain North African countries might become inclined to ignore this economic rationale. Already today, many MENA countries with rich hydrocarbon reserves subsidize energy inputs to their power systems (MED-EMIP, 2010), and some governments might regard future international electricity interconnectors as an additional channel to commercialize natural gas abroad –with the motive of stimulating local investments in the electricity sector and the creation of jobs. In order to get an idea of the impacts of such behavior, further simulations were carried out, assuming a scenario in which the three North African gas-heavyweights, Egypt, Algeria and Libya, ‘subsidize’ their domestic gas power plants with natural gas at prices 20% lower than the opportunity costs indicated in Table V-5. Even with this moderate price reduction³⁴, the model results are tremendous: all three countries would become major electricity exporters, not only supplying electricity to their North African neighbors, but also substantially increasing electricity transfers to the European and the eastern Mediterranean satellite regions. In 2030, the NTC capacity to the North would increase to 11 GW, with net exports totaling around 67 TWh. Although it seems unlikely that such a scenario would actually materialize to the extent shown, the results nevertheless indicate that gas-producing countries might become interested in using their market power to ‘export’ natural gas via electricity interconnectors. The question of whether this possible motivation – and maybe not so much the desire for renewable energies – could shape the transformation of electricity systems in North Africa and its interconnections with Europe requires additional research, very likely including a combined analysis of the EU-MENA electricity and gas markets.

³⁴ Given the actual natural gas production and transport costs (OME, 2010), this price level would still be largely profitable for the North African gas producers.

5.4.2. Impact of potential carbon leakage effects

A similar effect of variable cost disparities can be expected if the postulate of homogenous carbon costs for all model regions (see section 4.3.3) was removed. Currently, contrary to Europe, none of the North African or Middle Eastern countries applies carbon costs in their power systems. If no particular measures are taken, carbon leakage would likely occur on future interconnected EU-MENA electricity systems. This possibility was simulated by applying zero carbon costs for North Africa, the East Mediterranean model region, and Turkey, while carbon costs were held unaltered for the European model regions. As in the previous case, a substantial distortion of the results was observed: dramatically increasing fossil power generation in North Africa, the East Mediterranean countries and Turkey, and huge net electricity transfers (around 280 TWh per year) from these regions into Europe in 2030.

This drastic outcome underscores that measures to counter carbon leakage effects are actually imperative if stronger trans-Mediterranean electricity integration is envisaged. The European Commission already has taken a similar position with regards to the future evolution of the European and North African network, stating that “further interconnection must be accompanied by safeguards to prevent risks of carbon leakage through power imports to increase” (European Commission, 2010).

6. Discussion

Interconnecting different power systems with heterogeneous supply and demand structures generally offers important economic benefits and efficiency gains. This theorem also holds true for the North Africa power markets, as simulations with a linear optimization model demonstrate in this study. The five North African countries –currently feebly interconnected, and challenged with soaring electricity demand and ambitious renewable energy targets – could achieve important cost savings until 2030 if they were to set up more powerful transmission capacities among themselves and with their European and Middle Eastern neighbors. However, these economic benefits can only be realized if all countries show a true willingness to cooperate. Political concerns, e.g. of becoming

too dependent on foreign electricity generation, can impede the ambition to upgrade transnational interconnectors and hence reduce the economic gain of system integration. A noteworthy aspect is that the North African region, despite having more competitive RES-E resources, tends to remain a net importer of electricity from Europe at least until 2030 (the time horizon of this study). The reason for this import bias is that Europe's power mix still includes important shares of cheaper base-load generation, rendering marginal electricity production costs lower than in the gas-based North African power systems. Only by assuming a very high RES-E penetration can moderate exports to Europe be achieved. This finding indicates that the paradigm of integrated, competitive electricity markets alone might not be enough to trigger electricity exports to Europe in the short run. Additional incentives or regulatory measures are required to enable first RES-E transfers from North Africa to Europe. A further finding of the analysis is that important distortions of the North African and trans-Mediterranean power exchange patterns can be expected if certain countries demonstrate strategic behavior by strengthening their domestic power sector's competitiveness with artificially low-priced fossil fuel supplies. Similar imbalances also occur if carbon leakage, due to heterogeneous climate policies between different regions in the Mediterranean, could not be suppressed by regulatory measures. The promoters of large, integrated EU-MENA power supply schemes should consider these effects, and develop, to the greatest extent possible, a level playing field conducive to a sustainable expansion of renewable energies in the North African region.

Acknowledgement

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VI. Renewable energy perspectives for the North African electricity systems: A comparative analysis of model-based scenario studies.³⁵

Abstract

Prospects for the integration of power markets and the expansion of renewable energy have recently triggered a number of publications dealing with transformation scenarios of the North African electricity systems. This paper compares five studies using economic electricity supply- and demand models to assess possible development pathways of the North African power systems from today until 2030 and 2050. The analysis shows that distinct modeling methodologies as well as different approaches to scenario design and parameter assumptions can strongly influence the studies' results, leading to very heterogeneous projections of North Africa's power generation structures as well as the patterns of electricity exchange with other regions, like Europe. Common findings of the studies are that the surplus costs of capital-intensive renewable energy expansion in North Africa can in most cases be offset by avoided fuel costs and avoided investments in conventional power plants. All studies further agree that increased transnational cooperation, notably in terms of market integration and cross-border power exchanges, can bring about important economic advantages for the North African power sector. Renewable energy expansion could also drive electricity exports to Europe, but in integrated power market schemes, such exports only become viable with a very high share of renewable energy exceeding 60% of the North African power demand.

³⁵ This chapter is based on an article published in the journal *Energy Strategy Reviews*. Brand, B., Blok, K., Renewable energy perspectives for the North African electricity systems: A comparative analysis of model-based scenario studies. *Energy Strategy Reviews* 6 (2015) 1-11, doi: 10.1016/j.esr.2014.11.002.

1. Introduction

Electricity generation from renewable energy sources (RES-E) in North Africa has become a prominent topic in the research community. Without doubt, much of the attraction to the issue is owed to the high public profile of the ‘Desertec concept’ – the idea of large-scale solar power generation from the North African deserts to supply electricity to Europe. The concept, promoted by industry players like the Desertec Industrial Initiative (Dii, 2014), the Mediterranean power grid initiative Med Grid (2014), but also adopted by high-level policy institutions like the Union of the Mediterranean (UfM, 2014), however, remains a subject of controversial debate. Despite a general consensus among experts, that renewable energies will play a more important role in future North African power systems, the detailed scope of RES-E deployment remains ambiguous. Opinions particularly diverge about the following questions: What is the realistic expansion pathway for RES-E in North Africa, i.e. what level of RES-E penetration can technically and economically be achieved at which point of time, and how will a future technology mix – renewable, but also conventional/fossil – in electricity generation look like? To what extent and under which conditions can electricity exports from North Africa actually be realized?

In the last years, these questions have increasingly been taken up by scientific research, which led to the publication of a number of studies on North African power system scenarios. As will be shown in this paper, the studies’ approaches as well as their results are very heterogeneous, and therefore difficult to compare. This was the motivation to carry out a more thorough investigation and systematic appraisal of the publications. Five different studies were analyzed, ranging from relatively brief peer-reviewed research papers to large road-mapping reports from prominent organizations or institutions. In order to present an unbiased picture, a common, uniform comparison methodology was applied to assess the studies’ scenario building approaches, modeling methods and model outcomes.

2. History of the North African electricity supply

Before starting with the analysis of the different scenario studies, it is instructive to have a look at the historic development of the North African electricity systems in the past. Over several decades, the entire region, encompassing the countries Morocco, Algeria, Tunisia, Libya and Egypt, has been marked by ever-increasing electricity generation. Annual growth rates averaged 8.9% between 1980 and 1990; 4.5% between 1990 and 2000 and 6.4% between 2000 and 2012. As can be seen in Figure VI-1, by far the largest power producer among the five countries is Egypt, followed by Algeria, Libya, Morocco and Tunisia. The proportions of these five countries have not changed significantly over the years, although economic and demographic parameters as well as population growth and per-capita electricity demand vary substantially from country to country³⁶.

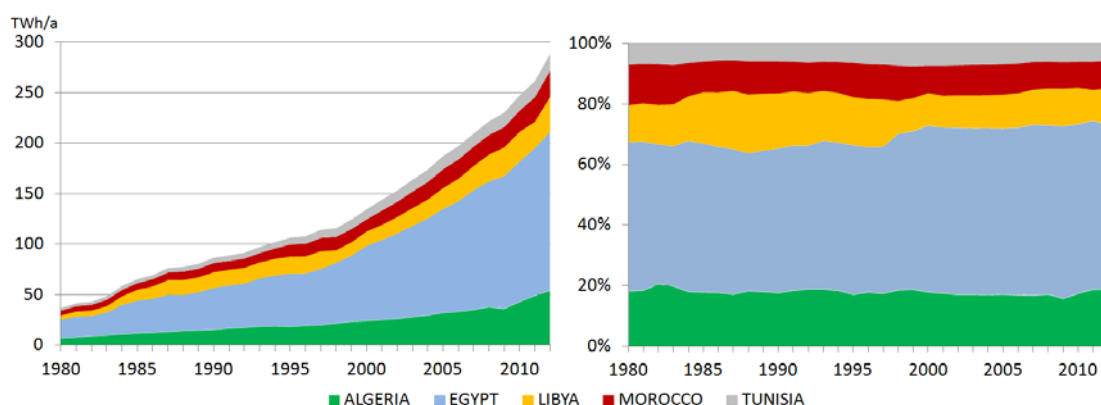


Figure VI-1 Electricity generation by country. Left: generation in TWh/year, right: percentage proportions. Source: IEA, 2013, annual reports of North African electricity utilities

Looking at the electricity mix by technology (see Figure VI-2), it becomes apparent that electricity generation in North Africa is still substantially based on fossil fuels. Today, the dominating source of power generation is natural gas (in

³⁶ Electric power consumption per capita (2011 data, World Bank): Morocco 830 kWh, Algeria 1090 kWh, Tunisia 1300 kWh, Egypt 1740 kWh, Libya 3930 kWh. Per-capita consumption in the EU is 6030 kWh.

2012: 75% of total North African generation). Since the 1980s, natural gas has continuously increased its share in the generation mix, pushing the formerly predominant, but inefficient and expensive oil³⁷ based generation to the second position (in 2012: 15%). All countries are striving to reduce oil usage in power generation, be it for reasons of cost-cutting (Morocco, Tunisia and Egypt are net petroleum importers) or to save greater quantities of this strategic fuel for export (in the cases of the net oil exporters Algeria and Libya). The shift in the fuel source, from oil to natural gas was eased further by important investments into gas extraction and transmission infrastructure that many gas-rich North African countries carried out in the 1990s in order to promote pipeline-based gas exports. These upgrades also helped to channel more natural gas into the domestic power sector, which has seen a boom in gas power plant projects in the last two decades. The share of coal in the North African power mix (in 2012: 4%) stems exclusively from Morocco, which, being the only North African country without any noteworthy hydrocarbon resources, had introduced steam coal power plants back in the 1980s, in order to diversify its electricity mix. Hydropower in North Africa (in 2012: 4%), for the most part, is supplied by Egyptian Nile river hydroelectricity and Moroccan hydro storage power plants. Due to lacking expansion potential, and as other energy sources accelerate production, the proportion of hydroelectricity in North Africa's power mix is set to decline. Non-hydro renewables, hardly noticeable in Figure VI-2, have only recently entered the North African power systems. Today, the bulk of non-hydro RES-E generation comes from wind farms, with Egypt, Morocco and Tunisia being the most important producers of wind energy. In 2012, their aggregated wind electricity production amounted to 2.8 TWh, representing about 1% of the total North African power generation of 289 TWh.

³⁷ In this article, the term 'oil' stands for all types petroleum-based liquid fuels (e.g. heavy fuel oil, light fuel oil, diesel) that can be used for power generation.

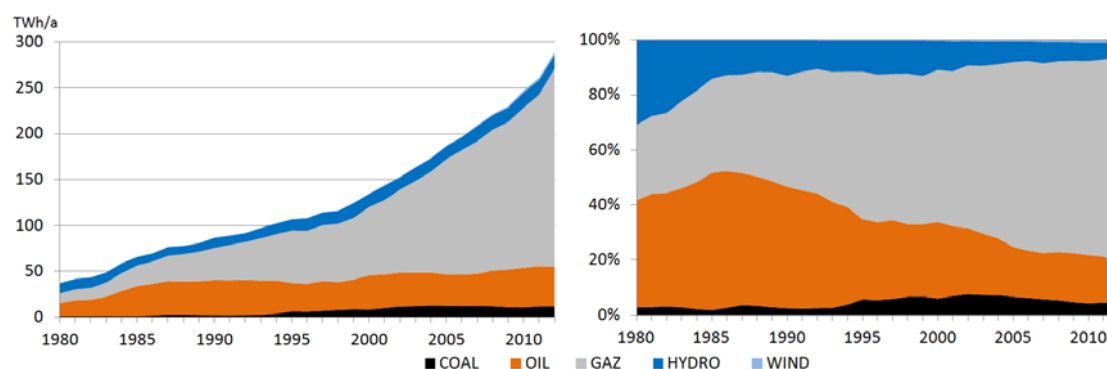


Figure VI-2. Electricity generation by source between 1980 and 2012. Left: TWh/year, right: percentage proportions. Source: IEA, 2013, annual reports of North African electricity utilities.

Today, as a result of nationally oriented energy policies, the North African power systems are fragmented into five separate electricity markets, mostly ruled by monopolistic state utilities. Despite certain efforts to increase regional cooperation and even the desire to form integrated power markets in the future (Ben Romdhane et al., 2013), the internal exchanges of electricity between the five North African countries are still hovering at a level close to zero (MED-EMIP, 2010). Somewhat more significant are the electricity exchanges between North Africa and its neighboring regions Europe and the Middle East. In 1997 and 1998, first electricity interconnectors were built between Morocco and Spain and between Egypt and Jordan. Over the past 15 years, the exchange of electricity across these power links has shown a net import balance of North Africa, which on average meets around 1% of its demand from abroad (see Section 5.4).

3. Trends, influencing factors and uncertainties of North African power system development

Forecasting the future of North Africa's electricity generation based on historic developments is certainly sketchy, but the observations outlined above nevertheless give some indications about likely trends in the near term. One of the trends with the highest inertia is certainly the growing electricity demand, which, as the developments of the past have shown, is still far from saturation in North Africa. Although exact growth rates are difficult to predict, the trajectory in Figure

VI-2, suggests a general continuation of the demand increase. A second, very certain pattern concerns the stagnation of hydro power in the North African power mix. Furthermore, liquid fuels (fuel oil, diesel) are also set to decrease their contribution to the North African electricity mix: Old, inefficient power plants of this type will continuously be replaced by more efficient gas power plants.

Much more uncertainty is associated with the question to what extent can the non-hydro renewable energy sources penetrate into the North African power systems. The favorable worldwide conditions for RES-E technologies (falling costs, improvements in technology) and the immense potential of solar and wind resources in the region speak for an accelerated RES-E expansion in the region, clearly going beyond the current 1% contribution to the North African power mix. The recent proliferation of renewable target announcements by North African governments (REN21, 2013) additionally underscores the generally positive prospects for RES-E in the region. However, doubts remain as to whether the ambitious policy targets can actually be realized as scheduled – especially in light of the current political and economic difficulties following the Arab Spring. A further open question concerns the choices of RES-E technologies. Wind power, photovoltaic (PV) and concentrated solar power (CSP) are expected to be the leading renewable technologies for bulk power generation in North Africa. However, their exact proportions in the future renewable mix are still a matter of speculation.

The same goes for the conventional fuel mix. While natural gas is set to gradually replace oil-based electricity generation, uncertainty remains about the future of coal in the North African power mix. Morocco already operates coal-fired power plants, but for instance Tunisia, suffering from declining natural gas reserves has also been deliberating the usage of coal for electricity generation (Wuppertal Instiut/Alcor, 2011) – although no final decision on this technology has been taken, yet. Whether carbon-intensive technologies will actually garner support in the region will certainly also depend on the future stance of national governments towards climate policies. Finally, many North African countries have also displayed nuclear ambitions in the past (Jewell, 2011) – which, despite the Fukushima disaster and the recent political upheavals, have never fully disappeared (Haddad, 2011). A likewise highly speculative field is whether the

prospects of large-scale intercontinental electricity exchanges, in particular with Europe, could shape the features of the North African power system. The current situation, as shown above, is mostly characterized by electricity supplied from Europe to North Africa. However, proponents of trans-Mediterranean, renewables-based electricity systems, postulate that under common political and regulatory frameworks prevailing over the entire EU-MENA region, North Africa could in the future become a major (green) electricity exporter.

Table VI-1. Summary of broad short-term trends, key uncertainties and the main factors influencing the development of the North African electricity systems

Broad short-term trends	Key uncertainties	Influencing factors
Continuation of growth in demand	Magnitude of non-hydro RES-E expansion	General political and economic situation
Stagnation of hydro power generation	RES-E technology choice (RES-E electricity mix)	Power plant technology costs Fuel costs
Decline of oil-based generation in the mix	Role of coal power Role of nuclear power	RES-E cost developments and economic potential
Expansion of non-hydro RES-E	Transnational/transcontinental electricity transmission. Detailed demand growth rates	Climate policies of North African countries Regulatory futures, market design EU-MENA electricity market integration Transnational cooperation. Social acceptance of electricity system transformation strategies.

Table VI-1 summarizes the contextual framework for building scenarios on the North African power systems. Broad trends for the near-term future – for instance the next decade – can be derived from the historic developments described in Section 2. When it comes to the mid- and long-term timeframe, e.g. as far as 2030 or 2050, any projection is subject to huge uncertainties given the multidimensionality of possible influencing factors. In this context, scenario approaches appear to be the only appropriate tool to imagine and understand potential futures of the North African power systems. At any rate, scenarios need

to be designed carefully to be sufficiently robust: This concerns the conception of reasonable political and regulatory frameworks for the scenarios, as well as an assessment of the technical and economical feasibility of the proposed power system transformation pathways. Our article places a particular emphasis on the latter requirement, by assessing publications and reports that use technoeconomic energy system modeling methods to portray North African power system transformations.

4. Overview of North African electricity system scenario studies

The literature on RES-E in North Africa has significantly increased in the past few years. Back in the 1990s, energy researchers had started to recognize the huge unexploited solar and wind potential in North Africa and developed strategies how these resources could be harnessed. Early studies were published on this topic in the 1990s by Staiss et al. (1994) and Knies et al. (1999), being the first to propose the exploitation of the North Africa's renewable energies to supply Europe. Czisch (2005) and a series of studies by the German Aerospace Center (DLR 2005, 2006 and 2009), explored large-scale renewable energy generation in the North African region in more detail, likewise including an assessment of the feasibility of RES-E exports to Europe. The leadership of mostly European (German) scholars with their strong stance on electricity exports, has been criticized, for instance, by Supersberger et al. (2010) who state that the 'Western' reasoning about North African electricity issues would exhibit a "too simplified understanding of the political dynamics in the region, largely disregarding the behavior and aims of local actors and stakeholders". Indeed, most of the above-mentioned studies – and many others that followed³⁸ – have been predominantly deliberating on North Africa's potential for exporting renewable electricity, but neglected the question how renewable energies can be integrated into domestic power generation schemes. The present analysis is intended to assess only studies that take North African electricity needs into consideration; if power exchanges between neighboring regions (Europe, Middle East) occur,

³⁸ For example May (2005), Trieb et al. (2012), and Brancucci et al. (2013) are export-focused studies.

they must take place in a generalized supply-demand regime of integrated electricity systems – not purely for export purposes. By and large, the study is therefore limited to publications that fulfill the following requirements:

- Modeling of supply and demand in integrated electricity systems. The model must consider the domestic power demand in North Africa. Not admitted are studies that only focus on renewable export options (e.g. May, 2005, Trieb et al., 2012, Brancucci et al., 2013) or treat North Africa as a simplified satellite region for the power supply of Europe (EWI/Energynautics, 2011) without sufficiently taking into account domestic electricity consumption patterns.
- Multi-regional modeling: The geographical scope of the studies must cover all five North African countries. Studies that focus only on single countries or smaller clusters of North African countries, for example the Maghreb region (Mercados, 2011; Brand and Zingerle, 2010), are not considered.
- Inter-temporal modeling: The model must be able to treat power system transformation in a dynamic manner over large timescales starting from the status quo of fossil fuel-based power system infrastructures in North Africa into the future. Not considered are static ‘greenfield approaches’ that assess the feasibility of electricity system infrastructures at a fixed target year, for example 2050, but omit modeling the respective investment pathways towards such system configurations over the intervening years (Dii, 2012; Boie et al., 2014 or Huber et al., 2012),.

A further requirement was that the studies must be sufficiently recent (not older than 5 years) in order to reflect latest developments on North African power markets and to ensure a minimum up-to-datedness of the modeling parameters, e.g. technology costs or fuel prices. At the time of writing, five publications were found that meet these requirements (see Table VI-2).

4.1. Dii (2013)

The study ‘*Desert Power - Getting Started*’, a non-peer-reviewed policy roadmap report, explores the transformation of the EU-MENA electricity system

to an unprecedented degree of comprehensiveness. Contrary to a previous publication (Dii, 2012), titled ‘Desert Power: 2050’, the updated study not only demonstrates the feasibility of an (almost) fully RES-E-based trans-continental electricity network by 2050 – it also provides insights into the details of the transformation processes needed to achieve this end-state. The authors use the cost-minimizing linear programming model PowerACE, developed by the Fraunhofer Institute for Systems and Innovation Research (ISI), to calculate the optimized power plant dispatch for all 8760 hours of the year across 42 interconnected countries in Europe and the MENA region. Investments into power generation assets and cross-country transmission capacities are endogenously optimized in 10-year intervals, while respecting technological boundary conditions, but also market diffusion constraints. The existing power plant fleet of all involved countries as well as their self-set renewable energy goals by 2020, were taken into consideration. Most essentially, by the year 2050, an overall emission cap of 194 Mt per year (amounting to average specific emissions of 24 g CO₂ / kWh) must be achieved by the entire EU-MENA power system, which is equivalent to approximately 90% emission reductions compared to 2010 levels. The model uses this central constraint to back-cast the power system transformation between 2020 and 2050 and to calculate the necessary additions in renewable power plant capacity. For the North African countries, eligible RES-E technologies are on-shore wind power, PV and CSP power plants. Dii’s report contrasts two basic scenarios against each other: One ‘Connected’ case, where the model is allowed to build additional transmission capacities between MENA power systems and the EU, and one ‘Disconnected’ scenario, where such interconnections are suppressed. An additional scenario variant is the ‘Inertia’ scenario. Here, connections between MENA and EU power systems may be realized, but in a less extensive manner, as the level of transnational cooperation is assumed to be low and climate action less ambitious.

4.2. Brand (2013)

The article ‘*Transmission topologies for the integration of renewable power in the electricity systems of North Africa*’, published in 2013, likewise employs a cost-minimizing linear dispatch- and investment modeling approach with an

hourly dispatch resolution to explore cost-optimized transformation pathways of the EU-MENA power systems. Having a clear focus on North African power systems, the study features a less refined geographical scope than Dii's Getting Started study (Dii, 2013), covering only the 5 North African countries in detail, while Europe, as well as the Middle East and Turkey are treated as simplified model regions. Moreover, the study is limited to mid-term expansion pathways with a time horizon of 2030. The scenario projections are based on a forecasting approach by which the national RES-E expansion histories, as well as the national renewable energy goals of the North African governments were extrapolated until 2030. The model calculated investments in generation capacities and transnational interconnectors in 5-year intervals, while the daily dispatch as well as the power flows between the countries were simulated on an hourly scale for 24 representative days (576 time segments or 'time slices' per year) in order to take account of the intermittency and seasonal variability of RES-E technologies. One central aim of the study was to analyze the benefits of transnational cooperation on electricity markets. Therefore, simulations were carried out for two different sub-scenarios distinguishing different levels of transnational cooperation and the willingness to set up and share transmission capacities between the North African countries.

4.3. Fragkos et al. (2012a)

Researchers at the University of Athens, Greece published a journal article in 2012 dealing with scenarios for the energy systems of the 'MED-9' countries (five North African countries, plus Jordan, Syria, Lebanon and Israel). Titled '*Model-based analysis of the future strategies for the MENA energy system*', the publication in large parts corresponds to a previous, EU-funded study by the same authors (Fragkos et al., 2012b) that explored the impact of four different high-level trends until 2030 for the South Mediterranean region: (1) A "Reference Scenario", characterized by no noteworthy climate action, a continuation of the present energy policies, and only moderate achievements in terms of RES-E expansion and international cooperation; (2) a scenario labeled "MED-EU Initiatives" where the MED-9 region – through EU support and integrated climate and energy policies – makes substantial progress in terms of RES-E expansion,

even resulting in electricity exports to Europe. The “Global Integration” scenario (3) also assumes enhanced efforts with regards to RES-E deployment and climate action, but cooperation mostly takes place in a global, multilateral context, without particular support from the EU. (4) The “Fragmentation” scenario is the most pessimistic case. It postulates a negative macro-economic future for clean energies: Political instability and conflicts hold sway in the region, leading to high risk premiums and a generalized capital shortage – ultimately constraining the countries to maintain and expand existing oil, natural gas or coal power plants instead of investing in renewable electricity generation technologies. All four scenario cases were modeled with the MENA-EDS model, a partial energy market equilibrium model, which had been tailored specifically to portray the MENA region’s energy system transformation³⁹. As the model covers the entire energy system, not only the electricity sector (which is regarded as a sub-sector of the energy system), the temporal resolution of the power plant dispatch modeling is comparatively low, with only 9 distinct time segments per year. However, the MENA-EDS model indirectly accounts for the interactions between the structure of the power generation system and the evolution of energy supply and demand. For example, massive penetration of PV and CSP technologies in power generation results in changes in generation costs and in electricity prices for end consumers, and thus to changes in the evolution of electricity demand over time.

4.4. Haller et al. (2012)

As the study’s title ‘*Decarbonization scenarios for the EU and MENA power systems*’ indicates, Haller et al. focus on climate policy-motivated scenarios. Similar to Dii’s Getting Started study, the authors proceed from the assumption that strong restrictions on carbon emissions – 90% reduction by 2050 compared to the 2010 levels – hold sway over the EU-MENA power systems. The research paper also resembles Dii’s study in other ways: Having the same time horizon (2050) and using the same modeling approach (linear optimization), both studies also discriminate between the same connection scenarios for the intercontinental

³⁹ The MENA-EDS system is basically a derivative of the PRIMES model, which had been used to simulate EU energy policy scenarios in a study by Capros et al. (2014)

transmission systems: A fully integrated, interconnected EU-MENA power system is compared against a scenario where both continents remain without any noteworthy transmission links. The interconnected scenario is labeled “PolGrid”, whereas “PolNoGrid” stands for the case where transcontinental power line projects do not materialize. Simulations were carried out with the LIMES-EU+ model, developed by the Potsdam Institute for Climate Impact Research (PIK). Within the geographical outreach of this model are 19 different EU-MENA regions; but only two that represent North Africa: Morocco, Algeria and Tunisia are aggregated into one single model region, Libya and Egypt into the other. The model’s temporal resolution considers 49 different time slices per year, consisting of 12 characteristic days in 6-hour intervals plus one additional ‘super peak’ time slice to cover the case where low renewable supply meets high electricity demand. Investments into power system infrastructure – power plants and transmission lines – were simulated in 10-year intervals.

4.5. OME (2011)

Authored by the France-based Observatoire Méditerranéen de l’Energie (OME), the non-peer-reviewed publication ‘*Mediterranean Energy Perspectives*’ presents mid-term energy system forecasts for the Mediterranean region up to 2030. The analysis covers a total of 24 countries adjacent to the Mediterranean Basin, including the five North African countries. OME simulates energy demand-supply equilibriums for the entire energy system by using an econometric model (called Mediterranean Energy Model, MEM). In OME’s model, the electricity system is – similar to Fragkos et al.’s study – being viewed as a sub-system of the whole energy system. Expansion of power plant capacity is optimized with a linear programming algorithm minimizing electricity generation costs. Electricity supply/demand is calculated in terms of total, aggregated values without particular consideration of the temporal dynamics (dispatch) of the power system. OME’s model considers the following generation technologies: coal, nuclear, natural gas, hydro, wind and solar power. No difference is made between CSP and PV technologies which are both subsumed under the term ‘solar power’. OME’s study compares two scenarios: A “Conservative” scenario, characterized by a continuation of current policies and rather low efforts towards renewable

energy deployment. The “Proactive” scenario, by contrast, stands for strong achievements in the area of energy efficiency and successful implementation of renewable energy projects. In both the Conservative as well as the Proactive scenario, OME’s study considers nuclear power a realistic option for the North African power systems, with nuclear power plants being deployed until 2030.

Table VI-2. Overview of the examined studies

Study	Scope of scenario exercise + Sub-scenarios	Num. of + scen.	Modeling approach	Time horiz.	Geogr. coverage	Model particularities / constraints	Temp. resolution (time- slices/y)	NA-EU interconne- ctors ^{***}
Dii (2013)	Decarbonization of EU-MENA power systems. <i>Interconnected / not interconn. / Inertia scenario (limited climate action)</i>	3	Lin. optimization of power system transformation	2050	EU-MENA + Saudi Arabia	CO ₂ emission cap at 194 Mt/year by 2050 Max. allowed interconnection capacity between two countries: 20GW Self-supply per country: min. 70%	8760	12
Brand (2012)	RES-E goals / different levels of cooperation and integrated system planning in NA <i>High / low level of cooperation.</i>	2	Lin. optimization of power system transformation	2030	EU-MENA	Achievement of national RES-E targets. Nuclear power expansion excluded	576	6
Fragkos et al. (2012a)	Examine different macro-economic and energy and climate policy frameworks in the MENA region. <i>Reference / MED-EU integration / Global Integration / Fragmentation</i>	4	Partial equilibrium model for MENA energy market ^{**}	2030	MED-9	CSP technology preferred for RES electricity exports Modeling of emission trading scheme (ETS) in the MED-EU integration scenario	9	7
Haller et al. (2012)	Decarbonization of EU-MENA power systems. <i>BAU / Climate Policies interconnected / not interconnected</i>	4	Lin. optimization of power system transformation	2050	EU-MENA	90% CO ₂ reductions compared to 2010 nuclear power expansion excluded, no modeling of oil-based power plants interconnector expansion rate <1 GW/y between two countries, AC transmission for submarine interconnectors.	49	3
OME (2011)	Examine different attitudes towards energy mix diversification, energy efficiency and RES-E deployment. <i>Conservative / Proactive</i>	2	Econometric supply ^{**} and demand model	2030	MED-24	Achievement of national RES-E targets (Proactive scenario) No distinction between CSP and PV technology	1	-

^{*} only main scenarios, no sensitivity cases

^{**} electricity system is modeled as sub-part of the entire energy system

^{***} used in modeling.

5. Comparison

This section summarizes the studies' most significant differences, starting with the assumptions on electricity demand, the generation mixes, carbon emissions, the economic costs and the prospect for electricity exports by North African countries.

5.1. Electricity demand

One of the most important factors influencing electricity system modeling is the electricity demand. While there appears to be a consensus that the historic trend of increase in demand will remain unbroken over the next decades, the actually forecasted projections drastically diverge from study to study (see Figure VI-3) – an indicator that the underlying mechanisms and drivers of the North African electricity demand are still not very well explored. To illustrate the high discrepancies, Figure VI-3 depicts projections of the Dii study 'Desert Power 2050' (Dii, 2012), the precursor of Dii's 'Getting Started' study (Dii, 2013), which exhibits a tremendously high differential of North African power demand forecasts, ranging from 511 TWh to 1273 TWh by 2050. The high demand scenario rests on historic statistical data of the Arab Union of Electricity, while the assumptions for the low demand result from extrapolations of the "450 Policies" scenario of IEA's World Energy Outlook (IEA, 2011). Both demand assumptions were later rectified in the "Getting Started" study, where a common value, 994 TWh/a, was used as the base for further electricity system modeling. Haller et al. (2012) propose a lower demand, 721 TWh/a, calculated on the basis of historic annual growth rates for the North African power demand⁴⁰.

⁴⁰ Haller et al. (2012) assume that historic demand growth rates linearly decrease to zero until 2050. This results in an average annual demand growth rate of 2.7% between 2010 and 2050. For the same period, demand in Dii Getting Started grows by 3.5% annually. Fragkos et al. (2012a) calculated an average demand growth of 5.8% per year between 2010 and 2030, while the OME (2011) Proactive scenario assumes an annual demand growth of 3.5% in the same time period.

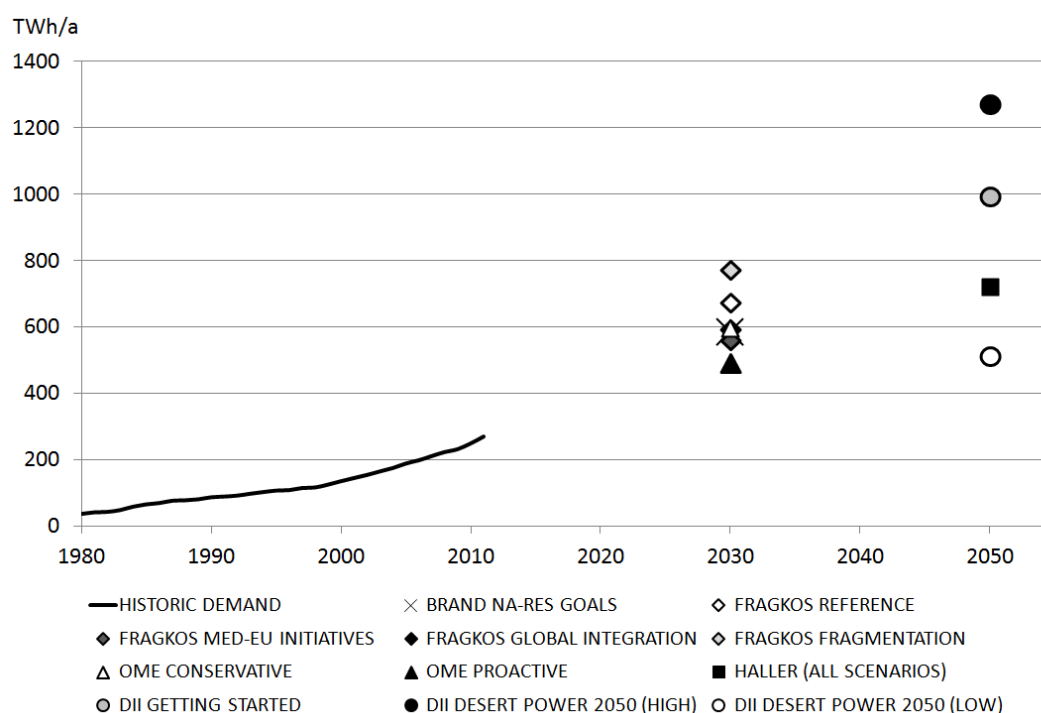


Figure VI-3. Cumulated net electricity demand (in TWh/a) of the five North African countries.

Contrary to the above-described extrapolation approaches applied by the 2050 scenario studies, the publications with a time horizon until 2030 appear to use more refined demand evaluation methods. In Fragkos et al. (2012a), for instance, electricity demand is endogenously calculated by a supply-demand equilibrium model based on demographic and macroeconomic forecasts for the North African economies. Here, the demand projections range from 560 to 771 TWh/a by 2030. OME (2011) likewise applies a bottom-up approach, based on GDP and population growth parameters, to calculate the future North African electricity demand. OME's baseline scenario ("Conservative Scenario") assumes a demand of 596 TWh/a by 2030, while the "Proactive Scenario", due to increased energy efficiency, suggests a lower demand of 491 TWh/a. Brand (2013) does not carry out its own demand assessments, but uses the demand of OME's baseline scenario (596 TWh/a by 2030) for power system modeling.

5.2. Electricity generation mix

It is insightful to compare the different studies' resulting power generation mixes. The proportions of the energy sources, as illustrated in Figure VI-4, give indications about the underlying scenario assumptions and model particularities – and even reveal preferences with regards to the technology choices for the North African power systems. The data displayed in Figure VI-4 was retrieved directly from the publications, or was kindly provided by the authors of the studies.

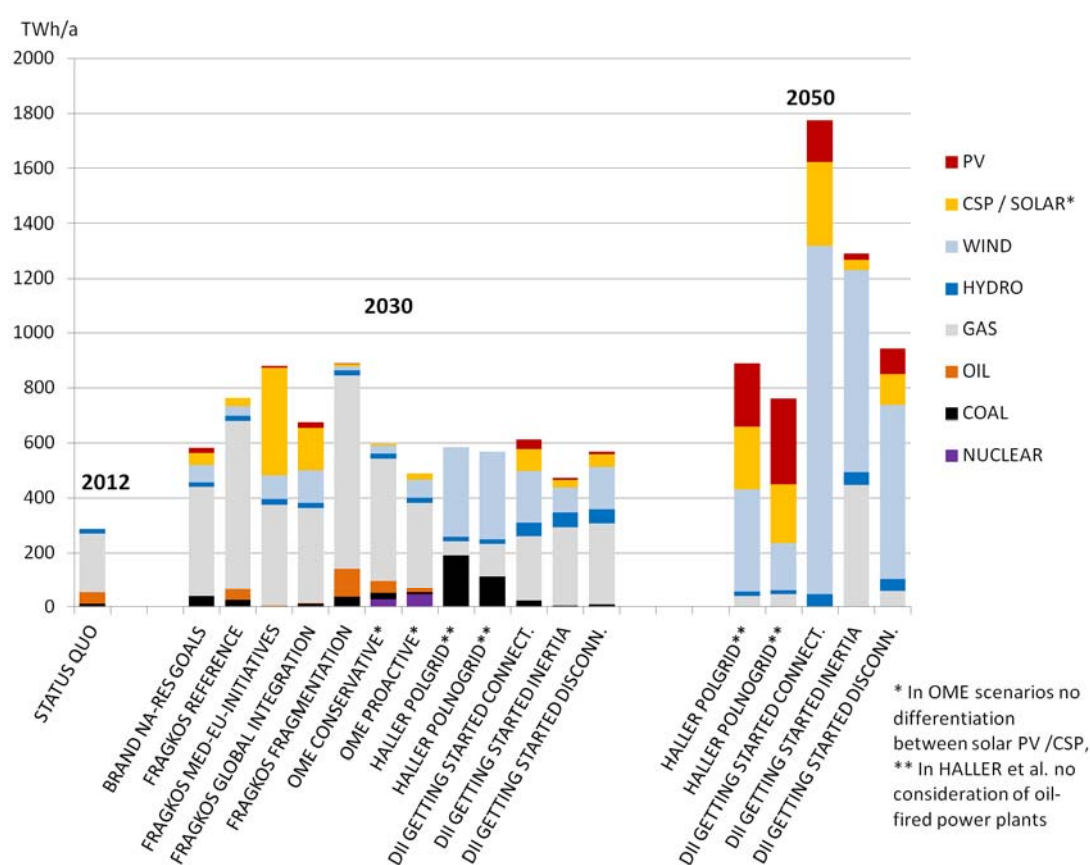


Figure VI-4. Electricity generation mix of all five North African countries by source, in the different scenario studies (values in TWh/a). Left: studies with 2030 time horizon, right: 2050 scenario studies.

5.2.1. Generation mix of the 2030 studies

For 2030, all studies forecast an increasing penetration of non-hydro renewable energies in the North African power generation mix. Even in the most pessimis-

tic case, the “Fragmentation” scenario of Fragkos et al. (2012a), the contribution of wind and solar power at around 25 TWh is significantly higher than the current production, which hardly reaches 3 TWh. The scenario with the highest projected non-hydro RES-E production (481 TWh in 2030) is the “MED-EU initiatives” scenario, also by Fragkos et al. The authors assert that the tremendous RES-E growth rates needed to reach this high level⁴¹ can be achieved by extensive efforts of North African governments towards climate action and also by massive foreign direct investments (FDI) into renewable power plant projects in North Africa. A major driver for these investments would be EU-supported large-scale green power export schemes from MENA countries to Europe. Most remarkable output in all four scenarios of Fragkos et al. is the strong expansion of CSP, clearly outrivaling wind and photovoltaic power production. The authors explain this to be the result of ‘special support’ CSP technology would receive in the MENA region, going much beyond the support for wind farms. Likewise, it is claimed that PV expansion is less systematically pursued, because PV could only be built in relatively small units and must be connected cost-inefficiently to the low and medium voltage grid. Although this argument is not fully valid – all over the world, an increasing number of PV power plants already reach the multi-MW scale, including high-voltage grid connection – the CSP bias exhibited by Fragkos et al.’s scenarios in a certain way reflects the current preferences of governments and financiers: Despite dropping prices for PV, international development banks tend to support CSP in the region, while MENA governments continue to stick to dispatchable, storage-based CSP technologies in their renewable electricity roadmaps (e.g., World Bank, 2011). OME (2011) and Brand (2013) forecast a relatively moderate expansion of solar electricity, giving clear priority to wind power as the least-cost RES-E supply option. The same is true for the studies by Haller et al. (2012) and Dii (2013), which likewise clearly favor an initial expansion of wind power on the pathway towards decarbonized North African electricity systems.

⁴¹ This expansion would require very high annual growth rates for RES-E of over 30%. These are not unrealistic, as similar growth rates have already been reached in the last two decades in other renewable energy markets, for instance in Germany or Spain.

Most scenarios agree about natural gas being the main energy carrier for the conventional part of the North African power mixes until 2030. An exception is the study of Haller et al., which projects a comparatively strong penetration of coal power (between 20% and 32% of the generation mix), combined with a large share of wind power (56%), but virtually no PV and CSP contribution. The reason for this surprising ‘wind-coal’ symbiosis (with a few amount of natural gas needed for balancing) is the still incomplete process of decarbonization by 2030. Carbon emissions are still allowed to a certain extent by the model constraints; therefore the model seeks a cost-minimizing combination consisting of coal power generation (due to low fuel prices) and wind power as the cheapest low-carbon technology. In the subsequent years, with the carbon constraint becoming more and more restrictive, electricity production using coal will gradually decline (see Section 5.2.2.). Somehow puzzling is the nuclear power supply put forth by OME (2011) in both the “Proactive” and “Conservative” scenarios. From the current perspective, looking at the long lead times for nuclear power plant construction and the instable political situation in the region as well as a rising public resistance against nuclear power even in MENA countries (see Haddad, 2011) it appears rather unlikely that nuclear power could come into gear in North Africa as early as 2030.

5.2.2. Generation mix in the 2050 studies

As already mentioned, the two studies with 2050 as the horizon, Haller et al. (2012) and *Dii Getting Started* (Dii, 2013), resemble each other in the modeling methodology (both employ linear programming techniques under similar exogenous carbon emission constraints) and with regard to the scenario design: Haller et al.’s “PolGrid” and Dii’s “Connected” scenario assume an interconnected EU-MENA electricity system, while in the “PolNoGrid”/“Disconnected” scenarios, transcontinental interconnectors are suppressed by model constraints. Despite these similarities, the results of the two studies show surprising divergences. The most distinguishing feature is the clear prevalence of wind production in Dii’s 2050 mix compared to Haller et al. In all of Dii’s scenarios (depicted in Figure VI-4), onshore wind takes the largest share of the total power generation of North Africa – between 57% and 72%; Haller et al.’s wind contribution of 23%

to 42% remains significantly below that. We explain these differences through a substantial cleavage in the studies' wind potential assumptions. In Dii's study, wind generation in North Africa features around 2700 average full load hours by 2050, while the wind prospects in Haller et al. are much less optimistic with an average of only 1660 full load hours⁴². It is obvious that under the regime of least-cost modeling, this difference renders wind power much more attractive in the Dii scenario, explaining the resulting higher wind penetration in Dii's study. In consequence, Dii's scenarios forecast a clearly more moderate expansion of solar power in the North African generation mix, reaching only a maximum of 26% in the "Connected" scenario. In Haller et al.'s scenarios on the other hand, CSP and PV make up particularly large portions of the power supply (51% to 68%) – which means a radical change compared to the situation in 2030 where wind still dominates (see Chapter 5.2.1). Likewise remarkable are the different proportions of PV and CSP in the electricity mixes. While in the paper of Haller et al. solar electricity production is mostly oriented towards PV, in Dii's study, a clear preference for CSP can be observed. CSP penetration is particularly strong in the "Connected" scenario, where the North African region exports huge amounts of electricity (see Chapter 5.4.). Apparently, the interconnected scheme favors the installation of dispatchable CSP plants offering a higher capability to cater for supply/demand balancing across the different regions of the EU-MENA electricity system.

5.3. Carbon emissions

Contrary to most western countries, where the question of carbon emissions reduction is raised in almost any debate about future energy supply schemes, this issue has so far been widely neglected in the approaches to electricity planning of the MENA countries. The scientific literature also barely mirrors the theme: Carbon emission abatement has so far been addressed only by a few MENA researchers (examples Chedid et al., 2001, El Fadel et al., 2012), while western scholars traditionally put more emphasis on this topic. This is also true for the

⁴²

A likely reason for this is that both studies use different sources and methodologies for wind power assessment (Haller et al.: NCEP/NCAR wind velocity data of 2009, calculated at 80m hub height, Dii: MERRA wind data of 2007, hub height 120m)

studies analyzed in this paper: Dii (2013) and Haller et al. (2012) explicitly utilize exogenous carbon constraints as the initial position of their modeling approaches for the EU-MENA electricity system; OME (2011) considers an increased decoupling of carbon emissions from economic growth in its “Proactive” Scenario, while Fragkos et al. (2012a) assume an emissions trading system (ETS) in the “MED-EU initiatives” scenario with carbon prices also applying to the MENA power sector. A good way to illustrate the consequences of these normative carbon reduction scenarios for the electricity systems is to compare the forecasted emissions of the North African power system with the historic values over the past three decades. Figure VI-5 depicts the specific carbon emissions in g/kWh from 1980 until 2010 (historic values based on IEA, 2012) as well as the projections of the different scenarios for 2030 and 2050.

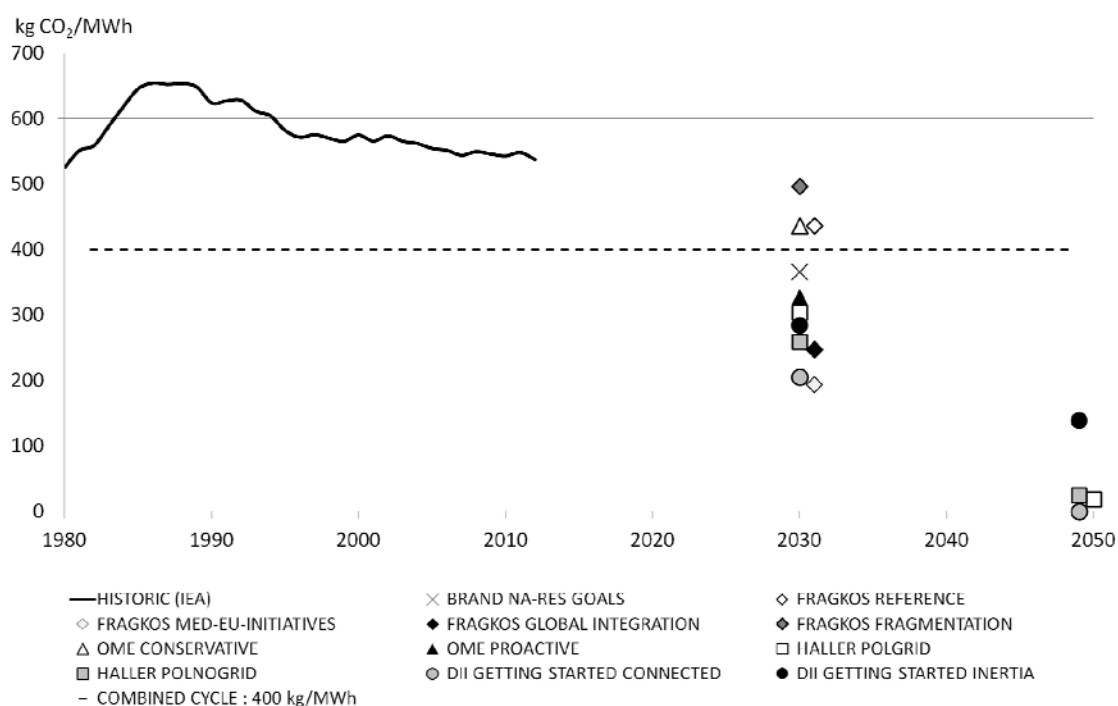


Figure VI-5. Specific direct CO₂ emissions of the North African power systems (kg/MWh). Values only include direct CO₂ emissions from fuel combustion in power plants, not life-cycle emissions.

Looking at the historic development, it can be stated that North Africa experienced an increase in specific CO₂ emissions until the mid-1980s, but later on succeeded in reducing the carbon intensity of its power system. The initial rise is

related to the increased penetration of carbon-intensive power plants (oil, open cycle gas turbine, and coal power) in the North African power mix: Back in 1980, Nile hydropower from Egypt still accounted for around 25% of the total North African power generation, keeping the specific emissions relatively low. The growing electricity demand in the subsequent years triggered the expansion of conventional fossil fuel power plants, but as no major hydropower generation capacity could be added, the specific emissions consequently increased. In a second phase, after having reached a peak in 1986, the per-kWh emissions of North African power generation decreased. Two factors explain this development: First, the reduced utilization and/or a gradual replacement of oil-fired power plants, and second, a shift to the less carbon-intensive natural gas combined cycle power plants, which are today very popular in North Africa. The average specific CO₂ emissions of a combined cycle power plant – 400 g/kWh – can be regarded as a ‘benchmark’ of what could theoretically be achieved in North Africa with an efficiency-maximized conventional power system. Set aside nuclear power and the (unlikely) diffusion of Carbon capture and storage (CCS) technologies in the region, this benchmark could only be undercut by a significant penetration of renewable electricity into the North African power mix. As depicted in Figure VI-5, most scenarios – except for OME’s “Conservative” and Fragkos’s “Reference” and “Fragmentation” scenarios – afford a very optimistic outlook in this regard, predicting a substantial acceleration of emissions reduction in 2030 and 2050, compared to the current trend.

5.4. Transmission / electricity exports

Very often, discussions about the expansion of renewable energy in the MENA region are motivated by the prospect of physical exports of green electricity to Europe. Advocates of this concept assert that the high solar and wind potential in the Southern Mediterranean would render renewable power generation in this region more competitive than in Europe; by importing cheap green electricity from North Africa, so the argument goes, European countries – especially those not so well-endowed with renewable resources – could achieve their RES-E targets more cost-efficiently. This rationale has even been taken up at the political level, for instance by the EU’s Directive on Renewable Energies (2009) which

explicitly allows EU member states to include RES-E of non-EU origin into their national renewable energy accounting. Despite the presence of this export-enabling framework, the actual reality of North-South electricity transmission looks completely different: For more than 15 years now, North Africa has been a net importer of electricity. At the moment, the main electricity import corridor is an interconnection between Spain and Morocco through which Morocco imported 4.9 TWh in 2012 (ONEE, 2012)⁴³, representing around 19% of Morocco's national electricity demand. The reason why imports from Europe are presently so attractive for North Africa, are price differentials between the North African and European electricity systems⁴⁴. Compared to the gas- and oil-dominated North African power systems, European countries can often produce electricity at lower costs, because their power plant fleets comprise comparatively more coal, lignite or nuclear power plants – all of which operate at low marginal costs. Additionally, the increased penetration of renewable energies into the European power markets brings down prices at the power spot markets (due to the 'merit order effect', Sensfuß et al., 2008), and thus increases the attractiveness of selling power abroad to North Africa. What could be done to reverse this picture, i.e. to increase the economic attractiveness of electricity transmission from North Africa to Europe? It seems obvious that to achieve a situation where the markets would favor power flowing from the South to the North, substantially more generation capacity at comparatively lower generation costs than in Europe (e.g., RES-E capacity) must be installed in North Africa. Theoretically, as the RES-E penetration increases, a point should be reached where the North African power generation costs could undercut the costs of European generation (including the costs of transcontinental electricity transmission). From the modeling perspective, it is particularly interesting to quantify what level of RES-E expansion would enable exports from North Africa.

⁴³ To a much lesser extent, electricity exchanges also take place at the eastern end of North Africa: In 2012 Egypt, exported around 780 GWh electricity to Jordan (NEPCO, 2012).

⁴⁴ For example, Morocco current procures electricity at the Iberian power exchange OMEL because prices are on average 9 €/MWh cheaper than the domestic power generation costs in Morocco (Thiam, 2013).

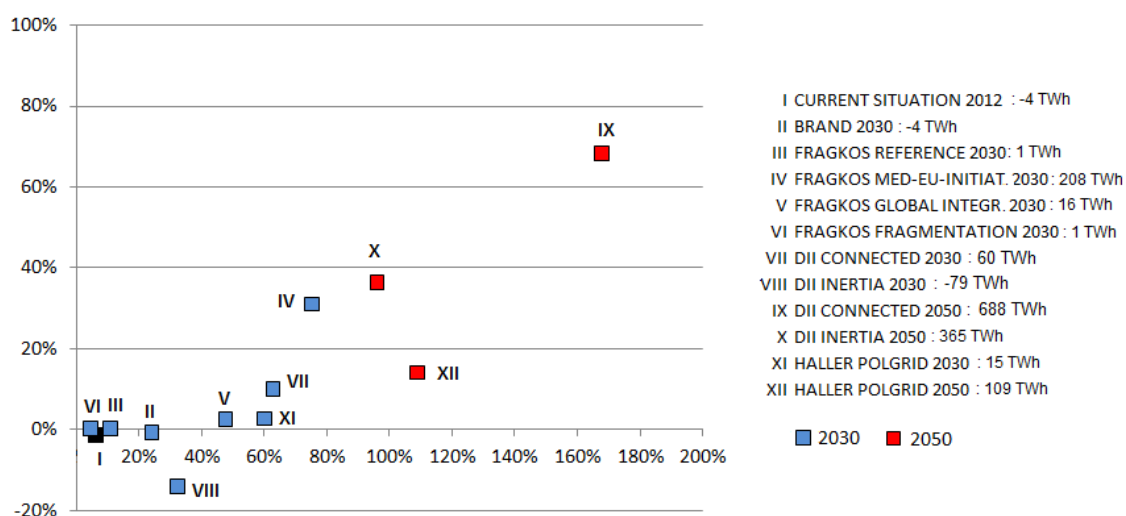


Figure VI-6. North African net power exports as a function of RES-E penetration (both in percent of the total electricity demand).

Figure VI-6 summarizes the power exchange projections of the different scenario studies. The net electricity exports of North Africa to its neighbor regions Europe and the Middle East are plotted against the renewable share of the North African power demand at different points in time. It can be observed that up to a level where RES-E covers only around 60% of the North African demand, all studies predict relatively low exports – or even net electricity imports. For instance in the Dii “Inertia” scenario (data point VIII), North Africa remains a net importer by 2030, with 14% of its electricity demand being imported. The paper by Brand (data point I) likewise forecasts net electricity imports by North Africa, though only amounting to less than 1% of its overall electricity demand by 2030. Noteworthy electricity exports out of North Africa start when the RES-E generation share in demand exceeds the 60% level: Beyond this threshold, electricity exports significantly increase. Dii’s “Inertia” scenario even performs a remarkable turnaround of the North African electricity transmission patterns: While still being a substantial importer of electric energy in 2030 (VIII), by 2050 the region becomes a net exporter (X), sending around 36% of its electricity demand abroad - mostly to Europe, but partially also to the Middle East region and to Saudi Arabia (which only Dii included in the modeling). The switch from imports to exports is realized once there is a reversion of the power generation cost differentials between North Africa and the adjacent regions. Among all

scenarios, the most substantial exports are forecast by Dii's "Connected" case (IX), where a very much larger amount of electricity, equivalent to 68% of North Africa's demand can be exported by 2050. A value rendering the picture of Figure VI-6 somehow ambiguous is the data point XII, where the "PolGrid" scenario of Haller et al. projects a significantly lower North African export rate (only 14% of the North African demand⁴⁵), despite a relatively high penetration of renewables. This can be explained with the relatively restrictive exogenous constraints concerning trans-Mediterranean grid expansion set by Haller et al. In the North-South direction, Haller et al.'s model only allows three transmission corridors, while Dii's model can add interconnectors at 12 different geographical locations across the Mediterranean. Additionally, Haller et al. limit the model's transmission options to alternating current transmission (HVAC) instead of high voltage direct current (HVDC) technologies – even for subsea interconnectors – which entails higher losses along the transmission path, and hence reduces the attractiveness of electricity exports from North Africa to Europe.

Generally, it can be stated that due to the heterogeneity of the assumptions and modeling approaches, it is rather difficult to draw final conclusions about the exact conditions needed for North African power systems to perform substantial electricity exports. Increasing the RES-E penetration to higher levels as shown above – certainly is one precondition, but definitely not the exclusive factor for leveraging electricity exports out of North Africa. Much will also depend on the power system transformations in the other regions adjacent to North Africa. This not only concerns Europe⁴⁶, but also the Middle East and even the Gulf region, which are striving for RES-E expansion as well. Scenarios examining the interaction of these power systems with the one in North Africa (e.g. through new interconnections between Egypt and Saudi Arabia⁴⁷) are at the moment not suf-

⁴⁵ From the European perspective, the electricity transfers from North Africa into 2050 account only for 2% of the European demand in Haller et. al's "PolGrid" scenario against 14% in Dii's "Connected scenario".

⁴⁶ Most studies assume that RES-E expansion in Europe follows the EU National Renewable Energy Action Plans (NREAP).

⁴⁷ A 3000 MW submarine interconnection project between Saudi Arabia and Egypt is currently under construction.

ficiently explored and further research in this area is undoubtedly needed. It goes without saying that the ‘export question’ cannot be answered with economic models alone, as it touches upon much more wide-ranging issues such as the political willingness of the involved countries to cooperate and the effectiveness of regulatory measures to enable transnational electricity exchange.

5.5. Economic costs

All five studies contain discussions of the economic costs associated with power system transformation. However, the results can only very roughly be compared, as each model uses its own cost metrics and its own assumptions for discount factors, technology costs and fuel price developments. Furthermore, the studies’ results are often displayed in such a way, that the cost incurred by the North African countries cannot clearly be discerned: Only Brand (2013) exclusively focuses on the five North African countries; OME (2011) and Fragkos et al. (2012a) link their cost results to the group of MED-24 or the MED-9 countries, while Dii (2013) and Haller et al. (2012) aggregate the costs of the entire EU-MENA region. Due to these differences, this chapter can only draw some general conclusions about the studies’ most significant cost patterns.

Most studies demonstrate that renewable energy integration in North Africa is an economically viable business case compared to strategies of conventional power system expansion. Avoided investments in conventional power plants and avoided costs for fossil fuels have the potential to offset the investments in capital-intensive RES-E expansion. For instance, OME’s study reveals that between 2012 and 2030, the renewable-friendly “Proactive” scenario brings about natural gas savings worth €100bn for the Southern Mediterranean region. These savings largely exceed the additional costs of renewable energy investments (€30bn, compared to the “Conservative” scenario) – resulting in net savings on the order of €70bn. Fragkos et al.’s study also confirms the finding that renewable energy expansion can be a financially attractive alternative, especially if seen in light of the heavy fossil fuel subsidies that many MENA economies are shouldering to keep energy prices low. Until the time horizon of 2030, the scenarios “MED-EU Integration” and “Global Integration”, both characterized by strong RES-E ambitions and an accelerated removal of energy subsidies, accumulate total system

cost savings for the MED-9 countries of more than €150bn compared to the fossil fuel-dominated “Reference” scenario that features only a partial removal of energy subsidies.

It must be mentioned that the above-described positive cost outlooks for renewable integration are only valid in the scenarios where renewable energy penetration in the electricity mix is still relatively moderate. If wind and solar technologies fully take over power generation (nearing 100% RES-E supply as in Dii, 2013 or Haller et al., 2012), increased costs for the overall system are the consequence due to higher requirements in transmission capacity, higher curtailment of intermittent power generation and a greater need for storage-bound RES technologies, such as cost-intensive CSP plants. Hence, the ‘last percentage points’ to reach a fully renewable-based power system are the most expensive – also for the North African power systems. In Dii’s study, the costs of reaching an almost 100% renewable energy supply (the “Connected” scenario by 2050) are higher by €23bn compared to the “Inertia” scenario with a relatively lower share of RES-E. Haller et al.’s decarbonization scenarios likewise demonstrate (by means of cumulative distribution functions of electricity prices) that very high RES-E penetration comes at additional costs for the EU-MENA region.

A second finding is that increased cooperation and integrated system planning generally reduces the costs of power system transformation. The effect of cooperation on interconnected power systems and market integration in North Africa is, for instance, analyzed by Brand (2013) who contrasted two cases of cooperation for the North African region: one with strong efforts to integrate power markets and the joint use of transnational interconnectors for electricity exchanges; the other with only low-profile cooperation. The results indicate that the effect of cooperative behavior and mutual use of transnational power transmission results in benefits of around €3.4bn by 2030. Haller et al. also examined the cost advantages of integrated power systems. Although the authors focus on the entire EU-MENA region and not exclusively the five North African countries, clear benefits of cooperation can be observed: In a cooperative environment (“PolGrid”), average electricity prices in 2050 are approximately 13% lower than in the “PolNoGrid” scenario where transcontinental transmission grid expansion is suppressed. Clear cost benefits of EU-MENA power system integration are also the result of Dii’s study *Getting Started* (Dii, 2013), which cal-

culated 5% lower electricity prices for the “Connected” compared to the “Disconnected” scenario, which disables interconnectors between Europe and the MENA region.

6. Conclusion

This paper has analyzed five different studies dealing with long-term renewable energy expansion scenarios for the North African electricity systems. All studies employ techno-economic energy system modeling techniques. This ensures that the proposed transformation pathways stay technically feasible and economically optimized over the long time scales of the scenarios. The appeal of such quantitative models is that they produce tangible, ‘calculated’ results – making them attractive for stakeholders to prove the cogency of their studies while conveying recommendations to policy makers. Weaknesses, however, lie in the manifold possibilities of influencing the model outcomes, e.g. by modifying parameter settings or by applying exogenous, normative constraints to the modeling process. Often, the high degree of technical sophistication of models conceals that scenario outcomes can be influenced by a priori assumptions of the modelers. As our paper shows, many studies build on constraints set by the authors. The studies with a 2050 scenario horizon, for example, assume the success of climate action in the North African region and the opening of cross-continental electricity markets to electricity transmission over large distances. Not surprisingly, such assumptions lead to model results suggesting large-scale RES-E expansion in the region – even exceeding the domestic power demand of North Africa itself. The three 2030 scenario studies exhibit more moderate perspectives for RES-E integration, because their forecasting approaches still consider an important contribution of conventional, fossil fuel-based, power generation. In these studies, diverging outlooks are observed for the future of coal and nuclear power in North Africa. In all studies, i.e. the 2030 as well as the 2050 scenarios, there also is no coherent picture of RES-E technology preferences in the North African electricity mix. The results diverge on the expected proportions of solar and wind technologies: While some studies favor wind power, the others give preference to solar PV and, particularly, CSP technologies. A high degree of heter-

ogeneity across all studies is also observed with regards to the options of North African countries to export electricity.

Despite these discrepancies, it can be stated that the five publications constitute a highly valuable contribution to the current research on North African electricity strategies. The studies demonstrate in a coherent and robust manner that RES-E integration into the regional power systems is technically and economically feasible – even up to very high degrees of penetration. By examining the conditions and limitations of these schemes within integrated power systems, the studies also shed more light on the viability of electricity exports to Europe. For instance, it is certainly helpful for North African energy planners to know that integrated EU-MENA electricity markets can only enable large-scale electricity exports to Europe, if very high levels of RES-E penetration are reached in North Africa. Consequently, a delayed RES-E expansion would result in a postponement of potential power exports. Thanks to the inter-temporal modeling approaches, these interdependencies are better understood.

Of course, techno-economic models can only describe system transformation pathways within specific scenario frameworks. For this meta-analysis, it was an opportune coincidence that the presented studies cover a broad variety of scenarios: ranging from optimistic, vision-driven scenarios to rather dismal frameworks characterized by political instability and regulatory constraints. It is true that at the moment, the situation mostly corresponds to the latter case, as many North African countries are still shaken by instability and economic downturn following the events of the Arab Spring. In this context it might be helpful to keep in mind that, setting aside the promise of large transcontinental export schemes, it would be economically beneficial for the North African power systems to pursue smaller, incremental efforts to integrate RES-E, and possibly focus on more locally based power system cooperation.

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VII. Summary and conclusion

There are many good reasons to believe in a renewable energy future for North Africa. The region is endowed with a high renewable energy potential, features relatively well-developed infrastructures, and has close ties with the energy sector of Europe, to which it supplies important quantities of natural gas and crude oil. The status as an energy exporter is certainly the reason why the perception of renewable energies in the region has been dominated by the concept of large-scale electricity transmission from North Africa to Europe. This export paradigm, however, conceals a much more important question: How can renewable energies grow endogenously (and cost-efficiently) into the existing North African electricity markets to satisfy domestic demand? An additional question concerns the impact and consequences of increasing renewable energy penetration. My thesis addressed these topics by means of five essays dedicated to different integration aspects of renewable energy sources into the electricity systems of the following countries: Morocco, Algeria, Tunisia, Libya, and Egypt. All five countries stand at the very beginnings of renewable energy expansion, with only few renewable power projects having been realized so far.

1. Summary of the results

My research followed a strongly quantitative approach, using models to simulate cost-efficient renewable energy transformation pathways and to assess the related economical and infrastructural consequences.

Chapter II started with an analysis of the electricity systems of the Maghreb region comprising Morocco, Algeria and Tunisia. These three countries were chosen because their governments pursue relatively progressive policies towards electricity market integration, and also have announced roadmaps for RES-E

expansion. Assuming that the official planning goals are realized⁴⁸, the article analyzed the impacts of RES-E integration, contrasting it with a Business-As-Usual (BAU) scenario, in which conventional (fossil-fuel) power plants remain the exclusive power generation technology. The results, obtained by linear optimization of power plant investment pathways until 2025, show that renewable energy expansion would actually have a significant impact on the main features of the Maghreb power system: the national electricity generation mix, the conventional power plant portfolio, the power plant dispatch, as well as system costs. Comparing the projected power plant portfolio of the RES-E scenario in 2025 with that of the BAU scenario, revealed that around 18% coal power capacity and 12% natural gas combined cycle power plant capacity can be avoided. The modeling results also show that the three countries under scrutiny need a more than doubled capacity of flexible open cycle gas turbines to cope with the intermittency of PV and wind power. Interestingly, the open cycle gas power plants only operate very scarcely in the daily dispatch, as their capacity is only required in times when high demand meets very low renewable feed-in. As to the electricity generation mix (in terms of energy production), the RES-E scenario entails - for all three countries - an overall decrease of electricity generation from coal power plants by 30% and a reduction of 21% of electricity production from gas power stations. The avoided consumption of hard coal and natural gas brings about a noticeable impact on the system costs - additionally to the savings related to avoided fossil power plant investments. In the overall picture, however, the net savings induced by the RES-E integration are not able to fully compensate the extra costs of the renewable targets. Nevertheless, they lead to important savings in the conventional system, amounting to approximately €2.8 billion (for all three countries until 2025). Interestingly, the largest contribution to these savings (up to 90%) stems from avoided fuel costs, while only a smaller part is related to avoided investments into new coal and gas power stations. A further observation of Chapter II was that RES-E integration is not equally efficient in the three countries. Tunisia, for instance, has a higher cost-efficiency of RES-E integration than Algerian or Morocco. This indicated that

⁴⁸ As already noted, these goals represent the status back in the year 2010 (time of writing of the article). Some of the goals were later adjusted and revised (mostly upward, particularly by Algeria and Tunisia).

there might still be room for an improvement of the current renewable goals, for instance by better adapting them (in conjunction with the conventional power system) to the specific renewable potentials of the Maghreb countries.

Chapter III scrutinized the role of concentrated solar power (CSP) in North African power systems. Despite high investment costs, CSP technology continues to be appreciated by many decision makers in the MENA region, mainly because of its ability to provide firm capacity to the power system, to store energy and to control the power output. Whether these qualities - summarized under the term “dispatchability” - actually have an economic value, was examined for CSP expansion scenarios until 2030 in Morocco and Algeria. This specific question required a redesign of the previous power system model in order to properly simulate CSP operation in the daily power plant dispatch and to calculate optimized technical power plant configurations. The analysis focused on storage-based parabolic trough technology, the currently most widespread type of CSP plants in North Africa. It could be shown that with rising CSP penetration, also the need for CSP thermal storage capacity increases. The economic “value of dispatchability” - calculated as the cost difference between storage-optimized and non-storage CSP expansion strategies - follows the same pattern: in Morocco the value of dispatchability is €39/MWh for 5% CSP penetration, but goes up to 55€/MWh with 30% CSP electricity in the generation mix. The corresponding values in Algeria are €29/MWh (5% penetration) and €35/MWh (30% penetration). The observation that CSP in Morocco features a higher value of dispatchability than in Algeria, can be explained with the better match of flexible CSP plants with Morocco’s relatively inflexible coal-based power plant environment. In Algeria, where the power system is already inherently flexible, due to the large number of natural gas-fired power stations, the need for integrating storage-based CSP plants is comparatively low. Chapter III also examined whether the economic advantages of dispatchability can outweigh the main disadvantage of CSP technology - its significantly higher investment costs compared to PV and wind technologies. By modeling competition between CSP, wind and PV, it was found that CSP integration only becomes economically advantageous in Morocco and Algeria, if high RES-E quotas are targeted in the electricity mix. From a purely economic viewpoint, it can therefore be stated

that renewable energy expansion pathways (which always start with low RES-E penetrations), should first address projects with “low-hanging renewable fruits” like wind and PV, before deploying the more costly CSP technologies, for which a need only may arise in later expansion phases when the electricity system has to cope with more intermittency due to higher wind and PV penetration.

Chapter IV demonstrated how electricity system models could also be used to support decision-making processes about power system strategies in North Africa. The background of this research was a participative stakeholder process in Tunisia, organized by the German Development Agency GIZ, the Tunisian energy conservation agency ANME and other Tunisian partners, to evaluate different electricity roadmaps for the country until 2030. Tunisia faces a soaring electricity demand and has additionally a very unbalanced primary supply because, presently, the power system is almost entirely run by natural gas. The study examined five different scenarios in which natural gas partially cedes generation shares to other technologies: (i) a strong coal power strategy in which power plants run by (imported) hard coal take over 60% of the national mix by 2030, (ii) a renewable diversification scenario with 30% renewable energy penetration (15% wind, 15% solar), (iii) a combined ‘coal-renewable’ diversification with 50% coal power and 15% renewables, (iv) a nuclear scenario with around 25% nuclear power, and (v) a business as usual (BAU) scenario foreseeing the continuation of natural gas supply, except for a small 5% contribution of wind power. By considering these proportions as exogenous constraints, the model calculated optimized capacity expansion pathways for the Tunisian power system between the years 2010 until 2030. The total system costs (investments, operation and fuel costs) over this timeframe amount to €13.1 billion for the coal power diversification scenario (the most cost-efficient expansion case) and a significantly higher €14.4 billion for the renewable expansion scenario (the scenario with the highest economic costs). At first glance, from an economic planning perspective, this would speak for the coal-based scenario as preferred expansion strategy. The objective of this project, however, was not to judge the scenarios exclusively on basis of economic cost, but also to look at aspects of supply security, environmental sustainability, and socio-economic benefits of the different expansion pathways. These aspects were assessed subsequently in a

second step after the modeling. Here, a multi-criteria decision analysis (MCDA) was applied on 13 different decision criteria that Tunisian energy experts deemed to be relevant for evaluation. The criteria pertain to the four previously mentioned key criteria groups: economic criteria (e.g., total system costs, electricity prices), security of supply (e.g. responsiveness to peak load events, natural gas dependency), socio-economic impact (e.g., job creation, social acceptance), and environmental impact (e.g. emissions of greenhouse gases, nuclear waste). The weighting parameters required for the MCDA calculations (which were performed with the TOPSIS method) originated from a survey among 37 Tunisian energy experts. The survey revealed that economic cost criteria have with 34% the highest priority, followed by concerns about security of supply (27%), socio-economic considerations (20%) and environmental aspects (19%). By introducing this prioritization in the MCDA process, a final ranking of the 5 scenarios was calculated. Compared with an economic optimization perspective, the MCDA results suggest completely different decision priorities: the scenario with the highest score is the diversification scenario with 30% renewable energy in the Tunisian electricity generation mix in 2030, while the coal diversification scenario, initially favored in an exclusively economic perspective, was placed at the fifth rank.

Chapter V differs from the previous chapters in a way that it also included the transmission system into the modeling process. The key research objective was to analyze how renewable power expansion would affect the regional transmission systems of North Africa. Additionally, the chapter explored the economic benefits of setting up transnational interconnectors for mutual sharing of generation capacity and electricity exchanges among the North African countries. A multiregional electricity market model, likewise based on linear optimization, was specifically developed to address these questions. It simultaneously optimized the generation system, as well as the expansion of transnational interconnectors between North African countries. By assuming a fulfillment of the North African renewable energy targets by 2030, the model results indicated that an efficient interconnection of the five countries could bring about net benefits of around €3.4 billion for the region. A second topic addressed in Chapter V was the economic feasibility of electricity exports from North Africa to Europe. The

model results suggest that on competitive, integrated electricity markets in the EU-MENA region, North Africa remains at least until 2030 a net importer of electricity from Europe. Only in the unlikely case that the North African countries were to outperform their own renewable energy goals (the initially planned renewable share must be doubled to over 50% of the total generation) can moderate electricity exports out of North Africa become viable.

Chapter VI concluded this dissertation with a meta-analysis of recent publications of other researchers on North African electricity systems. All studies had addressed the question of renewable energy integration into North Africa likewise by using models. The comparative analysis in this chapter featured two types of studies: those with a mid-term time horizon (2030), and long-term studies until 2050. The outcomes of the mid-term scenario studies were found to be more cautious with regards to RES-E expansion. In most cases, they only forecasted moderate RES-E shares in the power mix by 2030 (the average RES-E share was 25%, only in one case it reached 57% of total generation). The long-term scenarios were overtly more optimistic; here, the RES-E share in generation ranged in the ballpark of 95-100%. These high RES-E shares are mainly a result of strong normative constraints set by the authors, who assumed for 2050 an (almost) full decarbonization of the EU-MENA power systems. This inevitably leads to very high renewable power shares in the electricity mixes. A further finding of the meta-analysis was the heterogeneity of the projected wind and solar power proportions in the electricity mixes. While some studies expected a rather wind-based power expansion in North Africa, other studies' placed more optimism towards the solar future of North African electricity systems. A high degree of heterogeneity across all five studies was also observed in terms of electricity export options for the North African countries. Nevertheless, one general trend could be observed by comparing the studies' results: the realization of large-scale exports from North Africa to Europe depends strongly on the renewable energy penetration in the North African electricity mix. North African power systems roughly need to reach 60% RES-E penetration (of total demand) before noteworthy net electricity transfers can occur.

2. Conclusions

2.2. Cost-efficient integration of renewable energies

How can renewable energies be integrated cost-efficiently into the North African electricity systems? This central research question has been examined by quantitative methods - mostly based on economic power system modeling - throughout the five publications of my thesis. From the results, the following overall conclusions have been obtained:

Choosing the right renewable technology mix:

Onshore wind, photovoltaic, and CSP power plants are deemed to be the three most promising technologies for large-scale renewable power expansion in North Africa. Which is the “best” (i.e. economically most efficient) combination of these technologies? From a least-cost modeling viewpoint, it can be stated that, as long as renewable penetration is low and intermittency is not a significant factor, North African power systems should prioritize on renewable technologies with low investment costs. The current technology costs, but also future cost projections (technological learning) suggest that onshore wind power (as long as sufficiently good sites are available) should receive the greatest weight in the electricity mix, followed by PV and far ahead of CSP technology. Only when it comes to very high renewable penetrations (nearing 100% of the demand) may the cost-intensive CSP technology actually be required to a larger extent in North Africa. For example, according to the decarbonization scenarios presented in Chapter VI, CSP’s ability to store energy and to control the power output – dispatchability – is needed to ensure the balancing of very high amounts of intermittent generation (PV, wind) in the electricity system. Astonishing in this context is that the renewable energy targets of North African countries clearly contradict the above-outlined least-cost planning logic, as they place surprisingly high priority on CSP in the renewable expansion process (see the results of Chapters II, III, and IV). Indeed, many of the first large-scale solar power plants that have been installed, or are under construction in North Africa,

are CSP power plants⁴⁹. Believing that this observation merits special attention, one article of the thesis (see next paragraph) was exclusively devoted to the question of CSP integration.

The integration of CSP power plants.

Whenever it comes to discussions about renewable energy sources in North Africa, CSP power enjoys a particular high popularity and is often regarded as the key renewable technology for the region. Proponents assert that CSP has high potential for local manufacturing and job creation; moreover, it is suggested that its integration into the power systems would bring about important economic advantages because of the dispatchability of the technology. Indeed, outfitted with thermal storage systems, CSP plants can control their power output and adapt it to the demand patterns; additionally, they guarantee firm capacity to the power system – almost like conventional fuel-fired power plants. The economic value of these advantages, or the “value of dispatchability”, was quantified in Chapter III of this thesis. The study, carried out for Morocco and Algeria, demonstrated that the value of dispatchability depends on the specific configuration of the CSP plants (sizing of storage and mirror field), the flexibility of the surrounding conventional power system (e.g. coal or natural gas-based electricity systems) and, finally, the targeted renewable share in the electricity mix: the more renewables are to be integrated, particularly intermittent wind and PV power, the more advantageous CSP technology becomes. However, the key takeaway lesson from the study was that the value of dispatchability is hardly sufficient to justify CSP investment decisions, especially when still sufficient conventional power plants capacities (e.g. flexible natural gas power stations) are available to ensure the safe operation of the electricity system.

Transmission infrastructures.

Choosing the right transmission infrastructures between the North African countries (and their neighbors) can tremendously increase the efficiency of RES-E integration. Transnational power connectors bridge the heterogeneities between different regions, allow the dispatching of conventional power plants across

⁴⁹ Algeria and Egypt have 30 MW and 20 MW installed CSP capacity; Morocco has 20 MW CSP installed and 140 MW currently under construction (state: July 2014).

larger geographical areas, and can help level out variations of solar and wind generation. Chapter V, which examined the impact of North African renewable energy goals in an interconnected system until 2030, demonstrated that optimized transnational transmission infrastructures can bring about economic benefits in the billion-euro scale. Successful tapping of these benefits, however, depends strongly on the willingness of the involved countries to partially rely on their neighbor's electricity systems. In an environment of reduced cooperation, for example if North African countries are not willing to accept too much reliance on generation capacity and/or dependency on electricity imports from their neighbors, the economic benefits of power system integration quickly diminish. The importance of transmission systems and transnational cooperation becomes even more apparent when we assume the extreme scenarios of very high renewable penetrations. This is confirmed in Chapter VI, demonstrating that the realization of (almost) fully renewable-driven supply schemes becomes more expensive if no powerful transnational electricity grid is set up in parallel to renewable capacity expansion. Only embedded in a strongly interconnected transnational network, countries can harness the advantages of large-scale renewable electricity generation. In the wider context of fully renewable-driven EU-MENA power system scenarios (likewise discussed in Chapter VI) the competitive advantage of the North Africa's renewable energy potential becomes so high that even the construction of subsea interconnectors for electricity exports across the Mediterranean Basin is justified.

Of course, these points mentioned only highlight a few aspects of how the efficiency of renewable energy integration could be improved. Not covered in this thesis (but certainly worth a more profound investigation) is how energy efficiency (EE) measures and demand side management (DSM) strategies could impact the quality and costs of renewable energy integration into North Africa's power systems.

2.2. Consequences of renewable energy integration

Many decision makers in North Africa are concerned about the consequences and impact of the increased deployment of renewable energies into their elec-

tricity systems. This issue constitutes the secondary research question of this thesis. In Chapter IV, the example of a stakeholder inquiry carried out in Tunisia showed that, above all, the *economic costs* of renewable energy expansion is seen as a critical issue. Coming in second place are concerns about the *security of energy supply*, followed by the aspirations of a positive *socio-economic impact* of power system transformations in terms of job creation and local value generation for the domestic industry. The *ecological impact* is at present only of minor importance for North African decision makers, because this issue still receives less public attention than it does, for instance, compared to Europe. Nevertheless, it can be observed that this is about to change. Rising awareness of the population, environmentalist groups, and last but not least the international climate policy places increasing pressure on North African decision makers to look more carefully at the environmental aspects of their electricity system strategies. A further often raised question is whether renewable power integration in North Africa could also increase the region's prospects for *electricity exports* to Europe.

Economic costs.

Modeling the economic costs of renewable integration into power systems depends on a host of factors, particularly regarding the assumptions of input parameters for technology costs and fuel prices, including their projections into the future. Results of Chapters II and V indicated that, in the short- and mid-term, renewable integration in North Africa's power systems has the consequence of increased overall systems costs compared to business-as-usual conventional (fossil) electricity system scenarios. However, the costs of renewable integration can be reduced by choosing the right technology mix and by intelligently adapting it to the structures of the existing conventional power systems. By doing so, extra costs of renewable capacity additions can be partially offset by fossil fuel savings and avoided conventional power plant investments. One important finding, discussed in detail in Chapter II, is that the greatest savings stem from the avoided fuel costs and only to a lesser extent from avoided investments. This means that the cost-saving effect of renewable energy integration is particularly high for power systems that are being run by the combustion of more expensive fuels like oil or natural gas. This is the case for almost all North African coun-

tries, except for Morocco, whose primary fossil fuel for power generation is coal. It must be mentioned that this cost-saving logic implies that the concept of opportunity costs is followed, and that fossil fuel prices for the North African power sector are not kept artificially low by subsidies – a presupposition which, admittedly, is not met in all of the five countries under scrutiny⁵⁰. From a long-term renewable energy integration perspective, beyond 2030, the positive cost effects become more palpable because of the expected cost decreases for RES-E technologies (learning curves). Nevertheless, it should not be forgotten that, from the system planning standpoint, these cost benefits can only be fully reaped if the desired RES-E penetration stays at moderate levels. As it was shown in the comparative scenario analysis of Chapter VI (Meta Study), the ambition to reach very high RES-E penetrations, nearing 100%, inevitably comes with increasing system costs: the need for new transmission capacity to balance intermittency, costly storage (or storage-bound CSP technology), and higher inefficiencies due to curtailments of fluctuating wind and solar power.

Security of supply.

Like anywhere else in the world, also in North Africa electricity system planners are concerned about possible implications of RES-E integration for the safe operation of their electricity supply systems. The most frequently purported misgivings relate to the intermittency and limited predictability of wind and solar (PV) power production; moreover, there is the argument that the remote geographical locations of large-scale renewable power plants (often in desert regions and far away from the demand centers) might strain the transmission systems, especially in the large territorial states of Algeria, Libya, and Egypt, where the grids are not very densely meshed. With regard to the intermittency issue, it can be stated that the models in Chapters II-V of this thesis (as well as most models in the Meta-study of Chapter VI) have carried out simulations at 1-hour time intervals. This time resolution is deemed sufficient to address the key issues of variability renewable power sources and to give a rough first-order approximation of the system's adequacy requirements (e.g. need for backup capacity, CSP storage, curtailment of wind and PV power). Although fluctuations on

⁵⁰ Nevertheless, a recent trend to a (partial) lifting of fuel subsidies is observed in some countries, such as Egypt, Morocco, and Tunisia.

very short time scales and also forecast uncertainty could not be captured by these (deterministic) models, it can be stated with relatively high confidence that North African power systems can sustain relatively large RES-E shares without endangering the security of power supply. To this aspect it may be added that one particular advantage of North African power systems is its inherent flexibility due to the dominance of natural gas power plants in the existing capacity mix. The second issue, the question as to whether grid bottlenecks could have detrimental consequences for the security of supply in North Africa, could not be addressed by this thesis in an exhaustive manner. This limitation is related to the models' "copper plate approach", whereby electricity generation and consumption of one country are aggregated into a single node, neglecting internal congestion and grid losses. This simplified approach allows analyzing the impact of RES-E integration on cross-border transmission networks - but not the consequences for the internal grid on the national level. A further, very sensitive aspect of security of supply, might be the reservation of North African governments against (even partial) energy dependency from other countries. What the results of Chapters V and VI show, is that increasing renewable penetration in the region actually entails more cross-border exchanges of electricity, and consequently so also increases the mutual dependency of North African countries. It is obvious that for large-scale RES-E supply schemes to materialize, a certain level of cross-border market integration, regulatory rules, and finally also political trust among the North African countries is required.

Ecological and socio-economic impact.

The ecological and social impacts of renewable power integration were briefly taken up in Chapters IV and VI. As for the socio-economic consequences, Chapter IV roughly estimated the job potential and industrial value generation of renewable energies in Tunisia. The results indicate that renewable expansion can actually generate more jobs and more economic value for the local industries than conventional or nuclear power system expansion pathways. However, it must be mentioned that, in total numbers (the overall jobs created merely exceed a few thousand), these contributions remain negligible if compared to the much bigger sectors like agriculture, tourism, and manufacturing. As for the ecological impact, Chapter IV likewise quantified the consequences of renewable energy

integration in terms of avoided emissions (CO₂, SO₂, NO_x, fine dust, radioactive waste) in Tunisia, which were compared to alternative fossil or even nuclear expansion. Chapter VI – a meta-analysis of different model-based studies on North African electricity system scenarios – took a particular look at the history of specific carbon emissions between the 1980s and today, comparing it with future scenarios until 2030 and 2050. The results suggest that with RES-E integration CO₂ emissions can be brought down at reasonable costs, and that, with some effort, even a full decarbonization of the power system is feasible. What must also be mentioned, however, is that in the last two decades, North African countries have – without being motivated by climate policies – already performed impressive reductions of their power systems' specific carbon emissions, simply by gradually phasing out expensive oil-based power generation and by introducing more and more highly efficient natural gas combined cycle power plants (see Chapter VI). Additional carbon emission reductions are certainly laudable from the general viewpoint of international efforts to combat climate change; however, its direct ecological benefits in terms of global climate change mitigation can be considered negligible, as the North African economies generally are only responsible for a very small fraction (less than 1.5%) of the worldwide greenhouse gas emissions.

Electricity exports.

This topic merits some attention, because in the past years, discussions about large-scale transcontinental electricity supply schemes (Desertec) have raised hopes that electricity sales to Europe could one day become an important source of revenue for the North African economies. However, it is difficult to imagine that such schemes can quickly materialize. From the viewpoint of integrated electricity markets, power exchanges preferentially happen between areas of high and low electricity generation costs, but at present, North Africa's electricity generation is on the average clearly more expensive (in terms of marginal costs) than in Europe due to the dominance of relatively high-priced fuel inputs (mostly natural gas). Europe's power mix, contrariwise, still rests on power generation with low marginal costs – coal, nuclear – and even an increasing share of renewable generation. In Chapter IV it was shown that, at least until 2030, it is hardly possible to overcome these price differentials. To close the gap (i.e., to

increase the economic attractiveness to send electricity abroad to Europe), North Africa must install much more RES-E capacity with low marginal generation costs. From the results of the comparative analysis of different power system studies of Chapter VI, it can be derived that only very high RES-E penetrations exceeding 60% of the North African demand can trigger noteworthy electricity exports abroad.

3. Outlook

In the five essays of this thesis, I explored various aspects of renewable energy integration into the North African power systems. The key results – obtained by techno-economic modeling – demonstrate that renewable energy integration can offer considerable opportunities for the North African economies, provided that technology choices, capacity expansion strategies, and the associated (international) transmission infrastructures are planned in a coordinated and efficient manner. In addition to these technical and economic aspects, the findings of the thesis also suggest that renewable power expansion is beneficial from a wider decision-making perspective; e.g., if social, environmental, and energy security aspects were to be brought into account.

It remains surprising that, despite this multitude of promising perspectives, the region's actual achievements in terms of RES-E expansion remain relatively low. While other regions of the world can boast of important achievements in terms of renewable energy expansion, North Africa lags far behind, even behind its self-imposed official renewable energy targets and existing roadmaps. True, at the moment the region faces obvious obstacles due to political and economic instability and even security issues which partially explain the backlog. However, it can be assumed that there are likewise some other, more deep-rooted reasons explaining the weak uptake of renewables in the region. Exploring these should be a key priority of future research about North African electricity strategies.

The research for this dissertation was carried out between the years 2009 and 2013, a time when much of the debate was inspired by the Desertec concept and a somewhat “European” perception of renewable energy perspectives for the North African region. Many of the studies at that time (see chapter I.2.) largely

focused on the opportunities of North African renewable power on integrated and interconnected electricity markets with Europe. As outlined in the introduction of my thesis, there had been several attempts to support this on the political level. First, there was the European Directive 2009/28/EC (Article 9), penned with a clear intention to stimulate renewable electricity from North Africa to Europe by allowing EU member states to include these imports into their national renewable energy accounting. This opportunity, however, was not taken up by any of the EU countries so far. Also the second initiative, the Mediterranean Solar Plan (MSP), a political process initiated by the Union for the Mediterranean (UfM) to support a common framework for cross-border renewable power markets in the Mediterranean region, was stalled in late 2013, because it had not been embraced some Southern European countries. With these initiatives lying idle, it becomes increasingly doubtful whether top-down frameworks for the EU-MENA region are actually able to provide much support at all for renewable energy expansion in North Africa. At least in the short- and mid-run, there is certainly good reason to focus more on local and regional renewable energy integration - as it has been attempted in this dissertation.

What is needed at the current stage is a deeper understanding of what exactly impedes (and what could accelerate) the realization of large-scale renewable power expansion in North Africa. Here, on the national or regional level, interested researchers will find a slew of topics waiting for analysis: industrial politics, institutions and stakeholders, the role of the oil and gas sector, security aspects, transnational relationships, subsidies, and legal and regulatory aspects. Bringing these issues together, exploring their interdependencies, and possibly also drafting a model for improved energy governance structures to facilitate the integration of renewable energies in North Africa would certainly be welcomed not only by the scientific community, but also by political and industrial practitioners concerned with the development of the region's renewable energy future.

VIII. Samenvatting

In dit proefschrift worden de verschillende aspecten van de integratie van hernieuwbare energie in de elektriciteitssystemen van Noord-Afrika bestudeerd. De vijf landen in deze regio – Marokko, Algerije, Tunesië, Libië en Egypte – staan allemaal aan het begin van de ontwikkeling van hernieuwbare energie: tot nu toe zijn slechts enkele projecten rond hernieuwbare energie gerealiseerd. De hoofdvraag van dit proefschrift luidt: Hoe kunnen hernieuwbare energiebronnen voor het opwekken van elektriciteit (RES-E) kostenefficiënt worden geïntegreerd in de elektriciteitssystemen van deze landen? Dit leidt direct tot een tweede vraag naar de potentiële impact van de integratie van hernieuwbare energie – niet alleen in termen van economische kosten, maar ook met betrekking tot veranderingen in de infrastructuur, het elektrische transmissienet, de voorzieningszekerheid en milieueffecten. Beide vragen zijn onderzocht door modelleringstechnieken voor elektriciteitssystemen toe te passen.

Samenvatting

Hoofdstuk II richtte zich op de elektriciteitssystemen in de Maghrebregio. De effecten van de integratie van RES-E tot 2025 werden bestudeerd en vergeleken met het *Business As Usual* scenario (BAU), waarbij werd gefocust op de nationale doelstellingen voor hernieuwbare energie van de drie Maghreblanden in de regio (Marokko, Algerije en Tunesië). De resultaten, verkregen via lineaire optimalisatie van de investeringsroutes in de productiecapaciteit, tonen aan dat de uitbreiding van hernieuwbare energie de belangrijkste kenmerken van de elektriciteitssystemen in de Maghreb duidelijk zou veranderen: de nationale energiemix, het portfolio van conventionele energiecentrales, de inzet van energiecentrales, en de systeemkosten. Door de integratie van zonne- en windenergie zou de vraag naar kolengestookte centrales en aardgasestookte STEG-centrales afnemen, terwijl de vraag naar flexibele ‘open cycle’ gasturbinecentrales zou toenemen. De integratie van hernieuwbare energie levert bovendien belangrijke kostenbesparingen op voor het conventionele deel van het elektriciteitssysteem, met name vanwege het vermeden gebruik van steenkool en

aardgas. Deze besparingen kunnen de extra kosten van investeringen in zonne- en windenergiecentrales echter nog niet volledig compenseren.

Hoofdstuk II liet daarnaast zien dat de integratie van RES-E niet in alle drie de landen even efficiënt is. Zo leiden de doelstellingen voor hernieuwbare energie in Tunesië tot een meer efficiënte integratie dan in Algerije of Marokko. Deze vaststelling duidt erop dat nog steeds ruimte voor verbetering zit in de huidige nationale doelen voor hernieuwbare energie van de Maghreblanden. Bijvoorbeeld door deze doelen beter te laten aansluiten op de specifieke mogelijkheden voor hernieuwbare energie, maar ook op de specifieke kenmerken van de huidige conventionele elektriciteitssystemen.

In hoofdstuk III werd de rol van geconcentreerde zonne-energie (*concentrated solar power* of CSP) in Marokko en Algerije onder de loep genomen. Ondanks de hoge investeringskosten wordt CSP nog altijd gewaardeerd door vele beleidsmakers in de Midden-Oosten en Noord-Afrika (MENA)-regio, hoofdzakelijk vanwege het vermogen gegarandeerde capaciteit te leveren aan het elektriciteitsnet, om energie op te slaan en om de energie-output te beheersen. Of deze kwaliteiten – samengevat met het begrip ‘*dispatchability*’ – ook echt economische waarde hebben, is onderzocht in verschillende scenario’s voor de expansie van CSP tot het jaar 2030. De analyse richtte zich op opslaggebaseerde parabolische trogtechnologie, momenteel het meest voorkomende type CSP-centrale in Noord-Afrika. Er kon worden aangetoond dat met het verder doordringen van CSP ook de vraag naar thermische opslagcapaciteit stijgt. De economische ‘waarde van *dispatchability*’ – berekend als het verschil in kosten tussen uitbreidingsstrategieën voor CSP-systemen die geoptimaliseerd zijn voor opslag en systemen zonder opslag – varieerde tussen de € 29/MWh en € 55/MWh, afhankelijk van het land en het niveau van CSP-penetratie. CSP *dispatchability* heeft in Marokko in het algemeen een hogere waarde dan in Algerije, wat verklaard kan worden door de betere aansluiting van de flexibele CSP-centrales op een omgeving met relatief inflexibele kolencentrales in Marokko. Algerije werkt met een flexibel elektriciteitssysteem met veel aardgascentrales; hier zijn de voordelen van CSP-*dispatchability* minder noodzakelijk. In het derde hoofdstuk werd tevens onderzocht of de economische

voordelen van dispatchability opwegen tegen het grootste nadeel van CSP: de significant hogere investeringskosten in vergelijking met fotovoltäische (PV) en windtechnologieën. Door concurrentie tussen CSP, windenergie en PV te modelleren in een algemeen RES-E uitbreidingsscenario, bleek dat CSP-integratie alleen economisch voordelig wordt wanneer hoge RES-E-quota worden nagestreefd in de elektriciteitsmix van Marokko en Algerije. Vanuit een puur economisch oogpunt kan daarom worden gesteld dat uitbreidingsscenario's voor hernieuwbare energie zich in de eerste plaats zouden moeten richten op projecten rond 'laaghangend hernieuwbaar fruit', zoals wind en PV, en dat de implementatie van de duurdere CSP-technologieën moet worden uitgesteld tot latere fases van de ontwikkeling van de elektriciteitssystemen.

Hoofdstuk IV liet zien hoe de modellen voor elektriciteitssystemen ook gebruikt kunnen worden voor ondersteuning van de besluitvormingsprocessen rond strategieën voor elektriciteitssystemen in Noord-Afrika. Het in eerdere hoofdstukken gebruikte lineaire optimalisatiemodel werd hier gebruikt in combinatie met een multicriteria besluitvormingsanalyse (MCDA) om vijf verschillende diversificatiescenario's te evalueren voor de Tunesische elektriciteitsmix tot 2030: (i) een strategie sterk gebaseerd op kolenenergie (ii) een hernieuwbaar diversificatiescenario (iii) een gecombineerd kolen- en hernieuwbaar diversificatiescenario (iv) een kernenergiescenario en (v) een *business as usual* scenario (BAU) waarbij aardgas de energieopwekking blijft domineren. De MCDA is uitgevoerd voor 13 verschillende besluitvormingscriteria, ondergebracht in vier groepen: economische criteria (bv. totale systeemkosten en elektriciteitsprijzen), de voorzieningszekerheid (bv. vermogen om piekvermogens op te vangen, afhankelijkheid van aardgas), sociaaleconomische effecten (bv. creëren van werkgelegenheid, maatschappelijke acceptatie), en de impact op het milieu (bv. uitstoot van broeikasgassen, kernafval). De verschillende criteria zijn gekwantificeerd aan de hand van het elektriciteitssysteemmodel; de bepalende parameters voor de MCDA-berekeningen waren afkomstig van een onderzoek onder 37 Tunesische energie-experts. Het onderzoek liet zien dat de economische criteria met 34% de hoogste prioriteit hebben, gevolgd door zorgen over de voorzieningszekerheid (27%), de sociaaleconomische impact (20%) en milieuaspecten (19%). Door

deze uitkomsten in het MCDA-proces mee te nemen, is een rangorde aangebracht in de vijf diversificatiescenarios. Daarbij scoorde het hernieuwbare diversificatiescenario het hoogst (30% RES-E in de stroomopwekking in Tunesië in 2030); het scenario gebaseerd op diversificatie met kolenenergie – dat op basis van een puur economisch perspectief favoriet was – eindigde op de vijfde plaats.

De modelleringsmethode uit hoofdstuk V neemt, in tegenstelling tot de methodes in de eerdere hoofdstukken, ook de transmissiesystemen mee in het proces. Het belangrijkste onderzoeksdoel in dit hoofdstuk was te analyseren hoe de uitbreiding van hernieuwbare energie de regionale transmissiesystemen in Noord-Afrika zou beïnvloeden. Daarbij werd ook onderzocht wat de economische voordelen zijn van het opzetten van grensoverschrijdende interconnectoren voor het wederzijds delen van opwekcapaciteit en uitwisseling van elektriciteit tussen de Noord-Afrikaanse landen onderling. Om dit te onderzoeken is een multiregionaal model voor de elektriciteitsmarkt ontwikkeld, tevens gebaseerd op lineaire optimalisatie. Het optimaliseert zowel het systeem voor energie-opwekking als de uitbreiding van grensoverschrijdende interconnectoren tussen de Noord-Afrikaanse landen. Als wordt verondersteld dat de landen hun doel voor hernieuwbare energie in 2030 zullen hebben behaald, tonen de uitkomsten van het model dat een efficiënte verbinding tussen de vijf landen een economisch voordeel van circa € 3,5 miljard voor de regio oplevert. Maar dit voordeel is wel sterk afhankelijk van een grote bereidheid van de Noord-Afrikaanse landen om te vertrouwen op elkaars opwekcapaciteit en elektriciteitsimport. Zodra deze bereidheid afneemt, zal ook de economische winst afnemen.

Een tweede onderwerp dat in het vijfde hoofdstuk aan bod kwam, is de economische haalbaarheid van de export van elektriciteit van Noord-Afrika naar Europa. De uitkomsten van het model suggereren dat Noord-Afrika, op de competitieve en geïntegreerde elektriciteitsmarkten binnen de EU-MENA-regio, tot ten minste 2030 een netto-importeur van elektriciteit uit Europa zal blijven. Alleen in het onwaarschijnlijke geval dat de Noord-Afrikaanse landen hun doelen voor hernieuwbare energie voorbijstreven (het in eerste instantie geplande aandeel hernieuwbaar moet dan verdubbeld worden tot meer dan 50%

van de totale energieopwekking) kan een bescheiden elektriciteitsexport vanuit Noord-Afrika haalbaar worden.

Hoofdstuk VI rondde het proefschrift af met een meta-analyse van recente publicaties van andere onderzoekers over Noord-Afrikaanse elektriciteitssystemen. Alle bekeken onderzoeken benaderden het vraagstuk van de integratie van hernieuwbare energie in Noord-Afrika op een vergelijkbare manier met gebruikmaking van modellen. De vergelijkende analyse in dit hoofdstuk bevatte twee typen onderzoeken: die met een middellange tijdshorizon (2030) en die met een lange tijdshorizon (tot 2050). De resultaten van de onderzoeken met een middellange horizon bleken over het algemeen voorzichtiger in hun voorspellingen van de uitbreiding van RES-E. In de meeste gevallen voorzagen zij slechts bescheiden RES-E-aandelen in de energiemix tegen 2030 (het gemiddelde aandeel RES-E in heel Noord-Afrika was 25%). De lange termijnscenario's waren duidelijk optimistischer, met een RES-E-aandeel in de opwekking variërend tussen de 95% en 100%. Deze grote aandelen waren grotendeels te verklaren door de sterke normatieve beperkingen die de schrijvers zich hebben opgelegd; zij veronderstelden een (bijna) volledige decarbonisatie van de EU-MENA-elektriciteitssystemen in 2050. Dit leidt onvermijdelijk tot een groot aandeel van hernieuwbare energie in de stroomlevering. Een andere uitkomst van de meta-analyse in hoofdstuk zes was de heterogeniteit van de verwachte aandelen van wind- en zonne-energie in de energiemixen. Sommige scenario's voorspelden een vooral op wind gerichte productiegroei in Noord-Afrika, terwijl andere optimistischer waren over een toekomst met veel zonne-energie. De vijf studies verschilden ook veel van elkaar in de mate van de verwachte elektriciteitsexport voor de Noord-Afrikaanse landen. Desalniettemin is één trend te zien in de resultaten van de studies: de realisatie van grootschalige export van Noord-Afrika naar Europa is sterk afhankelijk van de mate waarin hernieuwbare energie weet door te dringen tot de Noord-Afrikaanse elektriciteitsmix. De Noord-Afrikaanse energiesystemen moeten grofweg een aandeel van 60% (van de totale vraag) aan RES-E bereiken voordat noemenswaardige transfers van stroom kunnen plaatsvinden.

Conclusie en aanbevelingen

In de vijf essays waaruit dit proefschrift bestaat, heb ik de verschillende aspecten van de integratie van hernieuwbare energie in de elektriciteitssystemen van Noord-Afrika onderzocht. De belangrijkste resultaten – verkregen door gebruik van technisch-economische modellen – tonen aan dat de integratie van hernieuwbare energie aanzienlijke kansen voor de Noord-Afrikaanse economie kan bieden, onder de voorwaarde dat technologische keuzes, strategieën voor capaciteitsuitbreiding en de gerelateerde (internationale) transmissienetwerken gepland worden op een gecoördineerde en efficiënte manier. In aanvulling op deze technische en economische aspecten, suggereren de uitkomsten van dit proefschrift dat de uitbreiding van hernieuwbare energie vanuit een breder perspectief van besluitvorming lonend is, bijvoorbeeld wanneer rekening wordt gehouden met sociale, milieu- en leveringszekerheidsaspecten.

Het blijft verbazingwekkend dat de regio relatief weinig heeft bereikt op het gebied van RES-E-uitbreiding, met name gezien de rooskleurige perspectieven. Terwijl andere regio's in de wereld belangrijke resultaten kunnen laten zien rond uitbreiding van hernieuwbare energie, loopt Noord-Afrika ver achter; zelfs ten opzichte van de doelen voor hernieuwbare energie die ze zichzelf hebben opgelegd. Natuurlijk, momenteel loopt de regio aan tegen overduidelijke obstakels door politieke en economische instabiliteit en zelfs veiligheidsproblemen, die de achterstand deels verklaren. Toch kunnen we aannemen dat er aan de zwakke implementatie van hernieuwbare energie in de regio dieper gewortelde problemen ten grondslag liggen. Het in kaart brengen daarvan zou de hoogste prioriteit moeten hebben in toekomstig onderzoek naar de elektriciteitsstrategieën in Noord-Afrika.

Het onderzoek voor dit proefschrift is uitgevoerd tussen 2009 en 2013, een periode waarin de discussie rondom hernieuwbare energie in Noord-Afrika grotendeels werd gedomineerd door het Desertec-concept en een nogal 'Europese' kijk op perspectieven voor hernieuwbare energie in de Noord-Afrikaanse regio. Veel van de studies die in deze periode zijn uitgevoerd (zie hoofdstuk I.2) concentreerden zich grotendeels op de kansen voor hernieuwbare Noord-Afrikaanse energie op een met Europa geïntegreerde en verbonden elektriciteitsmarkt.

Zoals uiteengezet in de inleiding van mijn proefschrift zijn er diverse pogingen gedaan om dit op politiek niveau te ondersteunen, bijvoorbeeld door de Europese Unie (in het kader van de *European Renewable Energy Directive* 2009/28/EC) en via het *Mediterranean Solar Plan* (MSP), een initiatief van de Unie voor de Mediterrane Regio (*Union for the Mediterranean*, UfM). Noch de EU, noch de MENA-landen hebben beide kansen tot nu echter aangegrepen. Als zulke initiatieven op de plank blijven liggen, wordt het steeds onzekerder of top-down initiatieven de uitbreiding van hernieuwbare energie in Noord-Afrika überhaupt kunnen ondersteunen. In elk geval op de korte en middellange termijn is er voldoende aanleiding om zich meer te concentreren op lokale en regionale integratie van hernieuwbare energie – zoals in dit proefschrift wordt gepoogd.

In dit stadium is het nodig om beter te begrijpen wat de realisatie van grootschalige hernieuwbare energie in Noord-Afrika precies belemmert, en wat haar juist kan versnellen. En dit niet zozeer op een technisch-economisch niveau, maar op het niveau van nationale en regionale energiepolitiek. Geïnteresseerde onderzoekers vinden hier legio onderwerpen voor verder onderzoek: industriebeleid, instituties en stakeholders, de rol van de olie- en gassector, veiligheidsaspecten, internationale betrekkingen, subsidies, en juridische en regelgevingsaspecten. Het samenbrengen van deze aspecten, het onderzoek naar hun onderlinge afhankelijkheden en het indien mogelijk schetsen van een model voor een betere bestuursstructuur om de integratie van hernieuwbare energie in Noord-Afrika te vergemakkelijken: dat zal niet alleen door de wetenschappelijke wereld met open armen ontvangen worden, maar ook door de politieke en industriële actoren die zich bezighouden met de ontwikkeling van hernieuwbare energie in de toekomst van deze regio.

IX. Curriculum Vitae

Bernhard Brand was born on June 6, 1974 in Würzburg. He finished high school in 1993 in Kempten, Southern Germany, where he also completed his national civilian service. From 1994 to 2000 he studied physics at the University of Heidelberg, graduating with a diploma thesis about neutron decay (nuclear physics). During his studies, he also became interested in renewable energies, but likewise discovered his passion for the Arab culture which motivated him to spend several months in Morocco for language trainings and internships in the sustainable energy sector. In 2002, he joined CIM (Center for International Migration and Development), a sub-organization of the German development agency GIZ, which deployed him for a long-term stay to Morocco as an integrated expert for a local engineering company, Afrisol S.A., based in Casablanca. In his function as assistant manager, he developed the solar photovoltaic branch for this company and carried out several off-grid power projects, including a large-scale rural electricity project on behalf of the Moroccan power company ONE for which two rural municipalities were electrified with photovoltaic power systems. Upon his return in 2005 to Germany, he became associate editor of the renewable energy publication Photon International for which he published various articles about technology and markets in the photovoltaic industry. Since 2007, Bernhard has been working as an independent consultant and solar expert for different engineering companies, e.g. Fichtner or Lahmeyer in Germany. Being involved in feasibility studies, technical advisory, and the roll-out of PV and CSP power plant projects in MENA countries, he quickly recognized the increasing importance of system integration issues when it comes to the large-scale integration of grid-connected renewable power systems. This became his motivation for the present PhD thesis. In order to strengthen his scientific skills in modeling and energy system analysis, he joined first (in 2009) the Institute of Energy Economics at the University of Cologne (EWI) as an associated researcher; later on, since 2011, the Wuppertal Institute for Climate, Environment and Energy.

X. Publications

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- Brand, B., Zingerle, J., The renewable energy targets of the Maghreb countries: Impact on electricity supply and conventional power markets. *Energy Policy* 39 (2011) 4411-4419, doi: 10.1016/j.enpol.2010.10.010.
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